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Earnings Adjustment Frictions: Evidence from the Social Security Earnings Test

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Abstract

We study frictions in adjusting earnings to changes in the Social Security Annual Earnings Test (AET) using a panel of Social Security Administration microdata on one percent of the U.S. population from 1961 to 2006. Individuals continue to "bunch" at the convex kink the AET creates even when they are no longer subject to the AET, consistent with the existence of earnings adjustment frictions in the U.S. We develop a novel framework for estimating an earnings elasticity and an adjustment cost using information on the amount of bunching at kinks before and after policy changes in earnings incentives around the kinks. We apply this method in settings in which individuals face changes in the AET benefit reduction rate, and we estimate in a baseline case that the earnings elasticity with respect to the implicit net-of-tax share is 0.23, and the fixed cost of adjustment is $152.08.

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1 Introduction

In a traditional model of workers' earnings or labor supply choices, individuals optimize their behavior frictionlessly in response to policies that affect their incentives. However, several recent papers have suggested that individuals face frictions in adjusting behavior to policy (Chetty, Looney, and Kroft 2009; Chetty, Friedman, Olsen, and Pistaferri 2011; Chetty, Guren, Manoli, and Weber 2012; Chetty, Friedman, and Saez 2012; Chetty 2012; Kleven and Waseem 2013). Costs of adjusting behavior help to govern the welfare cost of taxation (Chetty *et al.* 2009), and they also help to explain heterogeneity across contexts in the observed elasticity of earnings with respect to the net-of-tax rate (Chetty *et al.* 2011, 2012b; Chetty 2012).[1]

This paper develops evidence on the existence, nature and size of frictions in adjusting earnings in response to policy. The U.S. Social Security Annual Earnings Test (AET) represents a promising environment for studying these questions. This setting provides a useful illustration of many issues—such as the development and application of a methodology for estimating elasticities and adjustment costs simultaneously—that are applicable to studying earnings responses to policy more broadly. The AET reduces Social Security Old Age and Survivors Insurance (OASI) claimants' current OASI benefits as a proportion of earnings, once an individual earns in excess of an exempt amount. For example, for OASI claimants aged 62-65 in 2013, current OASI benefits are reduced by 50 cents for every extra dollar earned above $15,120. The AET may lead to very large effective benefit reduction rates (BRRs) on earnings above the exempt amount, creating a strong incentive for many individuals to "bunch" at the convex kink in the budget constraint located at the exempt amount (Burtless and Moffitt 1985; Friedberg, 1998, 2000).[2] Reductions in current benefits due to the AET sometimes lead to increases in later benefits; nonetheless, as we discuss in detail in Section 2, several factors may explain why individuals' earnings still respond to the AET.

The AET is an appealing context for studying earnings adjustment for at least three reasons. First, bunching at the AET kink is easily visible on a graph, allowing credible documentation of behavioral responses.[3] Second, the AET represents one of the few known kinks at which

[1] The net-of-tax rate is defined as one minus the marginal tax rate (MTR). Literature including Altonji and Paxson (1988) examines hours constraints in the context of labor supply.

[2] Other papers on the AET include Gruber and Orszag (2003) and Song and Manchester (2007).

[3] Other papers have examined bunching in the earnings schedule, including Blundell and Hoynes (2004) and Saez (2010). Saez shows that the amount of bunching can be related to the elasticity of earnings with respect to the net-of-tax rate.

bunching occurs in the U.S.; indeed, our paper represents the first study to find robust evidence of bunching among the non-self-employed at any kink in the U.S.[4] Third, the AET is important to policy-makers in its own right, as it is a significant factor that affects the earnings of the elderly in the U.S.

We make three main contributions to understanding adjustment frictions. First, we document that earnings adjustment frictions exist in the U.S., by showing that in some contexts individuals do not adjust immediately to changes in AET. We focus particularly on cases in which a kink in the effective tax schedule disappears, either because individuals reach an age at which they are no longer subject to the AET, or because legislative changes remove the AET for some groups.[5] We focus on the disappearance of kinks because in the absence of adjustment frictions, removal of a convex kink in the effective tax schedule should immediately lead to a complete lack of bunching at the earnings level associated with the former kink; thus, any observed delay in reaching zero bunching should reflect adjustment frictions. We observe clear evidence of delays in some contexts, consistent with the existence of adjustment frictions. Nonetheless, across several contexts—including both anticipated and unanticipated changes in policy—the vast majority of individuals' adjustment occurs within at most three years. Adjustment appears even faster in certain contexts.

Second, we assess the mechanisms that underlie the patterns of adjustment we observe, in order to build a model consistent with these descriptive patterns. We assess the extent to which employers play a role in coordinating individual responses to the AET by offering jobs with earnings at the AET exempt amount.[6] In our main period of study, we find little evidence that those too young to claim benefits (and therefore not subject to the AET) bunch at the kink, suggesting that the primary responses to the AET are mediated by employees' choices. We also find evidence that the individuals who respond to the removal of the AET are primarily those locating at the kink prior to its removal, suggesting that these individuals are particularly responsive. Others subject to the AET appear to be unresponsive, suggesting heterogeneity in

[4] The lack of bunching at other kinks is consistent with the existence of adjustment costs, although this finding could also be explained by other factors such as a low elasticity of earnings with respect to the net-of-tax rate. As we discuss in greater detail in Section 6, Chetty *et al.* (2012) do find evidence of more diffuse earnings responses to the Earned Income Tax Credit among the non-self-employed.

[5] For consistency with the previous literature on kink points that has focused on the effect of taxation, we sometimes use "tax" as shorthand for "tax-and-transfer," while recognizing that the AET reduces Social Security benefits and is not administered through the tax system. The "effective" marginal tax rate is affected by the AET BRR, among other factors.

[6] Due to interactions between adjustment costs for workers and hours constraints set by firms, some individuals may bunch at a kink even though they are not directly subject to the policy that creates the kink. Chetty *et al.* (2011) document that employers play such a role in Denmark.

adjustment costs or elasticities in the population.

Third, we specify a model of earnings adjustment consistent with the descriptive evidence that allows us to estimate a fixed adjustment cost and the elasticity of earnings with respect to the effective net-of-tax rate. Recent work demonstrating the importance adjustment costs has raised the question of how to estimate both the elasticity and adjustment cost simultaneously. We develop tractable methods that allow estimation of elasticities and adjustment costs with kinked budget sets. This complements Kleven and Waseem (2013), who develop a method to estimate related parameters in the presence of a notch in the budget set. Our method relies on the fact that the amount of bunching at a kink increases with the elasticity but decreases with the adjustment cost. This prevents estimation of both parameters using a single cross-section—since a small amount of bunching, for example, could be consistent with either a low elasticity or a high adjustment cost—but with with two or more cross-sections of individuals facing different tax rates in the region of the kink, we can specify two or more equations and find the values of two variables (the elasticity and the adjustment cost).[7] The model of Saez (2010) describes how bunching should vary between two different kinks in a frictionless setting, and the extent to which observed bunching deviates from this pattern is attributed to the adjustment cost. Intuitively, inertia due to an adjustment cost leads to an excess amount of bunching after a kink in the budget set becomes less sharply bent (or disappears altogether). Our primary estimation method uses the degree of such inertia (in combination with the initial amount of bunching at the kink) in estimating the size of the adjustment cost (and elasticity).[8]

We apply our method to data spanning the decrease in the AET benefit reduction rate from 50 percent to 33.33 percent in 1990 for those aged 66 to 69, as well as two settings in which the AET no longer applies for certain groups (at age 70 in the 1990-1999 period, and for ages 66-69 beginning in the year 2000). In a baseline specification examining the 1990 change, we estimate that the fixed adjustment cost is $152.08 (in 2010 dollars)—if the gains exceed this level, then the individual adjusts earnings—and that the earnings elasticity with respect to the net-of-tax share is 0.23. This specification examines data on individuals in 1989 and 1990; thus, our estimated adjustment cost represents the cost of adjusting earnings in the first year after the

[7] Under certain approximations that we later specify, this can yield a system of linear equations that can be solved in closed form. Though we employ a more general framework as our primary estimation strategy, the intuition in the simplified case helps in understanding the forces that drive our estimation.

[8] As we describe in detail later, this intuition applies to our primary empirical approach, the "Sharp Change" approach.

policy change. Other empirical strategies show results in the same range. By contrast, when we constrain the adjustment cost to be zero in 1990, we estimate a statistically significantly higher earnings elasticity of 0.39 in the baseline specification (69 percent larger than the unconstrained estimate).[9] These estimates suggest that while adjustment costs are modest in our setting, they have the potential to change elasticity estimates substantially, thus demonstrating that it can be important to incorporate adjustment costs when estimating elasticities. Nonetheless, our estimates are specific to our setting, and adjustment costs and elasticities may be substantially different (larger or smaller) in other contexts.

In the course of investigating these issues relating to frictions and earnings adjustment, we build on previous literature on the AET to provide new evidence that enriches our understanding of how the AET affects earnings. First, we systematically investigate each of the major AET policy changes since 1961. Second, we use SSA administrative data with a full sample of 13,612,313 observations on 619,580 individuals, building on certain previous studies that use survey data. Third, our study is the first to estimate bunching in the context of the AET through a method similar to Saez (2010). Fourth, we present evidence on individuals' earnings reaction to changes in the Delayed Retirement Credit. Fifth, we investigate whether mortality expectations help drive individuals' earnings responses to the AET by estimating the pattern of life expectancy around the exempt amount. Sixth, we investigate whether individuals change earnings in response to the AET by changing jobs or by changing earnings levels within a job, as well as whether employers coordinate employees on the AET exempt amount. Finally, we show that individuals serially bunch at the exempt amount.

The remainder of the paper is structured as follows. Section 2 describes the policies we examine. Section 3 presents a framework for analyzing the behavioral response to these policies and describes our empirical strategy for quantifying bunching. Section 4 describes our data. Section 5 presents empirical evidence on the earnings response to changes in the AET. Section 6 explores certain mechanisms underlying the behavioral responses. Section 7 specifies a tractable model of earnings adjustment and estimates the fixed adjustment cost and elasticity simultaneously. Section 8 concludes with discussion and avenues for future work.

[9] In our context, it makes sense that the estimated elasticity is higher when we do not allow for adjustment costs than when we do, as adjustment costs keep individuals bunching at the kink even though tax rates have fallen.

2 Policy Environment

Figure 1 shows key features of the AET rules from 1961 to 2009. The AET became less stringent over this period. The dashed line and right vertical axis show the benefit reduction rate. From 1961 to 1989, every dollar of earnings above the exempt amount reduced OASI benefits by 50 cents (until OASI benefits reached zero).[10] In 1990 and after, the benefit reduction rate fell to 33.33 percent for beneficiaries above the Normal Retirement Age (NRA).[11] The figure also shows that the AET applied to a narrower set of ages over time. In 1961, the AET applied to ages 62-71; starting in 1983, the AET was eliminated for 70-71 year-olds; and starting in 2000, the AET was also eliminated for those NRA and above. The solid line and left vertical axis show the real exempt amount. Between 1961 and 1971, the exempt amount rose with price inflation. Beginning in 1972, the exempt amount typically rose faster than inflation. Starting in 1978, the AET had different rules for beneficiaries younger than NRA and those at least NRA but younger than the maximum age subject to the AET. Subsequently, the exempt amount rose much faster on average for beneficiaries NRA and older than for younger beneficiaries.[12]

We later model the AET as creating a positive implicit marginal tax rate for some individuals, consistent with the empirical finding that some individuals bunch at AET kinks, certain theoretical considerations we describe below, and previous literature. In the empirical section, we explore evidence relating to certain mechanisms that explain this response.[13]

When current OASI benefits are lost to the AET, future scheduled benefits are increased in some circumstances. This is sometimes called "benefit enhancement." Benefit enhancement

[10] In addition to this threshold, until 1972 there was a second, higher earnings threshold over which the benefit reduction rate was 100 percent (Social Security Annual Statistical Supplement 2012). The second threshold is well above the first threshold, ranging from 25 percent to 80 percent higher depending on the year.

[11] The NRA, the age at which workers can claim their full OASI benefits, is 65 for those born 1937 and before, rises by two months a year for cohorts between 1938 and 1943, is constant at age 66 for cohorts between 1943 and 1954, and rises by two months a year until reaching age 67 for those born in 1960 and later.

[12] The exempt amount has not been a "focal" earnings level—such as $1,000, $5000, or $10,000—that could lead to bunching at the exempt amount even in the absence of AET. Indeed, in our main period of study we find no evidence of bunching at the exempt amount among those younger than the ages to which the AET applies. In 2000 and subsequently, those in the year of attaining NRA face the AET in the months prior to such attainment, but they are subject to a higher exempt amount and a benefit reduction rate of 33.33 percent.

[13] In this paper, we focus on the marginal incentives created by the AET and intensive margin responses, following previous literature based on the technique of Saez (2010). Other important decisions could include the choice of whether to earn a positive amount, or the decision to claim OASI. We abstract from the claiming decision by examining a sample of OASI claimants, following previous literature such as Friedberg (1998, 2000); however, it is worth noting that that if the AET affects the claiming decision, there is no *a priori* reason that this change in claiming should increase or decrease the magnitude of the bunching responses we document among claimants. Moreover, we add to previous literature by showing in Appendix Figure F.22 that the hazard of claiming at year $t+1$ is smooth around the exempt amount at year t, indicating no evidence that claimants come disproportionately from close to or far from the kink. We discuss the claiming decision further in the Appendix. We examine the extensive margin response in a companion paper (Gelber, Jones, and Sacks 2013). (Cogan 1981 is a classic reference on fixed costs of adjustment in the extensive margin choice.)

can reduce the effective tax rate associated with the AET, in particular for those individuals considering earning enough to trigger the enhancement in the post-1972 period, as we describe in detail in Appendix A and briefly in this section.

Prior to 1972, the AET caused a pure loss in benefits for those NRA and above, as there was no benefit enhancement for these individuals. For beneficiaries subject to the AET aged NRA and above, a one percent DRC was introduced in 1972, meaning that each year of benefits foregone led to a one percent increase in future yearly benefits. The DRC was raised to three percent in 1982 and gradually rose to eight percent for cohorts reaching NRA from 1990 to 2008 (though the AET was eliminated in 2000 for those above NRA). A increase in future benefits between seven and eight percent is approximately actuarially fair on average, meaning that an individual with no liquidity constraints and average life expectancy should be indifferent between either claiming benefits now or delaying claiming and receiving higher benefits once she begins to collect OASI (as Diamond and Gruber 1999 show with respect to the actuarial adjustment).

As we describe further in Appendix A, future benefits are only raised due to the DRC when annual earnings are sufficiently high that the individual loses an entire month's worth of OASI benefits due to the reductions associated with the AET (Friedberg 1998; Social Secuity Administration 2012). In particular, an entire month's benefits are lost once the individual earns $z^* + (MB/\tau)$ or higher, where z^* is the exempt amount, MB is the monthly benefit, and τ is the AET benefit reduction rate. With a typical monthly benefit of \$1,000 and a benefit reduction rate of 33.33 percent, one month's benefit enhancement occurs when the individual's annual earnings are \$3,000 (=\$1000/0.3333) above the exempt amount. For example, if an individual born in 1933-1934 earned at or just above this amount in years when she was subject to the DRC, future benefits were raised by 0.46 percent (but no increase occurs if the individual earns below this amount). As a result, at or just above the AET threshold, earning an extra dollar does not affect subsequent OASI benefits. Thus, benefit enhancement is only relevant to an individual considering earning substantially in excess of the exempt amount. Indeed, we later describe suggestive evidence of both little systematic bunching reaction to changes in the DRC and little relationship between bunching and life expectancy.[14]

[14]Later, our empirical specification alternatively assumes that benefit enhancement does not (or does) affect the AET implicit marginal tax rate, and we find similar patterns in both specifications.

Thus, the AET could affect the earnings decisions of those NRA and above for a number of reasons. As we have discussed, for those to whom benefit enhancement is effectively irrelevant (because they are only considering earning sufficiently near to the AET that they would not receive benefit enhancement through increasing earnings), the marginal incentives they face are not affected by benefit enhancement. For those to whom benefit enhancement is relevant (because they are considering earning in a region well above the AET exempt amount, thus triggering benefit enhancement), the AET could also affect decisions, for several reasons. First, the AET was on average roughly actuarially fair only beginning in the late 1990s. Indeed, prior to 1972, the AET represented a pure loss in benefits for those NRA and above. Furthermore, those whose expected lifespan is shorter than average should expect to collect OASI benefits for less long than average, implying that the AET is more financially punitive (though we ultimatley find no evidence consistent with this hypothesis). Liquidity-constrained individuals or those who discount faster than average could also reduce work in response to the AET. Finally, many individuals may also not understand many features of the AET or other aspects of OASI (Liebman and Luttmer 2011).

For beneficiaries under NRA, the actuarial adjustment raises future benefits whenever an individual earns any amount over the AET exempt amount.[15] Future benefits are raised by 0.55 percent per month of benefits withheld for the first three years of AET assessment. This creates a notch in the budget set at the AET threshold—as opposed to a simple kink, whose properties we explore in our theory sections. Our discussion of the effects of kinks therefore does not directly apply to pre-NRA ages. Thus, in our estimates of elasticities and adjustment costs, we limit the sample to ages NRA and above, for which the budget set (in the region of the exempt amount) is a kink rather than a notch.

3 Initial Bunching Framework

As a preliminary step, we begin with a model with no frictions. This model is well-known and described in detail elsewhere, but we briefly describe it here and in more detail in Appendix E.[16] After we have presented our empirical results, we specify a model with frictions that is

[15] Social Security Administration (2012), Section 728.2; Gruber and Orszag (2003). Formally, the number of months' worth of benefit enhancement received by OASI recipients is $floor(\tau \cdot (z - z^*)/MB)$ for those NRA and above, and $ceiling(\tau \cdot (z - z^*)/MB)$ for those below NRA. See Appendix A for more details.

[16] Saez (2010) describes this model in greater detail. This work follows earlier work on estimation of labor supply responses on nonlinear budget sets, including Burtless and Hausman (1978) and Hausman (1981). Moffitt (1990) surveys these methods.

consistent with the descriptive patterns we document.

Appendix Figure F.1 shows the budget constraint and incentives created by the AET for those NRA and above in the frictionless case. Start with a linear tax (Panel A) at a rate of τ. Now, suppose the AET is introduced (on top of pre-existing taxes), so that the marginal net-of-tax rate decreases to $1-\tau-d\tau$ for earnings above a threshold z^* (Panel B). For small $d\tau$, individuals earning in the neighborhood above z^* reduce their earnings. If ability is smoothly distributed, a range of individuals initially locating between z^* and $z^* + \triangle z$ (as depicted in the density in Panel C) will instead locate exactly at z^*, due to the discontinuous jump in the marginal net-of-tax rate at z^*. In fact, we find empirically that these individuals locate in the neighborhood of z^*, as shown in Panel D.

To measure the amount of bunching, we use a technique similar to Chetty *et al.* (2011) and Kleven and Waseem (2013), which we illustrate in Appendix Figure F.2 and describe further in the Appendix. The x-axis measures before-tax income, z, while the y-axis measures the density of earnings. In Panel A, we show that the ex-post density of earnings in the presence of a kink is comprised of a number of groups. Those in the region labeled X in the figure ("bunchers") have optimal earnings above z^* under the lower rate of τ and at z^* under the higher rate of $\tau + d\tau$. Those in the region labeled Y in the figure consist of individuals whose optimal earnings are below z^* under a lower marginal tax rate of τ, as well as other individuals whose optimal earnings are above z^* under the higher marginal tax rate of $\tau + d\tau$. Panel B shows that to estimate the size of region X, we must estimate the *ex post* density and subtract the mass associated with Group Y.

As described further in Appendix B, we divide the data into \$800 bins and estimate a seven-degree polynomial through the densities associated with the bins. In estimating this polynomial, we control for dummies for being in the seven bins nearest to the kink,[17] to capture the bunching near the kink that we wish to ignore when we estimate the counterfactual polynomial density. Our estimate of bunching, B, is the difference between the mass in these seven bins and the area under the polynomial in this \$5600-wide region. We estimate confidence intervals through a bootstrap procedure that we describe further in Appendix E.8 (and the results are similar under the delta method). We report our bunching amount, B, normalized by the share of

[17] This implies that our estimate of excess bunching is driven by individuals locating within \$2800 of the kink (as the central bin runs from \$400 under the kink to \$400 above the kink). We discuss this issue further in the Appendix. As we show in the Appendix, we have also experimented with other bandwidths, which yield similar results.

individuals in the neighborhood $[z^* - \delta, z^* + \delta]$ who belong to Group Y (which we approximate as the area under our polynomial over this range).[18]

Some apparent limitations of our approach are worth discussion. First, following previous literature on earnings responses to kinks, we do not take into account other choices that could affect earnings in the long run, such as human capital accumulation. However, human capital accumulation is likely to be less important for the older workers we study than it is for the population as a whole. Second, other programs—such as Medicaid, Supplemental Security Income, Disability Insurance, or taxes such as unemployment insurance payroll taxes—create earnings incentives near the bottom of the earnings distribution. While we acknowledge that other incentives represent a concern in principle—applicable to most of the literature on bunching at kinks—we also note that the kinks created by these programs are typically inapplicable or safely far away from the AET convex kink.[19] The results show very clear evidence of bunching at the AET kink and no visible, systematic evidence of bunching in other regions close to the AET kink. Third, we follow the previous work and largely do not distinguish among the potential reasons for a response to the AET. Following previous literature, our bunching framework presumes that consistent with the empirical evidence documenting clear responses to the incentives created by the AET, certain individuals treat the AET as creating *some* effective marginal tax rate above the exempt amount.

Finally, the results are specific to the AET and may not generalize outside of this context. We estimate the speed of adjustment among those initially bunching at a kink, a group that our empirical results suggest is more responsive to the AET than other groups. We therefore believe it is all the more interesting that we still find evidence of modest adjustment frictions among this group whose initial bunching indicates a substantial degree of flexibility (enough to locate at the kink initially). Furthermore, our estimation procedure relies on estimating bunching at more than one kink, and therefore it has the potential to incorporate information

[18] While we show this excess bunching at the kink as arising in a frictionless model here, this technique is also suited to measuring the excess bunching at the kink arising in a model with frictions (as the key in either setting is that there is excess bunching at the kink, which this technique can measure in either case).

[19] We have found that many other incentives, including income tax rates, are smooth on average around the AET convex kink. The AET also potentially creates other distortions that we discuss further in Appendix A, including: a slight notch for those NRA age and above every time an entire month's worth of benefits are lost due to the Delayed Retirement Credit; an additional non-convex kink in the budget constraint at the point at which OASI benefits are fully phased out; and a notch for those below the NRA for every month of withheld benefits that triggers the actuarially adjustment described above. However, in the case of those NRA and above that we focus on, these incentives are not likely to be relevant for potential bunchers and do not appear to be empirically relevant (as we discuss elsewhere). For these reasons, we abstract from these additional features but discuss such incentives further when we present our empirical evidence.

on the responses of individuals across a wide range of the income distribution (across multiple kinks).

4 Data

We primarily rely on the restricted-access Social Security Administration Master Earnings File (MEF) and Master Beneficiary Record (MBR), described more fully in the Appendix. The data contain a complete longitudinal earnings history with yearly information on earnings since 1951; the type and amount of yearly Social Security benefits an individual receives; year of birth; the year (if any) that claiming began; and sex (among other variables).[20] Separate information is available on self-employment earnings and non-self-employment earnings. Prior to 1978, the data measure annual Federal Insurance Contribution Act (FICA) earnings. Starting in 1978, the measure of earnings in the MEF reflects total wage compensation, as reported on Internal Revenue Service (IRS) forms. Our dataset is a one percent random sample of all Social Security numbers in the MEF, keeping all available years of data for each individual sampled.

Several features of the data are worth discussion. First, these administrative data allow large sample sizes and are subject to little measurement error. Second, earnings (as measured in the dataset) are the base for FICA taxes and are not subject to manipulation through tax deductions, credits, or exemptions. Third, because earnings are taken from the W-2 form, they are subject to third-party reporting among the non-self-employed; third-party reporting has been found in the literature to greatly reduce evasion (Kleven et al. 2011). This limits the degree to which observed bunching among the non-self-employed—to whom we limit our sample—could reflect reporting issues. Fourth, the data do not contain information on hours worked or amenities at individuals' jobs.

Table 1 shows summary statistics for the sample of individuals aged 18-75, and for the sample that we typically focus on, all those aged 62-69 who claimed by age 65.[21] In both samples, we exclude those with self-employment income. The larger (smaller) sample has 13,612,313 (1,595,139) observations on 619,580 (545,615) individuals. 56 percent (57 percent) of the sample is male. 50 percent (57 percent) of observations have positive earnings. Mean earnings in the

[20] However, we only use data since 1961; prior to 1961, the AET was substantially different, as an individual lost all of his OASI benefit when he earned above the exempt amount.

[21] In our main results, we use this fixed sample to hold the sample constant across ages or years; as we describe later, we also investigate a number of other samples as robustness checks, including a sample in which we examine only those who have claimed by the time we observe them.

sample (conditional on having positive earnings) is $37,492.28 ($29,485.08). Excluding those with self-employment income reduces the sample size (relative to the full population) by 18% among 18-75 year olds and by 12% among 62-69 year olds. Note that median earnings among our main sample of 62-69 year-olds ($17,739.68) is not far from the AET exempt amount; the population our study examines is in a range with a thick density of earnings that is not far from the median.

The second dataset we use is the Longitudinal Employer Household Dynamics (LEHD) dataset of the U.S. Census (McKinney and Vilhuber, 2008; Abowd et al., 2009), described further in the Appendix. The data are based on unemployment insurance earnings records and longitudinally follow workers' earnings over time. The data have information on around nine-tenths of workers in covered states and their employers, though we are only able to use data on a 20 percent random subsample of these individuals. We use these data primarily in order to link employees to employers, as the SSA data that we have access to have no information about individuals' employers. We secondarily use these data because the sample size we are able to obtain in the LEHD is much larger than the (large) sample size we obtain in the SSA data.[22]

5 Earnings Response to Policy

5.1 Descriptive Evidence from Policy Variation Across Ages

We first examine the pattern of excess bunching across ages, in order to determine how quickly individuals respond to changes in policy across ages and whether they face delays in responding (consistent with the existence of adjustment frictions). Empirical work often estimates only short-run responses to changes in policy (see Saez *et al.*, 2012, for a review of literature on earnings responses to taxation). If individuals are able to respond more (less) in the long run than in the short run, then this large body of work would under-estimate (over-estimate) long-run responses. Moreover, most empirical specifications have related an individual's tax rate in a given year to the individual's earnings in that year. In order to choose the appropriate time horizon over which to study behavioral responses to policy, it is necessary to establish how long

[22] The LEHD lacks information on whether a given individual is claiming OASI. Nonetheless, the ultimate importance of this shortcoming is limited. In our SSA data, 97 percent of people claim by age 69, so it is a safe assumption that the great majority of the individuals observed in the LEHD data of the ages we are interested in (primarily ages 69 and 70) have claimed OASI. The magnitude of the bunching we observe is likely to be slightly understated relative to the magnitude we would measure in the population of OASI claimants, as the results include non-claimants in the sample. However, our primary interest concerns the patterns of responses to the AET across ages and over time, which prove to be visually and statistically clear in the LEHD.

it takes to respond to policy changes.

Subsequent to 1982, the AET applies to ages 62-69. The policy changes at ages 62 and 70—the imposition and removal of the AET—are "anticipated," by which we refer to changes that would be anticipated by those who have knowledge of the relevant policies. We begin by examining the period 1990-1999. Figure 2 plots earnings histograms for each age from 59 to 73. Earnings are measured along the x-axis, relative to the exempt amount, which is shown by a vertical line.[23]

Figure 2 shows clear visual evidence of substantial excess bunching from ages 62-71.[24] Figure 3 plots the point estimates and 95 percent confidence intervals for bunching at each age. Bunching is statistically significantly different from zero at each age in the 62-71 range ($p < 0.01$ at all ages). We find no evidence of adjustment in anticipation of future changes in policy, as those younger than 62 do not bunch.[25]

We do find evidence that unbunching takes more than one year, however, as those ages 70 and 71 show modest bunching. Figure 2 shows that the density near the kink is raised at these ages, and Figure 3 shows that the estimates of bunching are statistically significantly different from zero. Three other considerations also indicate that this reflects excess bunching at these ages. First, we show later that the statistically significant positive estimates are robust to varying the degree of the polynomial, the excluded region, and the bandwidth used in the estimates. Second, the distributions at other ages not affected by the AET that represent reasonable counterfactuals (such as 61 or 73), show nearly perfectly smooth earnings distributions, suggesting that the excess mass near the kink at ages 70 and 71 would not arise in the absence of the AET. Third, Appendix Figure F.3 shows that the mean percentage change in earnings from age 70 to 71 shows a modest spike near the exempt amount, consistent with continued earnings adjustment from age 70 to age 71 among those near the kink at age 70. We find it striking that even among the group bunching prior to age 70—that (the data reveal) are able to adjust earnings to the kink—we still find evidence of modest adjustment frictions.

[23] For ages younger than 62, we define the (placebo) kink in a given year as the kink that applies to pre-NRA individuals in that year. For individuals 70 and above, we define the (placebo) kink in a given year as the kink that applies to post-NRA individuals in that year.

[24] As discussed above, in this period individuals aged 62 to 64 faced a notch in the budget set (due to the actuarial adjustment of benefits) at the exempt amount; thus, the incentives they faced were different than those for individuals aged 65 to 69. However, the histograms show no evidence of a spike in earnings just above the kink (as one would predict if they respond to the incentives created by this notch).

[25] If the cost of adjustment in each year rose with the size of adjustment and this relationship were convex, we would expect anticipatory adjustment.

Figure 3 shows that excess bunching is substantially lower at age 65 than surrounding ages. The location of the kink changes substantially from age 64 to age 65; as Figure 1 shows, during this period the exempt amount is much higher for individuals NRA and above than for individuals below NRA. Individuals may have difficulty adjusting their earnings to the new, higher kink within one year.[26] This suggests that individuals also face delays in adjusting in this context.[27]

Similar patterns of adjustment occur when looking at the periods 1972-1982, 1983-1989 and 2000-2006 (Appendix Figures F.5 to F.7). We find evidence of adjustment delays, as individuals continue to bunch at the kink at ages older than the highest age to which the AET applies. However, in no case does adjustment appear to take more than three years.[28]

5.2 Descriptive Evidence from Policy Changes Across Time

We next examine adjustment to a legislated change in AET policy. As shown in Figure 1, the AET was eliminated for those NRA and above in 2000. This policy change was unanticipated prior to the year 2000, as the legislation enacting the policy change was passed in April 2000 and applied to workers' earnings in October 2000, and discussions prior to 2000 did not widely anticipate these changes.[29]

Figure 4 shows the results for those aged 66-69. Bunching in the earnings distribution is easily visible in the years prior to 2000. In 2000, however, there is immediately little bunching visible, and this lack of bunching persists after 2000.[30] A very small bump in the earnings histogram is visible near the kink, but this proves to be insignificant in these data. Figure 5 also shows the amount of excess bunching estimated by year, along with 95 percent confidence

[26] Prior to the divergence of the exempt amount for those below and above NRA in 1978, we find no such dip in bunching at age 65. This "placebo" evidence further supports the hypothesis that the dip in bunching at age 65 arises from delayed adjustment to the increase in the exempt amount from ages 64 to 65 that emerges after 1978.

[27] In our context, the only "appearance" of a new kink that we observe is the appearance of a kink at age 62. The amount of time since the appearance of the kink at age 62 is correlated with age, and elasticities and adjustment costs could also be correlated with age—thus confounding analysis of the time necessary to adjust to appearance of a kink. While recognizing these caveats, it is worth noting that the amount of bunching slowly rises from age 62 to 63, which suggests gradual adjustment. In principle, this could also relate to the fact that these graphs show the sample of those who have claimed by age 65, and the probability of claiming at a given age (conditional on claiming by age 65) rises from age 62 to 63. To address this issue, in Appendix Figure F.4 we show the results when the sample at a given age consists of those who have claimed by that age, which still shows a substantial increase in excess bunching from age 62 to 63.

[28] It is possible that a small amount of excess bunching occurs at ages 72 or older, but this is statistically insignificant.

[29] The AET was also eliminated in 1983 for individuals aged 70 and 71. However, our results across ages show that individuals bunch at ages 70 and 71 in the 1990-99 period, so that persistent bunching at these ages in the 1983-89 cannot cleanly be interpreted as delayed adjustment to the 1983 change (as opposed to delayed reaction to the disappearance of the kink at age 70).

[30] For comparison, Appendix Figure F.8 shows that bunching stayed relatively constant for the 62-64 year-old group that experienced no policy change in 2000. While this group faces a notch at the exempt amount rather than a kink (as explained above), the relative comparison is instructive and suggests that the fall in bunching in the 66-69 year-old group in 2000 and subsequent years was due to the removal of the AET for this group in 2000.

intervals.[31] The amount of bunching is significantly greater than zero in all years prior to 2000, and estimates for 2000 and subsequent years show no significant bunching. Because the change was passed in April 2000 and implemented in October 2000—both after most salaried workers would have learned about their pay that year—the fairly fast reaction suggests that bunching is driven by workers with substantial flexibility in their earnings.[32]

Appendix Figure F.9 shows bunching in 1999, 2000, and 2001 in the LEHD. A spike at the kink is easily visible in 1999, and a small amount of bunching is visible in 2000 as the two bins on either side of the exempt amount are raised relative to the rest of the density (paralleling the small bump in the earnings histogram in 2000 in the MEF). In fact, excess normalized bunching proves to be significantly different from zero in 2000 in the LEHD ($p < 0.01$). By 2001, there is no clear visual evidence of bunching at the kink, and normalized excess bunching is insignificantly different from zero in the LEHD.[33]

In 2000, we find weaker evidence of a delay in adjustment—it only appears to occur among a small number of individuals, and it is only statistically significant in the LEHD. Moreover, bunching in the LEHD in 2000 is not necessarily immediately apparent in the earnings density and is therefore substantially less convincing than the residual bunching in the SSA data at ages 70-71. Thus, we do not wish to rely on the finding of residual bunching in the year 2000; instead, we consider this evidence to be merely suggestive of a small amount of residual bunching.

However, a number of facts are clear. First, in at least some contexts—*i.e.* when aging out of the AET at age 70, apparently after the policy change in 1990 that we discuss later, and quite possibly after the policy change in 2000 (though to a smaller extent)—earnings adjustment frictions prevent some individuals from reacting immediately to the removal of a kink. Second,

[31] We have estimated this amount of excess bunching using three ways of calculating "placebo" kinks in 2000 and after: 1) by adjusting the exempt amount in 1999 using the CPI-U; 2) by adjusting the exempt amount in 1999 using the Employment Cost Index; 3) by using the exempt amount applicable to individuals in the year of attaining NRA in a given year (which is the same as the exempt amount that had been scheduled prior to the 2000 legislation to apply in each year to those NRA and above). Figures 4 and 5 show the first of these methods, but all of these methods show no significant bunching in these years (which is unsurprising given the lack of bunching visible in the histograms in 2000 and after).

[32] Due to changes that raised the scheduled exempt amount beginning in 1996, the AET had been scheduled to increase from $15,500 in 1999 to $30,000 by 2002. In principle, this could have affected the amount of bunching in 2000, even absent the elimination of the AET in this year. Nonetheless, bunching is unlikely to have been zero in the absence of the AET elimination, as the quarterly LEHD data discussed above show substantial evidence of bunching in quarterly earnings data prior to the fourth quarter of 2000, when the AET was eliminated.

[33] Since the sample size is much larger in the LEHD than in the MEF, it makes sense that we could estimate a small but statistically significant amount of bunching in the LEHD but not in the MEF in 2000. In principle, residual bunching in 2000 could also reflect individuals who earned money until their earnings reached the exempt amount (in a month prior to October). However, we also investigated the speed of adjustment from quarterly earnings data in the LEHD. (We do not primarily rely on these quarterly data because the AET is assessed yearly, and thus individuals can appear to bunch at the quarterly kink—defined as one-quarter of the earnings level associated with the kink in each year—even though their yearly earnings does not put them at the kink, or vice versa.) These data show a small but significant ($p < 0.01$) amount of bunching in each quarter of 2000 and in the first two quarters of 2001 but no significant bunching in subsequent quarters.

both when changes are anticipated (*i.e.* the changes in policy across age) and unanticipated (*i.e.* the policy change in 2000), adjustment occurs fairly rapidly, with the vast majority occurring within a maximum of three years. It is interesting to note that adjustment appears to be faster in the case in which the change is unanticipated than in the anticipated case. While this may be surprising, many other differences between the two sets of changes—including differences in the degree to which the changes are publicized, the ages affected, the calendar year, and the distribution of individuals' earnings—could be responsible for the discrepancy in the speed of adjustment. As we observe only a small number of changes in AET policy and confront several candidate explanations for heterogeneity in the speed of adjustment, we do not explicitly try to distinguish among these explanations.

5.3 Other Evidence Relating to Bunching

Figure 5 shows that there is no sharp change in the amount of bunching around the increases in the Delayed Retirement Credit in 1972 or 1982. We consider this suggestive—but not definitive—evidence of little discernable reaction to policy changes in benefit enhancement (particularly in light of our other results suggesting fast adjustment). A general downward trend in the amount of excess bunching is discernable in the 1990s—with the notable exception of a number of years, including 1995—which is coincident in the rise in the DRC through this period. However, we cannot conclusively attribute this potential trend to the influence of the DRC, as it could be due to other factors that changed over this period.[34] We discuss adjustment to the decrease in the AET marginal tax rate from 50 percent to 33.33 percent in 1990 later.

We also conduct a variety of robustness tests. Appendix Figure F.10 uses a bandwidth of \$500 instead of \$800, which changes our estimates little (as have other bandwidths we have chosen). In Appendix Figure F.11, we vary the degree of the polynomial we use between 6 and 8, which shows similar results; other sufficiently rich polynomials we have tried have also shown similar results. In Appendix Figure F.12, we vary the region near the kink we exclude when estimating the amount of excess bunching (from \$2,000 to \$3,000 to \$4,000) and again estimate similar results. Limiting the sample to those who have substantial benefits (such as those with \$1,000 or higher in benefits)—so that they are safely far from the concave kink in the budget

[34] For example, the AET threshold amount rose much faster in the 1996-1999 period than in the previous period. It is possible that this helps to explain the decrease in the amount of bunching observed in these years, as individuals may find it difficult to adjust earnings to a rapidly-increasing kink. Meanwhile, as we discuss later, the fall in excess bunching after 1990 may relate to adjustment to the reduction in the benefit reduction rate in 1990.

set created when the AET reduces OASI benefits to zero—also yields very similar results.

Appendix Figure F.13 shows that both men and for women bunch at the kink (though interestingly, men show more bunching than women). Previous work has demonstrated very different patterns of bunching among the self-employed and non-self-employed (Chetty *et al.* 2011) and has shown that bunching at the kink in response to the Earned Income Tax Credit is primarily driven by the self-employed (*e.g.* Chetty *et al.* 2012b). Appendix Figure F.14 shows histograms for those with self-employment income in 1990-1999—who are excluded from our main sample—who also show an increase in the earnings density near the kink.

6 Mechanisms

This section probes the mechanisms that underlie patterns of adjustment to the AET, examining which parts of the earnings distribution adjust to AET changes, whether adjustment relates to age at death, and whether employers or employees drive responses to the AET.

6.1 Who Adjusts?

We investigate who adjusts to the AET using the large sample sizes in the LEHD data, which allow us to estimate parameters precisely in relatively small population groups. Specifically, we examine how earnings change as the AET is removed from age 69 to age 70 during 1990-1999, when the AET applied to individuals aged 62-69. As in Appendix Figure F.3, Appendix Figure F.15 shows the mean percentage change in earnings from age 69 to age 70 (y-axis), against earnings at age 69 (x-axis). The graph shows a large spike at the kink: individuals locating near the kink at age 69 on average increase their earnings substantially from age 69 to age 70.[35] Recent literature has documented responses to kinks not captured by bunching, including Chetty, Friedman, and Saez (2012) in the context of the EITC and Kline and Tartari (2013) in the context of the Connecticut Jobs First program. In the context of the AET, our evidence shows that responses to incentives appear to be concentrated among a group of bunchers at the kink, with little apparent response among others (though we cannot rule out that such changes occur in ways we do not capture, such as responses over a longer time frame).[36]

[35] The increase near the kink is significantly higher than that in adjacent bins ($p < 0.01$). This spike in earnings growth is interesting in part because it directly documents responses to policy along the intensive margin, which is often found to be very inelastic (*e.g.* Eissa and Liebman, 1996; Meyer and Rosenbaum, 2001). When we examine earnings growth in year $t+1$ by earnings at year t, for ages t younger than 69 we do not observe such a spike at the kink.

[36] We further partially addressed the possibility of bunching over a different range by varying the bandwidth that we chose for estimating excess bunching.

This finding suggests that individuals locating near the kink at age 69 are different than other individuals at the same age. Indeed, the AET applies not only to claimants locating at the kink, but also to claimants initially locating above the kink. Thus, if those initially locating at the kink had the same elasticity and adjustment cost as others, we might have expected to see a large increase in earnings in a substantial range of earnings above the kink, as well. The fact that we do not observe this pattern is suggestive of heterogeneity in adjustment costs or elasticities.[37] For example, those initially locating at the kink may have low adjustment costs and react to the AET removal quickly, but those who do not bunch at the kink to begin with may have higher adjustment costs.[38]

In Appendix Figure F.16, we show that individuals at the kink tend to follow the kink from year to year. We graph the probability of being at the kink in year $t+1$, as a function of earnings in year t. There are clear spikes at the kink for ages 62-63 and ages 65-68, showing that individuals at the kink in year t are disproportionately likely to be at the kink again the next year.[39] We interpret this as further suggestive evidence that certain individuals are particularly responsive to the incentives created by the kink, in the sense that they serially bunch at the kink.

To understand which part of the earnings distribution is affected by the AET, we examine more closely how the distribution of earnings differs across adjacent ages that face different AET incentives. Appendix Figure F.17 stacks the distributions of earnings at ages 60, 61, and 62, as well as 69, 70, and 71. The earnings distribution changes modestly from year to year due to factors unrelated to the AET, as shown in the Figure from ages 60 to 61. However, the age-62 distribution shows a sharply different pattern than the age-60 or 61 distributions, with a sharp spike at the kink (particularly to the left of the kink), a higher density immediately to the right of the kink, and generally a lower density at earnings levels starting several thousand

[37] The income effect of the AET also rises with income, which would also lead the mean percentage earnings change to fall as income rises (under the assumption that leisure is a normal good). However, the income effect rises only gradually, whereas the mean percent earnings increase quickly falls just to the right of the exempt amount and remains relatively constant at this lower level as earnings rises—consistent with the hypothesis that those initially locating at the kink are more responsive to the AET. In fact, the data suggest that income effects (if any) are sufficiently small that they do not cause a noticeable systematic decrease in the mean percentage change in earnings as we move increasingly far to the right of the kink.

[38] The observed pattern is also consistent with such heterogeneity in elasticities or income effects.

[39] The probability of being near the kink in year $t+1$ is significantly higher ($p < 0.01$) for those near the kink in year t than in adjacent bins of the year t earnings distribution. For those aged 58-60, who should not be affected by the incentives to bunch at the kink, no such spike occurs—demonstrating that the spike at the kink for ages subject to the AET is not simply an artefact of the natural evolution of the earnings distribution (absent the AET). We define the "placebo" kink for individual aged 58-60 as the kink affecting those aged 62-64.

dollars above the kink.[40] Similarly, the age-69 earnings distribution shows a sharply higher earnings density than the age 70 distribution in the immediate region of the kink (particularly to its left) but shows a lower density than age 70 at higher earnings levels, eventually reaching a similar earnings density starting around $6,000 above the kink.[41] We return to this pattern of adjustment when discussing our model of fixed adjustment costs below.

Finally, it is possible that those with short expected lifespan could disproportionately bunch near the kink: the DRC should increase lifetime benefits more for claimants with longer life expectancy, which could lead the AET to be a larger effective tax on those with shorter lifespans (though as we note, the DRC only takes effect at earnings substantially above the exempt amount). In Appendix Figure F.18, we show graphs illustrating that life expectancy is smooth near the kink (not significantly different from adjacent bins), suggesting no evidence for such a mechanism.

6.2 Employers and the AET

We use the LEHD data to investigate whether employers play a role in mediating responses to the AET. Chetty *et al.* (2011) argue that employers drive a significant share of the bunching at kink points observed in Denmark. In their context, some individuals bunch at kinks even though they are not directly subject to the policy that creates the kink. Chetty *et al.* conclude that these individuals bunch at the kink because employers create jobs that have those earnings levels. In other words, some individuals bunch at kinks because their employers present them a limited equilibrium menu of earnings levels (including the kink earnings level), and they would face costs of adjusting earnings to a different level.

We explore this possibility by testing for bunching among workers who are too young to claim OASI benefits and are therefore unaffected by the AET. Above, we have presented evidence indicating that in 1990-1999, individuals at ages earlier than those subject to the AET show little evidence of bunching at the AET kink. Thus, during this period, the evidence is consistent with the hypothesis that responses to the AET are driven by employees' choices.[42]

[40] The age-61 distribution of earnings conditional on locating in the vicinity of the kink at 62, and the age-70 distribution of earnings conditional on locating in the vicinity of the kink at age 69, show similar patterns.

[41] Some adjustment to the removal of the AET continues to occur after age 70; the evolution of the income distribution from, for example, age 69 to 72 shows similar patterns.

[42] It is possible that employers drive some of the bunching at ages older than those subject to the AET, but we might then expect some degree of employer earnings coordination on the AET exempt amount for ages younger than 62 (and older than 70, including ages 72 and above).

We extend this analysis by estimating bunching over the entire age distribution in the pre-1972 period, when the DRC did not exist, as Appendix Figure F.19 shows. Appendix Figure F.19 shows a small amount of statistically significant excess bunching at some ages younger than those subject to the AET (though not at other ages), suggesting that some employers do coordinate employment responses in this way in the pre-1972 period—though this behavior is small in the aggregate.[43]

In Appendix Figure F.20, we graph the probability that individuals change at least one employer from age t to age $t+1$, against earnings at age t (during 1990-1999 period). At ages when people face the AET, the probability that individuals change jobs across employers is sharply lower for individuals locating near the kink at age t than for individuals with other initial earnings levels.[44] The probability of changing employers is also sharply lower at the kink when individuals transition from being subject to the AET at age 69 to being no longer subject at age 70, and this is true when we limit the sample to those who increase their earnings from age 69 to age 70. Those initially locating at the kink evidently have sufficiently flexible pay arrangements that they can change their earnings from age 69 to age 70 while typically staying at the same employer.[45]

7 Estimating Elasticities and Adjustment Costs

The results thus far suggest a role for adjustment frictions in individuals' earnings choices in some contexts. As a first step in incorporating frictions into an estimable model of earnings supply, we build upon the Saez (2010) model (described briefly in the first sections of Appendix E), which uses bunching to identify the elasticity of (taxable) earnings with respect to the net-of-tax rate.[46] We extend this model to allow for a cost of adjusting to tax changes.[47] We first develop the theory graphically to show how adjustment costs affect bunching. Next, we show

[43] Figure F.19 bins three adjacent years of ages (e.g. 18 to 20); doing so in the post-1971 period shows no statistically significant bunching in pre-62 age bins. While individuals could in principle be locating at the kink prior to age 62 in anticipation of facing the AET later, it seems unlikely that they would do so as early as their late 30s, when a small amount of statistically significant bunching first appears—around 25 years before they are first eligible for OASI.

[44] The probability of changing employers near the kink is significantly lower than that in adjacent bins ($p < 0.01$).

[45] Those locating at the kink might be different from other individuals for reasons—such as different demographics—that lead them to switch across employers less frequently. It is worth emphasizing that we attempt only to document the descriptive pattern that they change employers less frequently. We have also found that the graph of the probability of changing employers at age 59-61 against earnings at age 69 is smooth near the kink, suggesting that absent the AET incentives these individuals do not display noticeably different behavior in this regard.

[46] Formally, the elasticity of earnings with respect to the net-of-tax rate is defined as $\varepsilon \equiv -\left(\partial z / z\right) / \left(\partial \tau / \left(1-\tau\right)\right)$.

[47] Following recent literature on bunching—including Saez (2010), Chetty *et al.*, 2009, 2011, 2012a,b, Chetty (2012), and Kleven and Waseem (2013)—we specify a static model of earnings choice in each period. As we discuss further in the Conclusion, dynamic considerations represent an important topic for future research.

that using data on bunching at multiple kinks associated with different jumps in the net-of-tax rate, we can jointly estimate elasticities and adjustment costs. As discussed in Chetty *et al.* (2009), these parameters are jointly sufficient for welfare calculations in many applications.

Our model relies on features of the empirical results that we have documented in the previous two sections. We find evidence of adjustment frictions, which we model through a cost of adjusting earnings. The empirical results also suggest that employees' choices are primarily responsible for patterns of bunching in the main period we study; this motivates a model in which employees choose their earnings rather than a model in which employers coordinate responses.

As described further in Appendix E, agents maximize utility $u(c, z; n)$ over consumption and earnings (where greater earnings are associated with greater disutility at the margin), subject to a budget constraint $c = (1 - \tau) z + R$, where R is virtual income and the parameter n reflects the tradeoff between consumption and earnings supply.[48] We assume that in order to change earnings from an initial level, individuals must pay a fixed utility cost of ϕ^*. This cost could represent the information costs associated with navigating a new tax regime if, for example, individuals only make the effort to understand their earnings incentives when the utility gains from doing so are sufficiently large (*e.g.* Simon 1955; Chetty *et al.* 2007; Hoopes, Reck, and Slemrod 2013). Alternatively, this cost may represent frictions such as the cost of negotiating a new contract with an employer or the time and financial cost of job search, assuming that these costs do not depend on the size of the desired earnings change.

We model a fixed cost in order to build on recent literature that has focused on fixed costs (*e.g.* Chetty *et al.*, 2011; Chetty, 2012). The distribution of earnings at ages 62 or 69 is higher in a region surrounding the kink but lower in a region substantially above the kink than at ages 61 or 70, respectively, which is consistent with a simple model with fixed adjustment costs that lead to a region of inaction and a region of adjustment.[49] In Appendix E.6.2, we extend our model to a case in which the cost of adjustment is linear in the size of the adjustment.

[48] We describe the model in more detail in Appendix E.

[49] However, even with a fixed adjustment cost, the AET could in principle cause some individuals to reduce their earnings to levels just above the kink, which in principle could lead to a rise in the density to the right of the kink due to the imposition of the AET. Moreover, the shape of the distribution of earnings at age 70 conditional on locating at the kink at age 69 cannot be predicted *a priori*, as it should depend among other things on the correlation of the fixed cost of adjustment with the elasticity of earnings with respect to the net-of-tax rate. For example, if individuals with low fixed costs of adjustment tend to have low elasticities, then the conditional earnings distribution at age 70 should be closer to the kink on average than if individuals with low fixed costs of adjustment tend to have high elasticities. As a result of these factors, we cannot use the effect of the AET on such moments of the earnings distribution in estimating elasticities and adjustment costs without making more stringent assumptions.

We develop two different approaches for estimating elasticities and adjustment costs. Our first approach, which we call the "Comparative Static" method, relies on comparing bunching in two separate cross-sections of data. Our second approach, which we call the "Sharp Change" method, additionally relies on attenuation (due to adjustment costs) in the change in bunching among individuals who face a change in the size of the kink over time. As we explain, the Sharp Change method relates more directly to our observation that bunching persists among individuals who formerly faced a kink. We begin by describing the Comparative Static approach because it introduces concepts that the Sharp Change method builds upon.

7.1 Estimation: Comparative Static Approach

The Comparative Static approach is best suited to estimating elasticities and adjustment costs from two cross-sections of different individuals who face different policies. Figure 6 Panel A illustrates how a fixed adjustment cost attenuates the level of bunching. Recall that our frictionless model predicts that bunchers have initial earnings (i.e. earnings in the absence of a kink) in the range $[z^*, z^* + \Delta z_1]$. Consider the person with initial earnings $\underline{z}_1$ (on the linear budget constraint with tax rate τ_0). This individual faces a higher marginal tax rate τ_1 after the kink is introduced, which increases the marginal tax rate to τ_1 above earnings level z^*. Because she faces an adjustment cost, she could decide to keep her earnings at $\underline{z}_1$ and locate at point 1. Alternatively, with a sufficiently low adjustment cost, she would like to pay the adjustment cost and reduce her earnings to the kink at z^* marked by point 2. We assume that the benefit of relocating to the kink is increasing in distance from the kink for initial earnings in the range $[z^*, z^* + \Delta z_1]$.[50] These assumptions imply that above a threshold level of initial earnings, $\underline{z}_1$, individuals adjust their earnings to the kink, and below this threshold individuals remain inert. We have drawn this individual as the marginal buncher who is indifferent between staying at the initial level of earnings $\underline{z}_1$ (at point 1) and moving to the kink earnings level z^* (point 2) by paying the adjustment cost ϕ^*.

In Panel B, we show that the level of bunching is attenutated due to the adjustment cost: only individuals with initial earnings in the range $[\underline{z}_1, z^* + \Delta z_1]$ bunch at the kink (areas ii, iii, iv, and v)—whereas in the absence of an adjustment cost, individuals with initial earnings

[50] In general, this requires that the size of the optimal adjustment in earnings increases in n at a rate faster than the decrease in the marginal utility of consumption. This is true, for example, if utility is quasilinear. We explore the implications of this assumption in Appendix E.5.

in the range $[z^*, z^* + \Delta z_1]$ bunch (areas i, ii, iii, iv, and v). The amount of bunching is equal to the integral of the initial earnings density over the range $[\underline{z}_1, z^* + \Delta z_1]$:

$$B_1(\boldsymbol{\tau}_1, z^*; \varepsilon, \phi^*) = \int_{\underline{z}_1}^{z^*+\Delta z_1} h\left(\zeta\right) d\zeta. \tag{1}$$

Bunching therefore depends on the preference parameters ε and ϕ^*, the tax rates below and above the kink, $\boldsymbol{\tau}_1 = (\tau_0, \tau_1)$, and the exempt amount z^*. The lower limit of the integral, $\underline{z}_1$, is implicitly defined by the indifference condition drawn in Figure 6, Panel A:

$$\phi^* \equiv u\left((1-\tau_1)z^* + R_1, z^*; \underline{n}\right) - u\left((1-\tau_1)\underline{z}_1 + R_1, \underline{z}_1; \underline{n}\right) \tag{2}$$

where R_1 is virtual income, and $\underline{n}$ is the "ability" level of this marginal buncher. If the marginal tax rate above z^* were instead τ_2, where $\tau_0 < \tau_2 < \tau_1$, then bunchers would be comprised only of individuals with initial earnings in the range $[\underline{z}_2, z^* + \Delta z_2]$ (area iii), which is again attenuated relative to bunching under a frictionless model (areas i, ii, and iii). This generates a second expression for bunching and an indifference condition analogous to 1 and 2, respectively.

When we later perform our estimates, we make use of a minimum distance estimator described in Appendix E.8 to solve this nonlinear system of equations. The key assumption underlying that method is that utility is quasi-linear and isoelastic, which is common in the bunching literature (see Saez 2010, Chetty *et al.* 2011, Kleven, Landais, Saez, and Schultz 2012 and Kleven and Waseem 2013, for example).[51] If we were to relax the assumption of quasilinearity, we would need to observe wealth, which is not available in the data.

7.1.1 Intuition and Tractable Approximation

To build intuition regarding our minimum distance estimation procedure, and to derive an expression relating the elasticity and adjustment cost to the level of bunching that can be easily solved in closed form, we can use a series of approximations to specify a simple system of linear equations. Let $b \equiv B/h(z^*)$, *i.e.* the amount of bunching scaled by the density of earnings at z^* when there is no kink. Also assume that $h(z)$ is uniform and equal to $h(z^*)$ in the range between $\underline{z}$ and $[z^* + \Delta z_1]$. We show in Appendix E.6 that scaled bunching is

[51] As explained in Appendix E.8, in a baseline we also assume that the density of initial earnings $h\left(z\right)$ is uniform over the range $[z^*, z^* + \triangle z^*]$ (as in Chetty *et al.* 2011 or Kleven and Waseem 2013), but we alternatively use a lognormal distribution of earnings based on those aged 61 (who are similarly aged but do not face the AET) and find similar results.

approximately:

$$b_1(\boldsymbol{\tau}_1, z^*; \varepsilon, \phi) = \varepsilon \left(z^* \frac{d\tau_1}{1-\tau_0} \right) - \phi \left(\frac{1}{d\tau_1} \right), \tag{3}$$

where $d\tau_1 = \tau_1 - \tau_0$ and $\phi = \phi^*/u_c$ is the dollar equivalent of the adjustment cost. This equation shows intuitive comparative statics: All else equal, bunching is increasing in the elasticity, decreasing in the adjustment cost, and increasing in the size of the tax change at the kink. This generalizes and nests the formula developed in Saez (2010), which is equivalent in the case in which there is no adjustment cost. Because the amount of bunching is decreasing in the adjustment cost, constraining $\phi = 0$ and using the Saez (2010) will in general weakly underestimate the elasticity in a single cross-section, since attenuation in bunching is attributed to a small elasticity rather than to the adjustment cost. Note that the expository derivation in (3) does not impose quasilinearity but uses the uniform density assumption and a first-order approximation for utility in the neighborhood of the kink.

Equation (3) also shows the features of the data that allow us to identify ε and ϕ. We need to observe bunching at two or more kinks, with variation in the change in tax rate $d\tau_1$. If we observe bunching at exactly two kinks of different sizes, then we can solve for ε and ϕ exactly, as we then have a system of two equations and two variables. More generally, we could estimate a regression of b on $z^* \left(d\tau_1 / \left(1 - \tau_0 \right) \right)$ and $-1/d\tau_1$, with the constant omitted. The coefficient on the first term is ε, and the coefficient on the second term is ϕ.

Intuitively, with only a single cross-section of data, the amount of excess bunching increases in the elasticity and decreases in the adjustment cost, and thus it is not possible to identify both. Suppose that instead we have two cross-sections of data featuring different changes in marginal tax rates at the kink. The difference in the amount of bunching from one cross-section to the other will also depend on the elasticity and adjustment cost.[52] Phrased differently, the Saez (2010) formula describes how bunching should vary between two different kinks in a frictionless model, and the extent to which observed bunching deviates from this pattern is attributed to the adjustment cost. Let K_1 and K_2 be two kinks that involve jumps at z^* in the marginal tax rate of $d\tau_1 = \tau_1 - \tau_0$ and $d\tau_2 = \tau_2 - \tau_0$, respectively, and assume $d\tau_2 < d\tau_1$. Relative to the frictionless case represented by the Saez model, under the Comparative Static method, the change in bunching from K_1 to K_2 is larger. In the frictionless model, bunchers comprise

[52] Under the approximations above, (3) implies that $\frac{\partial b^2}{\partial \phi \partial (d\tau)} = \frac{1}{\tau^2} > 0$; as ϕ increases, the marginal impact of $d\tau$ on b increases.

areas i, ii, iii, iv, and v in Figure G.2 under K_1 and areas i, ii, iii under K_2 (thus decreasing by areas iv and v). Under the Comparative Static method, bunchers comprise areas ii, iii, iv, and v under K_1 and area iii under K_2 (thus decreasing by areas ii, iv, and v, rather than by only areas iv and v in the frictionless case).

7.1.2 Heterogeneity in Elasticities and Fixed Costs of Adjustment

Our empirical results suggest heterogeneity in the elasticity and the fixed cost of adjustment, as some individuals are more responsive to removal of the AET than others. Let $(\varepsilon_i, \phi_i, n_i)$ be jointly distributed according to a smooth CDF $G(\cdot)$, which translates into a smooth, joint distribution of elasticities, fixed costs and earnings $\tilde{H}(z, \varepsilon, \phi)$. As shown in Appendix E.6.1, assuming that the density of earnings, $\tilde{h}(z, \varepsilon, \phi)$, is again constant over the interval $[\underline{z}, z^* + \Delta z^*]$, we derive a generalized formula analogus to equation (3):

$$b = \bar{\varepsilon}\left(z^* \frac{d\tau_1}{1-\tau_0}\right) - \bar{\phi}\left(\frac{1}{d\tau_1}\right),$$

where $\bar{\varepsilon}$ and $\bar{\phi}$ are the average elasticity and adjustment cost for those who bunch at the kink.[53]

7.2 Estimation: Sharp Change Approach

The Sharp Change approach is best suited to estimating elasticities and adjustment costs when we are examining a constant population that experiences a change in the marginal tax rate at a kink (which may involve the kink disappearing), as we observe in our empirical applications. Suppose we observe a population that moves from facing a more pronounced K_1 to facing a less pronounced kink K_2 (as defined above). Adjustment costs prevent some individuals from "unbunching" from the kink, even though they would prefer to move away from the kink in the absence of an adjustment cost. The fixed adjustment cost therefore attenuates the change in bunching between two cross-sections in response to a reduction in the size of the kink, relative to the Comparative Static approach.[54]

The first source of attenuation in the change is driven by individuals in area ii of Panel B. They bunch under K_1 and continue to bunch after transitioning to K_2. The reason is that the

[53] We are grateful to Henrik Kleven for suggesting the approach that led to this derivation.

[54] If $d\tau_2 > d\tau_1$ instead – *i.e.* the kink becomes larger – then additional individuals will be induced to bunch, but the change in bunching will in general be attenuated in the Sharp Change approach relative to the Comparative Static approach (due to the adjustment cost). This is governed by an analogous set of formulas to the case $d\tau_2 < d\tau_1$ that we explore.

frictionless optimum under K_2 is z^* for everyone initially earning in the range $[z^*, z^* + \Delta z_2]$.

The second source of attenuation in the change is driven by individuals in area iv of Panel B. Panel C of Figure 6 demonstrates this. At point 0, we show an individual's initial earnings $\bar{z}_0 \in [z^*, z^* + \Delta z_1]$ under a constant marginal tax rate of τ_0. We now introduce the first kink, K_1. The individual responds by bunching at z^* at point 1. Next, we transition to the muted kink K_2. Note that since $\bar{z}_0 > z^* + \triangle z_2$, this individual would have chosen earnings $\bar{z}_2 > z^*$ (marked as point 2) under τ_2, if we had gone directly from no kink to K_2.[55] However, in order to move to point 2, this individual must pay a fixed cost of ϕ^*. We have drawn this individual as the marginal buncher who is indifferent between staying at z^* and moving to $\bar{z}_2$. All individuals with initial earnings in the range $[z^* + \Delta z_2, \bar{z}_0]$ will remain at the kink.

Thus, bunching under K_2 is:

$$B_2(\boldsymbol{\tau}_2, z^*; \varepsilon, \phi^*) = \int_{\underline{z}_1}^{\bar{z}_0} h\left(\zeta\right) d\zeta. \tag{4}$$

It follows that the absolute value of the change in bunching from K_1 to K_2 under the Sharp Change approach (area v in Panel B) will be smaller than under the Comparative Static approach (areas ii, iv and v). As discussed in Appendix E.7, ε is still identified by the adjustment of the top-most buncher: $\varepsilon = \frac{\bar{z}_0 - \bar{z}_2}{\bar{z}_2} \frac{(1-\tau_0)}{d\tau_2}$. The critical earnings level $\bar{z}_2$ is defined implicitly by the indifference condition in Panel C:

$$\phi^* \equiv u\left((1-\tau_2)\bar{z}_2 + R_2, \bar{z}_2; \bar{n}\right) - u\left((1-\tau_0)z^* + R_0, z^*; \bar{n}\right). \tag{5}$$

As before, our estimates use the minimum distance estimator described in Appendix E.8 to solve the system of nonlinear equations defined by (1), (4), and (5). Intuitively, we rely on a before-and-after comparison of bunching at the same kink, once the jump in marginal tax rates has been reduced. Inertia generates an excess amount of bunching in the period after the policy change. In the extreme case in which a kink has been eliminated, we can attribute any residual bunching to adjustment costs. The amount of residual bunching at the kink, in combination with the amount of bunching prior to the change in the jump in MTRs at the kink, therefore helps to identify both the elasticity and the adjustment cost.[56]

[55] Note that in general $\bar{z}_2$ may be different from $\underline{z}_2$.

[56] An approximation explained in the Appendix also helps to build intuition.

Relative to the frictionless case represented by the Saez model, the change in bunching from the larger kink K_1 to the smaller kink K_2 is attenuated under the Sharp Change method by the adjustment cost (in contrast to the Comparative Static method). As noted above, in the Saez model, bunching decreases by areas iv and v in Figure G.2 when moving from K_1 to K_2. Under the Sharp Change method, areas ii, iii, iv, and v bunch under K_1, whereas areas ii, iii, and iv bunch under K_2. Thus, bunching decreases only by area v in the Sharp Change method, rather than both areas iv and v in the frictionless case. The absolute value of the decrease in bunching from K_1 to K_2 is decreasing in the adjustment cost—$\bar{z}_0$ is increasing in the adjustment cost, and therefore area v is decreasing in the adjustment cost—helping to provide further intuition for our estimation procedure.[57]

7.3 Estimates of Elasticity and Adjustment Cost

To estimate ε and ϕ, we separately examine several changes in the AET benefit reduction rate, including a reduction in the rate in 1990; the elimination of the AET from ages 69 to 70; and the elimination of the AET in 2000. Our Sharp Change method is applicable in all of these contexts, as we observe a group from before to after these changes in policy. By contrast, the Comparative Static method is more applicable to analyzing changes in bunching when comparing two different groups each with positive (but different) marginal tax rates.

Estimating ε and ϕ requires estimates of the implicit marginal tax rate that individuals face. This requires estimates of both the "baseline" marginal tax rate, τ—the rate that individuals near the AET threshold face in the absence of the AET due to federal and state taxes—and estimates of the implicit marginal tax rate associated with the AET. We begin by using a marginal tax rate that incorporates the effects of the AET BRR, as well as the average marginal income and FICA tax rates (including federal and state taxes).[58] These estimates are predicated on correctly specifying the marginal tax rate (net of benefit enhancement), but recall that Appendix A shows that benefit enhancement is not relevant to an individual's marginal incentives for earning an extra dollar near the AET exempt amount. This assumption is also consistent with the methodology of Friedberg (1998, 2000), who treats the AET as a pure tax.

[57] As in the Comparative Static approach, the amount of bunching at K_1 is increasing in the elasticity (*ceteris paribus*) under the Sharp Change approach.

[58] Using TAXSIM and the Statistics of Income individual tax return files, we calculated the mean of the sum of federal and state marginal income and FICA tax rates for people with positive Social Security benefits and earnings within $2000 of the kink, in the same years as the data we examine. For example, when we examine data from 1989 and 1990, we calculate marginal tax rates in these years. The results are not senstive to other such choices.

Morever, as we have noted, the evidence shows little systematic bunching reaction to changes in the DRC. We vary these assumptions in various dimensions, which show similar results to the baseline: we exclude FICA taxes in the calculation of the baseline tax rate,[59] and we alternatively assume that the benefit enhancement corresponds to a reduction in the effective marginal tax rate.[60]

Before turning to our empirical estimates, we begin with graphical depictions of the patterns driving the estimates. Figure 5 shows excess bunching among 66-69 year-olds, for whom the BRR fell from 50 percent to 33.33 percent in 1990.[61] Excess bunching fell slightly from 1989 to 1990 but fell more subsequent to 1990.[62] For comparison, Appendix Figure F.8 shows that bunching stayed relatively constant—both in 1990 and subsequently—for the 62-64 year-old group that experienced no policy change in 1990. While this group faces a notch at the exempt amount rather than a kink (as explained above), the relative comparison is instructive.

Appendix Figure F.21 shows that the elasticity we estimate using the Saez (2010) method—constraining the adjustment cost to be zero—rises sharply from 1989 to 1990. This relates directly to our Sharp Change theory, which predicts that following a reduction in the change in the MTR at the kink, there may be excess bunching due to inertia (corresponding to area iv in Figure 6, Panel B).[63] Once we allow for an adjustment cost, this excess bunching is attributed to optimization frictions. Indeed, in the Appendix we explain that the rise (from just before to just after the policy change) in the elasticity that we estimate using the Saez (2010) method is a telltale sign that we face an adjustment cost as modeled by the Sharp Change method.[64] In a context in which individuals have not yet had a chance to adjust (and the effective marginal tax rate has fallen), frictions may lead to larger elasticity estimates. Interestingly, this is in some sense the opposite of the usual presumption that adjustment frictions should lead to attenuation of elasticity estimates. Our finding also contrasts with the usual presumption that in the presence of adjustment frictions, smaller variation in taxes (i.e. smaller kinks) yield

[59] Liebman, Luttmer, and Seif (2009) show that labor supply reacts to OASI benefit rules, suggesting that individuals may not perceive FICA taxes as pure taxes.

[60] Recall that in 1990, the DRC was not yet actuarially fair.

[61] The patterns around 1990 are extremely similar for the 67-68 year-old group that we focus on in our estimates.

[62] The Delayed Retirement Credit changed from 3 percent to 4 percent over this period, which we take account of in the specification in which we account for benefit enhancement.

[63] In the Conclusion, we discuss dynamic considerations that might subsequently cause residual bunching to disappear. Our focus in the Sharp Change estimates is limited to the period just after a policy change, before residual bunching has dissipated.

[64] Specifically, we show that if we actually face the Sharp Change model but mis-specify the model as a frictionless (Saez 2010) model, and we face a decrease in the jump in the marginal tax rate at the kink, the estimated Saez (2010) elasticity will weakly rise from just before to just after this policy change.

smaller elasticity estimates.

Table 2 presents our results from the Sharp Change method, examining the 1990 change. We estimate an elasticity of 0.23 in Column (1) and a positive adjustment cost of \$152.08 in Column (2), both significantly different from zero ($p < 0.01$). This specification examines data in 1989 and 1990; thus, our estimated adjustment cost represents the cost of adjusting earnings in the first year after the policy change. When we constrain the adjustment cost to zero using 1990 data in Column (3), as most previous literature has implicitly done, we estimate a substantially larger elasticity of 0.39.[65] Consistent with our discussion above, it makes sense that the estimated elasticity is higher when we do not allow for adjustment costs than when we do, as adjustment costs keep individuals bunching at the kink even though tax rates have fallen. The difference in the constrained and unconstrained estimates of the elasticity is substantial (69 percent higher in the constrained case) and statistically significant ($p < 0.01$).

We also consider alternative specifications. Using a lognormal earnings density rather than a uniform density (as described in Appendix E.8) changes the results little. Adjusting the marginal tax rate to take account of benefit enhancement (applicable to those individuals to whom benefit enhancement is relevant to their earnings choices) raises the estimated elasticity but yields similar qualitative patterns across the constrained and unconstrained estimates. This makes sense: for the same behavioral response, if we assume a less pronounced percentage change in the net-of-tax rate, we infer a larger elasticity. The next rows show other specifications: excluding FICA taxes from the baseline tax rate; other bandwidths; and other years of analysis. Our results are similar under these and other variations.

In Appendix Table G.1, we apply the Sharp Change method to the disappearance of the kink at age 70 (in which context we find residual bunching in Figure 2) and find similar (slightly higher) elasticity estimates and somewhat lower adjustment costs (though still in the same range). The constrained estimate of the elasticity is smaller than the unconstrained estimate; this makes sense, because we use data from age 69 to perform the constrained estimate, and adjustment costs attenuate the constrained estimate at those ages because they reduce bunching.

Appendix Tables G.2, G.3, and G.4 present further specifications. In Appendix Table G.2,

[65] Friedberg (2000) finds uncompensated elasticity estimates of 0.22 and 0.32 in different samples. However, differences in the estimation strategies imply that these results are not directly comparable to ours.

we apply the Sharp Change approach to the 1990 policy change but assume that bunching in 1989 is not attenuated by adjustment frictions (under the rationale that bunching could have reached a "steady state" in 1989 that is not attenuated by adjustment frictions).[66] We estimate results similar to the baseline. In Appendix Table G.3, we use the Sharp Change approach to estimate elasticities and adjustment costs using the disappearance of the kink in the year 2000 for those NRA and above. Given our small point estimate of residual bunching in 2000, it is unsurprising that we find small (though marginally significant) adjustment costs.[67]

In Appendix Table G.4, we apply the Comparative Static to the 1990 policy change. As noted above, the Comparative Static method is most applicable to performing estimates on unrelated cross-sections of individuals—rather than just before and just after a change in policy as in our results on residual bunching.[68] We use years 1989 and 1993 to make sure that none of the 66-69 year-olds in 1993 were observed in 1989 (to avoid the possibility of residual bunching at the kink among those initially bunching there).[69] We again find elasticity estimates in the same range, though with lower and typically insignificant adjustment costs. The linear approximation (3) shows higher (but still modest) adjustment costs, and higher elasticities than in the baseline specification (which makes sense because higher elasticities are needed to reconcile the higher estimated adjustment cost with the observed change in excess normalized bunching).

8 Conclusion

In the context of the Social Security Annual Earnings Test, we investigate the existence, nature and size of earnings adjustment frictions. We develop several related findings. First, we examine the speed of adjustment to the disappearance of convex kinks in the effective tax schedule. We document evidence of delays in adjustment in certain contexts, consistent with the existence

[66] The notion that individuals are initially in a steady state in which they have been able to make desired adjustments to the frictionless level of bunching is consistent with the observation that complete adjustment occurs within a few years of policy changes—prior to 1990 the AET policy parameters had last changed in 1983—and with the observation that they face frictions in un-bunching despite initially bunching. Thus, this estimate of the adjustment cost represents the cost of adjustment in the first year after a policy change, under the assumption that individuals are able to make adjustments within a few years of the previous policy change in 1983.

[67] We find smaller elasticities than in the baseline specification in part because normalized bunching was relatively small (though still positive and significant) in 1999, as shown in Figure 5.

[68] The Comparative Static approach is inapplicable to years in which we observe residual bunching at a former kink; in assuming zero marginal tax rate after a kink disappears, the Comparative Static method effectively cannot explain this residual bunching. Thus, the Sharp Change method is more directly motivated by our primary empirical observations.

[69] Because the Comparative Static approach examines separate cross-sections in 1989 and 1993 in our application, the interpretation of the estimated adjustment cost is different than in the Sharp Change approach (when it represented the cost of adjusting earnings within one year). Thus, there is no reason the estimated adjustment cost should be equal in the Comparative Static and Sharp Change applications we examine.

of earnings adjustment frictions in the U.S. Nonetheless, we find that adjustment to both anticipated and unanticipated policy changes is quite rapid, as the vast majority of adjustment occurs within at most three years of budget set changes. This suggests that in this context, long-run elasticities are similar to those estimated in a medium-run time frame of a few years.

Second, we investigate mechanisms that underlie the patterns of adjustment. Adjustment to removal of kinks occurs primarily through substantial earnings growth among those initially locating at the kink, suggesting that they are more responsive than others (due to some combination of different elasticities or adjustment costs). We additionally investigate the extent to which firms may help coordinate bunching. The responses appear to be driven mainly by employees, as those under the minimum age subject to the AET do not bunch at the AET kink during the primary period we examine. Additionally, the bunchers are disproportionately likely to remain with the same employer while under the AET and when responding to the removal of the AET. This combination of evidence suggests that the bunching primarily results from the choices of certain particularly responsive employees who choose themselves to vary their earnings, generally within the same firm.

Third, we specify a model of employees' earnings adjustment consistent with these findings and use it to estimate the earnings elasticity and the fixed adjustment cost. When we consider the change in bunching associated with the reduction in the AET benefit reduction rate from 50 percent to 33.33 percent for those above NRA from 1989 to 1990, we estimate that the elasticity is 0.23 and the adjustment cost is \$152.08. The results are typically similar with other populations, time periods, and methods. When we constrain adjustment costs to zero in the baseline specification, the elasticity we estimate in 1990 (0.39) is substantially (69 percent) larger, demonstrating the potential importance of taking account of adjustment costs. Our estimates demonstrate the applicability of the methodology and the potential importance of allowing for adjustment costs when estimating elasticities. The modest adjustment cost we estimate parallels our empirical observation that bunching responds rapidly to changes in policy.

The analysis leaves open a number of avenues of further inquiry. First, it would further enrich the framework to extend the static analysis here to a dynamic model. We consider our static framework for estimating elasticities and adjustment costs to be a natural first step in understanding estimation of these parameters (in the spirit of other static papers such as Saez 2010, Chetty *et al.*, 2009, 2011, 2012a,b, Chetty 2012, and Kleven and Waseem 2013),

but incorporating dynamic considerations is an important next step. The speed at which individuals respond to changes in the AET, the nature of income effects, and the distinction between anticipated and unanticipated changes—all of which we have begun exploring in this paper—are three of several possible pieces of evidence that could help in specifying a model of this sort (perhaps including a stochastic wage arrival process, and incorporating the benefits over time to adjustment at a given time).

Second, further work distinguishing among the possible reasons for reaction to the AET (such as misperceptions) remains an important issue. Third, further investigation of extensive margin and claiming responses to the AET would be valuable. Fourth, following most previous literature, we have treated the adjustment cost as a "black box," without modeling the process that underlies this cost, such as information acquisition or job search. Future research could model such processes and distinguish these explanations using data. Finally, the AET policy environment provides a useful illustration of many issues—such as a methodology for estimating elasticities and adjustment costs simultaneously—that should be applicable more broadly to studying adjustment to policy. As elasticities and adjustment costs may be substantially different in other contets, studying earnings adjustment to other policies is a high priority.

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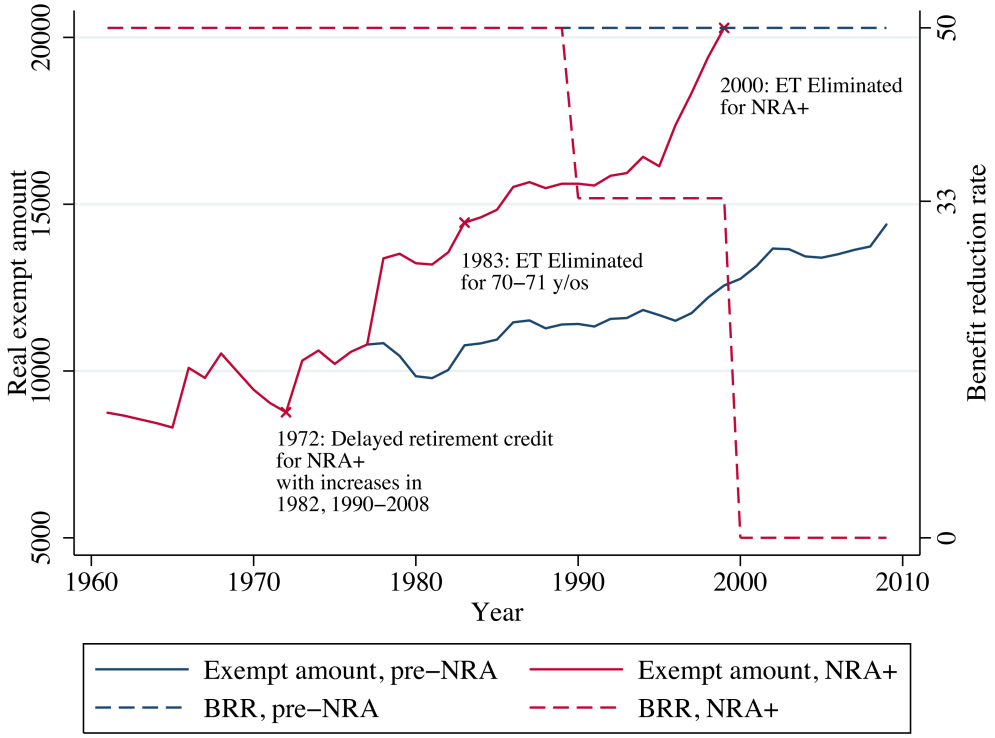
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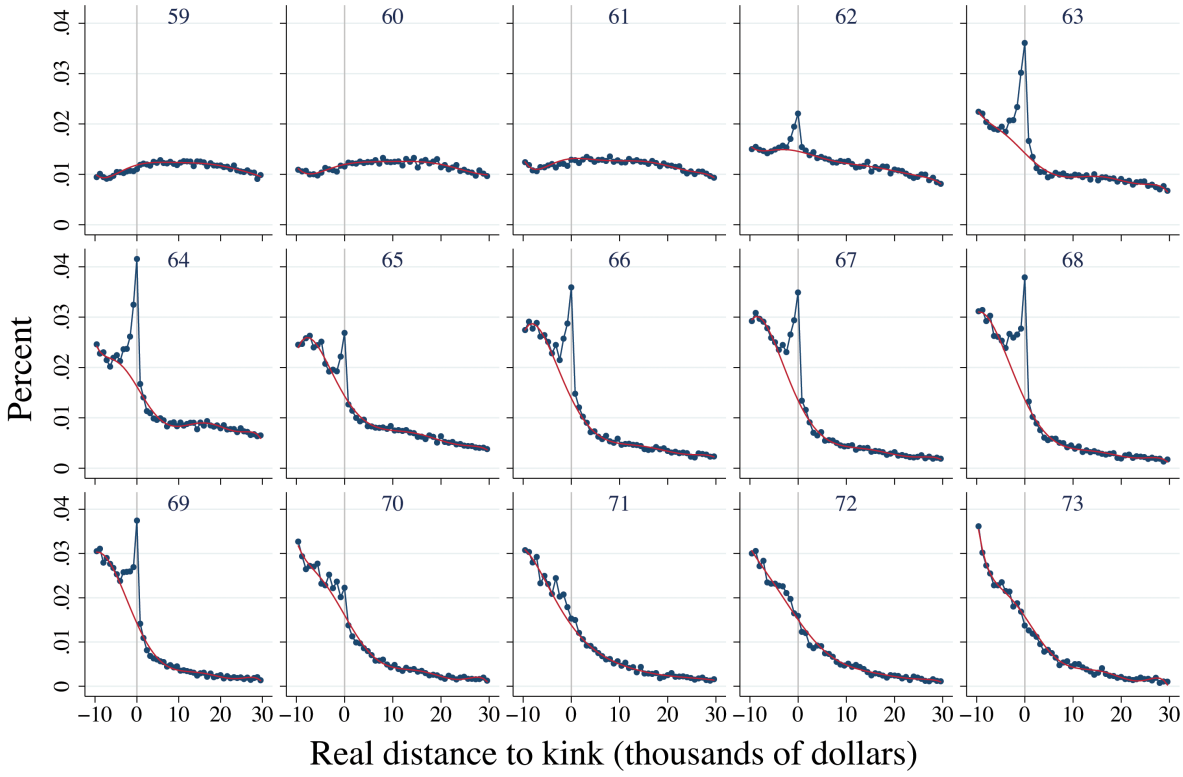
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Figure 1: Key Earnings Test Rules, 1961-2009



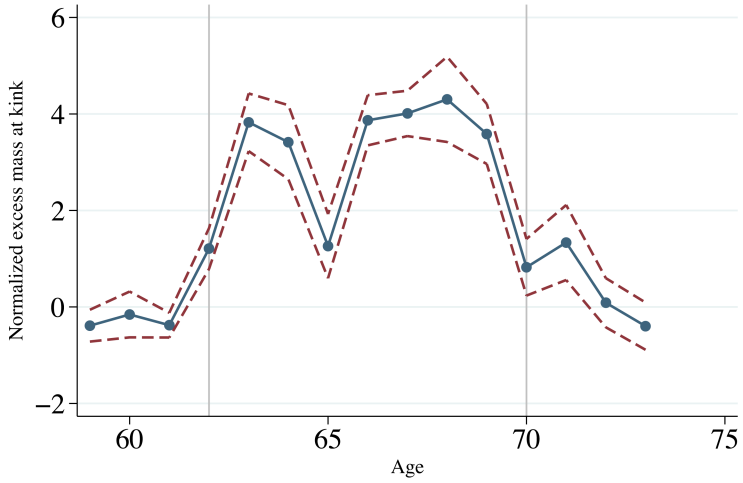
Note: The right vertical axis measures the benefit reduction rate in OASI payments for every dollar earned beyond the exempt amount. The left vertical axis measures the real value of the exempt amount over time.

Figure 2: Histograms of Earnings, 59-73-year-olds Claiming OASI by Age 65, 1990-1999



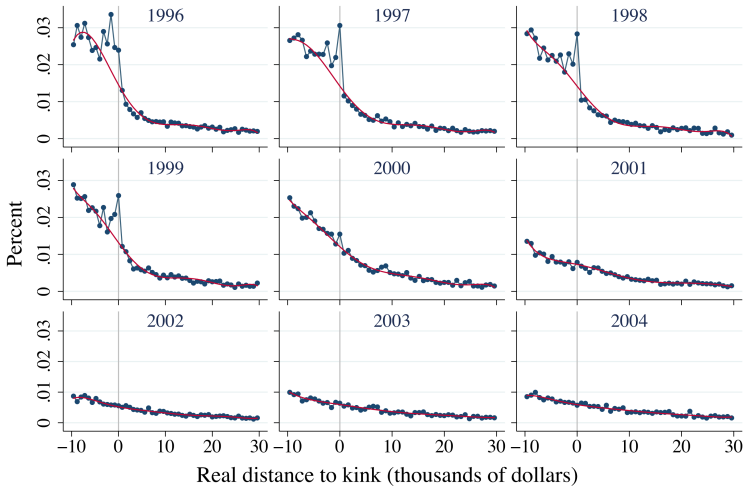
Notes: The bin width is $800. The earnings level zero, shown by the vertical lines, denotes the kink. "Claimant" refers to an individual who has claimed by age 65.

Figure 3: Adjustment Across Ages: Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 1990-1999



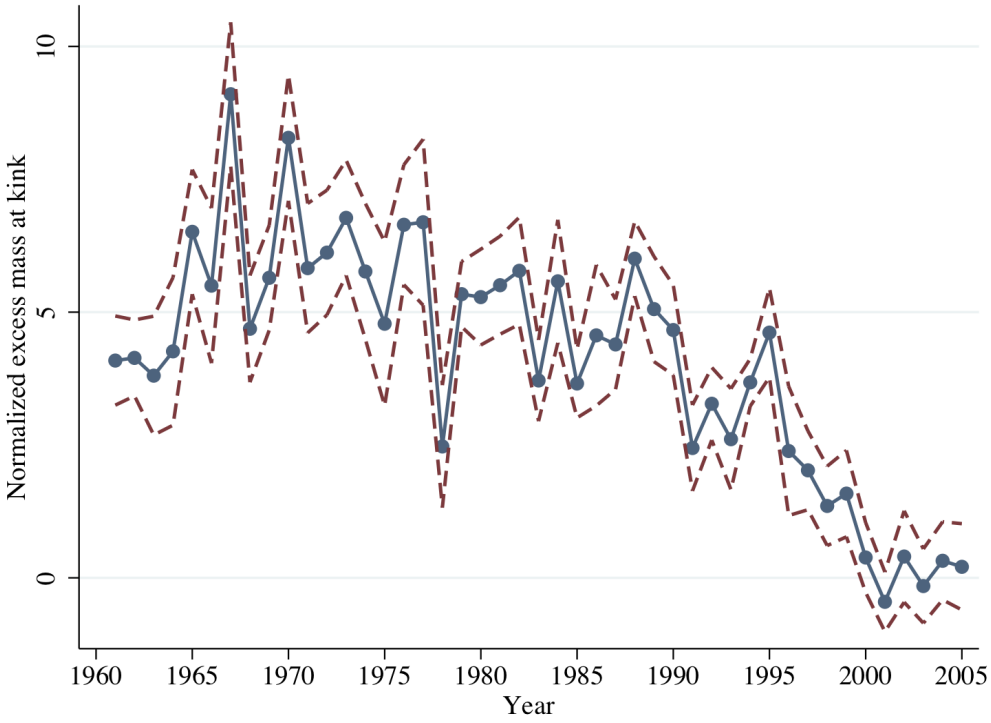
Note: The figure shows normalized excess bunching from a one percent random sample of SSA administrative data on Social Security claimants aged 59-73 between 1990 and 1999 (inclusive). Normalized excess bunching is calculated as described in the text. The vertical lines in Panel B show the ages at which the AET first applies (62) and the age at which the AET ceases to apply (70). See other notes to Figure 2.

Figure 4: Adjustment Across Years: Histograms of Earnings and Normalized Excess Mass, 66-69 year olds Claiming OASI by Age 65, 1996-2004



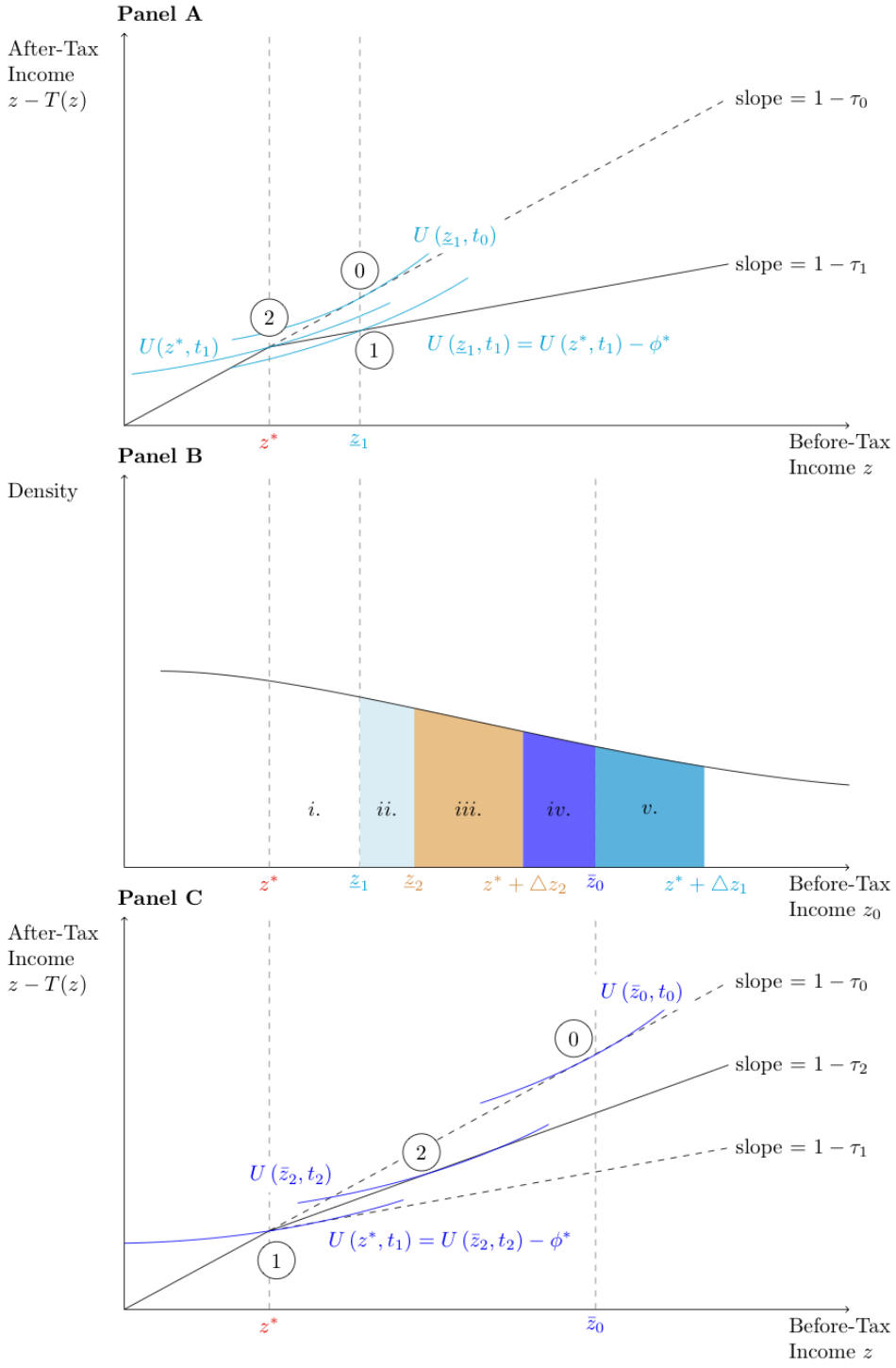
Note: The figure shows histograms of earnings from a one percent random sample of SSA administrative data on Social Security claimants aged 66-69 in each year from 1996 to 2004 (inclusive). In 2000 and after, the (placebo) kink is defined as the kink applying to those in the year of attaining NRA; as we mention in the text, the results are robust to two other ways of defining this placebo kink. See other notes to Figure 2.

Figure 5: Normalized Excess Bunching by Year, 1961-2005



Note: The figure shows normalized excess bunching from a one percent random sample of SSA administrative data on Social Security claimants aged 66-69 in each year between 1961 and 2005. See other notes to Figure 3.

Figure 6: Bunching Responses to a Convex Kink, with Fixed Adjustment Costs



Note: See Section 7 for an explanation of the figures.

Table 1: Summary Statistics, Social Security Administration Master Earnings File

	(1)	(2)
	Ages 18-75	Ages 62-69
Mean Earnings	37,492.28	29,485.08
	(282,940.03)	(783,897.87)
10th Percentile	5,234.74	1,957.78
25th Percentile	15,291.11	7,291.21
50th Percentile	32,982.17	17,739.68
75th Percentile	46,949.86	38,149.90
90th Percentile	66,370.18	58,417.99
Fraction with Positive Earnings	0.50	0.57
Fraction Male	0.56	0.57
Observations	13,612,313	1,595,139
Individuals	619,580	545,615

Note: The data are taken from a one percent random sample of the SSA Master Earnings File and Master Beneficiary Record. The data for ages 18-75 cover those in 1961-2005 who claim by age 65, who do not report self-employment earnings, and who have positive earnings. (However, the fraction with positive earnings is calculated by including those who have zero earnings.) Column 2 covers the same sample but limits the ages to 62-69, the group we examine most often. Earnings are expressed in 2010 dollars. Numbers in parentheses are standard deviations. The standard deviations are large because of very rare, abberrant large values of earnings (as documented in Utendorf 2001/2); these do not affect our estimates in the figures or tables because they are far above the AET exempt amount. These abberant values affect mean earnings far less than they affect the standard deviation. The results are robust to winsorizing.

Table 2: Estimates of Elasticity and Adjustment Cost Using Sharp Change Method

	(1)	(2)	(3)	(4)
	ε	ϕ	$\varepsilon \vert \phi = 0$	
			1990	1989
Baseline	0.23	\$152.08	0.39	0.22
	[0.20, 0.27]***	[49.54, 382.94]***	[0.33, 0.48]***	[0.18, 0.26]***
Lognormal density	0.26	\$165.76	0.43	0.24
	[0.22, 0.30]***	[45.57, 473.52]***	[0.35, 0.54]***	[0.19, 0.29]***
Benefit Enhancement	0.39	81.52	0.59	0.37
	[0.33, 0.45]***	[18.11, 245.88]***	[0.49, 0.73]***	[0.30, 0.43]***
Excluding FICA	0.32	\$129.14	0.50	0.30
	[0.27, 0.36]***	[34.18, 360.97]***	[0.42, 0.62]***	[0.25, 0.35]***
Bandwidth = \$500	0.25	\$90.65	0.38	0.24
	[0.21, 0.30]***	[6.04, 319.69]***	[0.30, 0.50]***	[0.19, 0.30]***
1988 to 1990	0.32	\$114.33	0.50	0.31
	[0.29, 0.35]***	[26.49, 307.96]***	[0.42, 0.62]***	[0.27, 0.34]***

Note: The table shows estimates of the elasticity and adjustment cost using the Sharp Change method described in the text. We report bootstraped confidence intervals shown in parentheses. We investigate the 1990 reduction in the AET BRR from 50 percent to 33.33 percent. The baseline specification assumes a uniform density, calculates the effective MTR by including the effects of the AET BRR and federal and state income and FICA taxes, and uses data from 1989 and 1990. Alternative specifications deviate from the baseline as noted. The estimates that include benefit enhancement use effective marginal tax rates due to the AET based on the authors' calculations relying on Coile and Gruber (2001) (assuming that individuals are considering earning just enough to trigger benefit enhancement). This translates the BRR before and after the 1990 policy change to 36% and 24%, respectively. Columns (1) and (2) report joint estimates with $\phi \geq 0$ imposed (consistent with theory, as described in the Appendix), while Columns (3) and (4) impose the restriction $\phi = 0$. The constrained estimate in Column (3) only uses data from 1990, whereas that in Column (4) uses only data from 1989 (except in the row where we investigate data from 1988 and 1990, in which case Column 4 uses data from 1988). *** indicates that the left endpoint of the 99 percent confidence interval is greater than zero; ** indicates that this is true for the 95 percent confidence interval; and * for the 90 percent confidence interval.

A Appendix: Additional Features of Annual Earnings Test[70]

When current benefits are lost to the AET, future scheduled benefits are increased in some circumstances. This is sometimes called "benefit enhancement." As we describe, for workers NRA or older in the pre-2000 period (when they faced the AET), benefit enhancement attenuates the effective AET BRR for individuals considering earning enough to trigger the benefit enhancement, but it does not attenuate the effective AET BRR for those considering earning less than this amount.

The benefit enhancement rules have varied over time, and they depend on whether the beneficiary is above or below NRA. Prior to 1972, there was no benefit enhancement for people aged NRA and older. In these years, the AET represented a pure loss in benefits for those NRA and above (equivalent to a pure tax). For beneficiaries NRA and older, a one percent Delayed Retirement Credit (DRC) was introduced in 1972. The DRC was intended to compensate beneficiaries who delayed claiming beyond age 65, but they also apply to earnings lost to the AET. For individuals above NRA, benefits are increased 1/12 of 1 percent for each month between ages 65 and 72 for which no benefits received after 1972 (Social Security, 2012, Table 2.A.20).[71]

This language indicates that each month's worth of foregone benefits—either because of delayed claiming or because of the AET—results in increased future benefits. A beneficiary has to forego an *entire* month of benefits in order to receive the DRC; if, for example, she earns slightly over the exempt amount and loses only a small amount of benefits to the AET, then her future benefits are not adjusted. Thus, the DRC provides no marginal relief from the AET for a claimant who is considering earning near the exempt amount: no benefit enhancement occurs when she earns a marginal dollar at or near the AET earnings threshold. Meanwhile, if she earns enough to forego an entire month's worth of benefits (but not when a smaller amount of benefits is lost due to the AET), future benefits are increased by 1/12 of 1 percent.

As a result of these rules, future benefits are enhanced when the individual's yearly earnings are over $z^* + (MB/\tau)$, where z^* is the exempt amount, MB is the monthly benefit, and τ is the AET benefit reduction rate.[72] For example, with a typical monthly benefit of \$1,000 and a benefit reduction rate of 33.33 percent, benefit enhancement occurs when the individual's yearly earnings are \$3,000 (=\$1000/0.3333) above the exempt amount. Benefit enhancement corresponding to one more month of reduced earnings occurs once annual earnings reaches \$6,000 above the exempt amount, and so forth. Thus, benefit enhancement is only relevant to an individual considering earning substantially in excess of the exempt amount and is therefore not relevant to marginal earnings decisions at the exempt amount. Indeed, this theoretical presumption is consistent with suggestive evidence we describe that indicates little systematic bunching reaction to changes in the DRC and that mean age at death is smooth near the

[70] This section is based on table 2.A.20 of the Annual Statistical Supplement of the Social Security bulletin, as well as extensive email correspondence with numerous officials at the Social Security Administration.

[71] The size of the DRC was increased to three percent per year in 1982, and then increased steadily throughout the 1990s, reaching eight percent for each year of foregone benefits in 2008. Starting in 1983, benefit enhancement only applied through age 69.

[72] Another month's benefit enhancement would occur if the individual earns more than $z^* + (2MB/\tau)$; a third month's benefit enhancement would occur if she earns more than $z^* + (3MB/\tau)$; and so on. Note that this creates 12 notches in the budget set, the final one at $z^* + (12MB/\tau)$.

exempt amount.

The AET is implemented in a number of stages. First, SSA must determine that a claimant is expected to exceed the exempt amount, or that she has already done so. Claimaints can notify SSA in advance if they expect to exceed the exempt amount, or they can report their earnings *ex post facto* at any point in the year. In addition, SSA uses W-2 records at the end of the year to determine if the AET threshold has been crossed (for those who have a W-2). Second, SSA withholds OASI benefits in monthly increments, until enough benefits have been withheld to cover the AET penalty amount. For example, assume an individual aged 66 with a monthly benefit of \$1,200 earns \$1,800 dollars beyond the AET exempt amount in 1992, when the benefit reduction rate was 0.3333. This individual should receive a yearly benefit reduction of \$1,800 $\times$ 0.3333 = \$600. SSA withholds an entire month's check, \$1,200, in order to collect the \$600. Finally, at the end of the year, SSA refunds any overwithheld benefits. In the same example, at the end of the year SSA would return \$600 in overwithheld benefits. Importantly, the DRC is not applied to future benefits in this case—less than a month's worth of benefits, \$600, was ultimately collected by SSA, after factoring overwithholding and refunds. After considering both withholding and refunds, the AET is ultimately applied at a yearly level—much in the same way that the Earned Income Tax Credit (EITC) is applied at the annual level but the receipt of the credit depends on one's withholding patterns and when the income tax return is filed.

In sum, for people NRA and older, the AET effectively acts as a kink for those earning close enough to the exempt amount, and benefit enhancement does not attenuate the marginal work disincentives associated with the AET in this range of earnings. However, the DRC is relevant to an individual considering earning enough to reduce her OASI benefit by at least a month's worth (*i.e.* at least $z^* + (MB/\tau)$). Empirically, we find that limiting the sample to those with substantial OASI benefits—for whom this earnings level is several thousand dollars above the AET exempt amount, and for whom the notch created by the DRC is therefore less relevant—yields very similar results to those we have shown. Our empirical specification alternatively assumes that benefit enhancement does not affect the AET implicit marginal tax rate (or does), and we find similar patterns in both specifications.

Note that in our empirical estimation, the region we "dummy out" near the kink z^* is $[z^* - 2800, z^* + 2800]$. Thus, if all of the "bunchers" arrived at the kink from initial earnings levels between z^* and $z^* + 2800$, we could find zero bunching despite substantial actual bunching. However, the earnings densities clearly show that the polynomial does not substantially overpredict bunching in the region above the exempt amount. Moreover, the densities also show no evidence of bunching near notches in the budget set created by the DRC. We alternatively use a bandwidth of \$500 and find similar results.

Individuals younger than NRA are subject to different rules for benefit adjustment, called "actuarial adjustment." The rule for this younger group was introduced in the legislation allowing people younger than 65 to claim early benefits (in 1956 for women and in 1961 for men). For those younger than NRA, future benefits are reduced 5/9 of 1 percent for each month under age 65 in which an individual claims benefits (Social Security, 2012, Table 2.A.20). This implies that if a beneficiary has any income withheld under the AET in a given month, then she receives a full benefit enhancement for that month. On the other hand, if a beneficiary does not have any income withheld under the AET in a given month, then she receives no benefit enhancement for that month. This creates a notch at the exempt amount—a discontinuous increase in future benefits when moving from just under the exempt amount to just over it—creating incentives

to bunch just above the exempt amount in order to receive the monthly (or yearly) benefit adjustment. We find no evidence for this kind of behavior; in fact, as we document in Gelber, Jones, and Sacks (2013), people tend to have earnings just below the exempt amount, exactly opposite the behavior we would expect if people were responding to the notch just described. Because benefit enhancement occurs at exactly the same earnings level that the AET begins to apply for those under NRA, we focus on the group NRA and above.

Formally, the number of months' worth of benefit enhancement enjoyed by OASI recipients is therefore $floor(\tau \cdot (z - z^*)/MB)$ for those NRA and above, and $ceiling(\tau \cdot (z - z^*)/MB)$ for those below NRA.

Benefit enhancement is actuarially fair if the net present value of the benefit enhancement equals the benefits lost due to the AET. The actuarial adjustment is approximately actuarially fair *in the sense that delaying OASI claiming an extra year is approximately actuarially fair*; however, this does *not* imply that benefit enhancement is actuarially fair when an additional dollar of benefits is withheld due to the AET. For example, actuarial adjustment is not actuarially fair for (among others) those with positive OASI benefits considering earning an additional amount above the AET exempt amount, because this does not result in additional benefit enhancement. Similar considerations apply to the DRC: additional marginal increments of earnings are *not* compensated through benefit enhancement (except in the case when an individual goes from earning just under to just over $z^* + (MB/\tau)$ (or one of the other 12 thresholds)).

The AET applies to an individual's earnings; spouses' earnings do not count in the earnings total to which the AET is applied. For a retired worker (*i.e.* primary) beneficiary whose spouse collects spousal benefits, the AET reduces the family's OASI benefit by the amounts we have described. The family benefit is also reduced when the spouse (separately) earns more than the AET threshold. For a retired worker beneficiary whose spouse is collecting benefits on his or her own earnings record, the AET reduces the retired worker beneficiary's benefits by the amounts described while not affecting the spouse's benefits. Thus, following previous literature (*e.g.* Friedberg 1998, 2000), we model the AET as creating the MTRs associated with the BRRs described, because the AET reduces family benefits by these amounts (all else equal). Our data do not contain the information necessary to link spouses (except when one spouse is claiming OASI benefits on the other spouse's record).

It is also worth noting how the actuarial adjustment and DRC interact with incentives for claiming OASI. Under the actuarial adjustment, the full benefit enhancement occurs when the individual earnings over the threshold level. Thus, the individual could in principle claim OASI; earn just over this threshold level; collect nearly her entire OASI benefit in this year (since the AET only reduces current OASI benefits at the margin); and later enjoy full benefit enhancement. This illustrates the more general point that it can be in an individual's interest to claim OASI even if the individual faces the AET. More generally, for individuals for whom the AET reduces OASI benefits sufficiently little, and for whom current OASI benefits are sufficiently important, it can be in their interest to claim OASI even if they face the AET. Appendix Figure F.22 shows that among the sample of individuals who have not claimed by year t, the hazard of claiming at year $t+1$ is smooth near the kink, indicating no evidence that claimants come disproportionately from close to or far from the kink.

B Appendix: Procedure for Estimating Normalized Excess Bunching

In order to estimate excess normalized bunching, we use the following procedure. For each centered-earnings bin z_i, we calculate p_i, the proportion of all people with earnings in the range $[z_i - k/2, z_i + k/2)$ (in a given time period and for a given age group). For example, underlying the first panel in Figure 2 is the probability p_i of earning in various bins z_i for 62 year-olds in the 1990-1999 period. The earnings bins are normalized by distance-to-kink, so that for $z_i = 0$, p_i is the fraction of people with earnings in the range $[-k/2, k/2)$. To estimate bunching, we assume that p_i can be written as

$$p_i = \sum_{d=0}^{D} \beta_d (z_i)^d + \sum_{n=-k}^{k} \gamma_k 1\{z_i = k\delta\} + \varepsilon_i \tag{B.1}$$

and run this regression. This equation expresses the earnings distribution as a degree D polynomial, plus a set of indicators for each bin within $k\delta$ of the kink, where δ is the binwidth. In our empirical application, we choose $D = 7, \delta = 800$ and $k = 3$ (so that seven bins are excluded from the polynomial estimation, including the bin centered at the kink). We show that our results to alternative choices of D, δ, and k.

Our measure of excess mass is $\hat{EM} = \sum_{n=-k}^{k} \hat{\gamma}_k$, the estimated excess probability of locating at the kink (relative to the polynomial term). This measure depends on the counterfactual density near the kink, so to obtain a measure of excess mass that is comparable at the kink, we scale by the predicted density that would obtain if there were no bunching. This is just the constant term in the polynomial, since the z_i is distance to zero. So our estimate of normalized excess mass is

$$\hat{B} = \frac{\hat{EM}}{\hat{\beta}_0}. \tag{B.2}$$

We consider two approaches for constructing standard errors. First, from Equation (B.2), it is straightforward to apply the Delta method. Second, we employ the parametric bootstrap procedure of Chetty, Friedman, Olsen and Pistaferri (2011). This bootstrap draws with replacement from the estimated distribution of errors ε_i from Equation (B.1). For each set of draws, we get a new value of p_i and use these new values to re-estimate B. The standard deviation across draws of B is our measure of the standard error $\hat{B}$. In practice these two procedures produced extremely similar results, so we only report standard errors from the bootstrap.

C Appendix: Social Security Data

Our data come from the Social Security Master Earnings File (MEF), which is described more extensively in Song and Manchester (2007). The MEF is a longitudinal history of Social Security taxable earnings for all Social Security Numbers (SSNs) in the U.S. Our data are a one percent random sample of SSNs; we randomly extract SSNs from the database and follow each of these individuals over the full time period. The AET is based on earnings as measured in this dataset. Prior to 1978, the data have information on annual FICA earnings; since 1978, the data have information on uncapped wage compensation. Before 1978, the data do not clearly distinguish between earnings from self-employment and non-self-employment earnings, but we are able to

distinguish them in the data starting in 1978. The data also contain information on date of birth, date of death, and sex.

We supplement the MEF with information from the Master Beneficiary Record (MBR) file, which contains data on the day, month, and year that people began to claim Social Security (and other variables). The majority of workers excluded from OASDI coverage are in four main categories: (1) federal civilian employees hired before January 1, 1984; (2) agricultural workers and domestic workers whose earnings do not meet certain minimum requirements; (3) individuals with very low net earnings from self-employment (generally less than $400 per year); and (4) employees of several state and local governments. However, civil service and other government workers are covered by Medicare and are therefore present in the MBR.

In choosing our main sample, we take into account a number of considerations. It is desirable to show a constant sample in making comparisons of earnings densities. Meanwhile, the AET only affects people who claim OASI, and thus we wish to focus on claimants. However, many individuals claim OASI at ages over the Early Entitlement Age (62), implying that they have not claimed at younger ages but have claimed by older ages. This implies that to investigate a constant sample, we cannot simply limit the sample to claimants at each age (because many people move from not claiming to claiming). To balance these considerations, our main sample at each age and year consists of individuals who ultimately claim in the year they turn 65 or earlier. We show that the results are robust to other sample definitions. Because we focus on the intensive margin response (consistent with Saez (2010) and subsequent papers on bunching), we further limit the sample to observations with positive earnings in our main analysis.

Information on AET parameters is from table 2.A.20 and 2.A.29 of the Annual Statistical Supplement to the Social Security Bulletin. Friedberg (1998, 2000) provides a thorough description of these rules. All dollar amounts are deflated to 2010 dollars using the CPI-U.

The standard deviation is large because of very rare abberrant large values of earnings (as documented in Utendorf 2001/2); these do not affect our estimates in the figures or tables because they are far above the AET exempt amount, and they affect mean earnings far less than they affect the standard deviation. The results are robust to winsorizing.

In 1983-1999, the AET is assessed on earnings until the month in which the individual turns age 70. For simplicity, in our baseline sample we measure age as calendar year minus year of birth. Thus, if an individual turns age 70 later in the year—in the extreme case, on December 31—she will have had an incentive to bunch at the kink during nearly the entire year when she is classified as age 70 in our data. As a result, her yearly earnings may appear to be located at or near the kink even though she is bunching at the kink applicable to 69-year-olds through almost all of the calendar year over which her earnings are observed. However, the figure shows that significant bunching occurs at age 71, which cannot be due to this coarse measure of birth dates. Thus, the results do show a delay in complete adjustment. We have also found substantial and significant ($p < 0.01$) bunching at age 70 among those born in January, who no longer face the AET immediately in January of the year they turn 70 and therefore should not show excess bunching at this age in the absence of adjustment frictions. Likewise, we find a spike in mean earnings growth from age 70 to age 71 among those born in January. In our sample period, the AET applied to ages 62-71 before 1983, and it applied to ages under NRA in 2000 and after. In these time periods, examining only those born in January also shows a delay in responding to the removal of the AET.

Since 1978, the earnings test has been assessed on yearly earnings, implying that we analyze the appropriate time period, *i.e.* earnings in a calendar year. Prior to 1978, the earnings test

was assessed on quarterly earnings. While there is likely some error in measuring the amount of bunching pre-1978, we believe that this is not a major issue: the patterns of bunching in the pre-1978 period are visually clear and appear unlikely to be changed in a qualitative sense by an examination of quarterly data. Moreover, Figure 5 shows that the amount of excess bunching falls from 1977 to 1978 and subsequent years, rather than rising as we might expect if we hypothetically measured bunching more accurately starting in 1978.

D Appendix: Longitudinal Employer Household Dynamics

We use the Longitudinal Employer Household Dyanmics (LEHD) dataset, which contains wage data available from state-level unemployment insurance (UI) programs. These data measure uncapped quarterly earnings for employees covered by state unemployment insurance systems, estimated to cover over 95 percent of private sector employment. Although coverage laws vary slightly from state to state, UI programs do not cover federal employees, the self-employed, and many agricultural workers, domestic workers, churches, nonprofits, and state and local government employees. We examine a 20 percent random sample of the original LEHD file, as this was the largest amount of data that our available server space could handle.

These administrative earnings records are linked across quarters to create individual work histories. In addition to earnings, information on gender and date of birth are available. The data on employees are linked to data on firms. Each firm at which an individual works in a given quarter is identified through a firm identifier. We consider an employee to have changed employers from year t to year $t+1$ if at least one of the federal employer IDs at which the employee works in year t is different in year $t+1$. However, when the individual works at one or more employer in year t and does not work at any employer in year $t+1$, we drop this individual from the sample. The results are similar when we treat these individuals as if they changed employers.

We select data from 1990-1999. During this period, the AET explicit benefit reduction rate was constant. 1990-1999 also represent natural years to investigate because large sample sizes are not available in the LEHD prior to 1990. When we include other years and age groups in the LEHD sample, we find similar results to those reported here. Note that the population we investigate is not constant over this period, because (among other reasons) an increasingly broad set of states is included in the LEHD over time. Data are available on 13 states in 1990, climbing to 28 states by 1999. In a given quarter, we include in our sample all states whose data are available. Holding the sample constant yields very similar results.

E Appendix: Model of Earnings Response

E.1 Baseline Model

We start with a baseline, frictionless model of earnings, following Saez (2010). We briefly sketch the key features of this model for comparison to our model with a fixed cost of adjustment. In Saez (2010), individuals maximize utility over consumption, c, and costly earnings, z:

$$u\left(c, z; n\right)$$

Heterogeneity is parameterized by an "ability" parameter n, which is distributed according to the smooth cdf $F(\cdot)$. Individuals maximize utility subject to the following budget constraint:

$$c = (1-\tau)\, z + R$$

where R is virtual income. This leads to the first order condition:

$$-\frac{u_z(c,z;n)}{u_c(c,z;n)} = (1-\tau),$$

which implicitly defines an earnings supply function: $z(1-\tau, R, n)$.

When necessary, we will use a quasi-linear and iso-elastic utility function:

$$u(c,z;n) = c - \frac{n}{1+1/\varepsilon}\left(\frac{z}{n}\right)^{1+1/\varepsilon}$$

Under this assumption, the first order condition simplifies to:

$$(1-\tau) - \left(\frac{z}{n}\right)^{\frac{1}{\varepsilon}} = 0,$$

which implies this earnings supply function:

$$z = n(1-\tau)^{\varepsilon}.$$

E.2 Linear Tax Schedule

Consider first a linear tax schedule with a constant marginal tax rate τ_0. Observe that with a smooth distribution of skills n, we have a smooth distribution of earnings that is monotonic in skill, provided we make the typical Spence-Mirlees assumption. Let $H_0(\cdot)$ denote the cumulative distribution function (CDF) of earnings under the constant marginal tax rate, and let $h_0(\cdot) = H_0'(\cdot)$ denote the density of this distribution. Under quasilinear utility, we have:

$$H_0(z) = F\left(\frac{z}{(1-\tau_0)^{\varepsilon}}\right).$$

Define $H_1(\cdot)$ and $h_1(\cdot)$ as the smooth CDF and density of earnings under a higher, constant marginal tax rate τ_1; H_1 is defined similarly as a function of τ_1.

E.3 Kinked Tax Schedule

Now consider a piecewise linear tax schedule with a convex kink: the marginal tax rate below earnings level z^* is τ_0, and the marginal tax rate above z^* is $\tau_1 > \tau_0$. Given the tax schedule, individuals bunch at the kink point z^*; as explained in Saez (2010), the realized density in earnings has an excess mass at z^*. Denote the realized distribution of earnings once the kink has been introduced at z^* as $H(\cdot)$:

$$H(z) = \begin{cases} H_0(z) & \text{if } z < z^* \\ H_1(z) & \text{if } z \geq z^* \end{cases}$$

Denote the density of this realized distribution as $h(\cdot) = H'(\cdot)$. In general there is now a discrete jump in the earnings density at z^*:

$$h(z) = \begin{cases} h_0(z) & \text{if } z < z^* \\ h_1(z) & \text{if } z > z^* \end{cases}$$

The share of people who relocate to the kink is:

$$B = \int_{z^*}^{z^* + \triangle z^*} h_0(\zeta)\, d\zeta$$

These "bunchers" are those whose *ex ante* earnings lie in the range $[z^*, z^* + \triangle z^*]$, who are induced to locate at the kink by the rise in the MTR above the kink point. For relatively small changes in the tax rate, we can relate the elasticity of earnings with respect to the net-of-tax rate to the earnings change $\triangle z^*$ for the individual with the highest *ex ante* earnings who bunches *ex post*:

$$\varepsilon = \frac{\triangle z^*/z^*}{d\tau_1/(1-\tau_0)}$$

where $d\tau_1 = \tau_1 - \tau_0$.[73]

E.4 Graphical Exposition

Appendix Figure F.1 depicts a setting in which a kinked budget set is introduced. In Panels A and B, the x-axis shows before-tax-and-transfer income, z, and the y-axis shows after-tax-and-transfer consumption, $z - T(z)$. Consider first a linear tax (Panel A) at a rate of τ. An individual optimally locates at a point of tangency, where the marginal rate of substitution (MRS) between earnings and consumption equals the net-of-tax rate, $1-\tau$. The figure shows indifference curves and earnings levels for low- and high-earning agents (labeled L and H, respectively). The low earner has an earnings level of z^*, while the high earner receives $z^* + \triangle z$.

Suppose the AET is introduced (on top of pre-existing taxes), so that the marginal net-of-tax rate decreases to $1 - \tau - d\tau$ for earnings above a threshold z^*. For small $d\tau$, individuals earning in the neighborhood above z^* will reduce their earnings. If ability is smoothly distributed, a range of individuals will locate exactly at z^*, due to the discontinuous jump in the marginal net-of-tax rate at z^*. In Panels A and B, individual L has the lowest *ex ante* earnings among those who bunch at z^*, individual H has the highest *ex ante* earnings in this group, and all others previously earning between z^* and $z^* + \triangle z$ also bunch at z^*. Those with *ex ante* earnings higher than $z^* + \triangle z$ reduce their earnings to a level greater than z^*.

Panels C and D of Figure F.1 depict densities of earnings we would expect to observe in the absence and presence of the AET, respectively. The x-axis shows before-tax earnings, z, and the y-axis measures the density of earnings. In Panel C, the density is continuous at z^*, reflecting a smooth distribution of ability. The blue region represents the set of individuals who bunch at z^* in the presence of the AET, *i.e.* those earning in $[z^*, z^* + \triangle z]$ in the absence of the AET. Panel D shows that once the AET is introduced, these individuals locate in the neighborhood of z^*. However, rather than depicting a mass point exactly at z^*, we have shown bunching in the region at and surrounding z^*, reflecting the fact that individuals often cannot

[73] This formula holds if there is a single elasticity ε in the population. Under heterogeneity, the method returns $\bar{\varepsilon}$, the average elasticity among bunchers. We investigate cases with heterogeneity below.

bunch exactly at the kink point (as discussed, for example, in Saez, 2010).

E.5 Fixed Adjustment Costs

We now extend the model to include a fixed cost of adjusting earnings. We assume that the adjustment cost reflects a disutility of ϕ^* of increasing or decreasing earnings from some initial earnings level. We begin by analyzing the response to a change in the marginal tax rate from τ_0 to τ_1, where the tax schedule is linear in both cases, in order to build intuition for the case with a kinked budget set. We assume that following a change in tax rates from τ_0 to τ_1, the gain (absent adjustment costs) to reoptimizing is increasing in n. In general, this requires that the size of the optimal earnings adjustment increases in n at a rate faster than the decrease in the marginal utility of consumption.[74] This is true, for example, if utility is quasilinear.

If the gain in utility is monotonically increasing in initial earnings, and the cost of adjustment is fixed, there exists a unique level of initial earnings at which the agent is indifferent between adjusting and staying at the initial earnings level. We formally state the implications in the following result:

Remark 1 (Linear Tax Change and Adjustment Costs)

After a change in linear tax rates from τ_0 to τ_1, if there is a constant adjustment cost of ϕ^* and the size of the optimal earnings adjustment increases in n at a rate faster than the decrease in the marginal utility of consumption, then:

1. *There is a unique threshold of initial earnings, $z_{0,\phi}$, above which all individuals will adjust their earnings in response to the tax change. Those initially locating below the threshold will not adjust.*

2. *The threshold level of earnings satisfies the following identity:*

$$u\left(\left(1-\tau_1\right)z_{1,\phi}+R_1, z_{1,\phi}\right)-u\left(\left(1-\tau_1\right)z_{0,\phi}+R_1, z_{0,\phi}\right)\equiv\phi^*$$

   where $z_{1,\phi}$ is the ex post earnings level of the individual who initially locates at $z_{0,\phi}$. In other words, at the threshold level, the gain in u from adjusting earnings is exactly equal to the adjustment cost ϕ^*.

3. *In the case of quasilinear utility, the threshold level of earnings is:*

$$z_{0,\phi}=\frac{\phi^*}{\alpha\left(\varepsilon,\tau_0,\tau_1\right)}$$

   where

$$\alpha\left(\varepsilon,\tau_0,\tau_1\right)\equiv\frac{1-\tau_1}{1+\varepsilon}\left[\left(\frac{1-\tau_1}{1-\tau_0}\right)^{\varepsilon}-1+\varepsilon\left(\frac{\tau_1-\tau_0}{1-\tau_1}\right)\right].$$

[74] To see this, note that the utility gain from reoptimizing is $u\left(\left(1-\tau_1\right)z_1+R_1, z_1; n\right) - u\left(\left(1-\tau_1\right)z_0+R_1, z_0; n\right) \approx u_c\cdot\left(1-\tau_1\right)\left[z_1-z_0\right]+u_z\cdot\left[z_1-z_0\right] = u_c\cdot\left(\tau_1-\tau_0\right)\left[z_0-z_1\right]$, where in the first expression, we have used a first-order approximation for utility at $\left(\left(1-\tau_0\right)z_0+R_0, z_0\right)$ and in the second expression we have used the first order condtion $u_z=-u_c\left(1-\tau_0\right)$. The gain in utility is approximately equal to an expression that depends on the marginal utility of consumption, the change in tax rates, and the size of the earnings adjustment. The first term, u_c, is decreasing as n (and therefore initial earnings z_0) increases. Thus, in order for the gain in utility to be increasing in n, we need the size of earnings adjustment $[z_0-z_1]$ to increase at a rate that dominates.

4. *The ex post distribution of earnings is:*

$$H(z) = \begin{cases} H_0(z) & \text{if } z < z_{1,\phi} \\ H_0(z) + H_1(z) - H_0(z_{0,\phi}) & \text{if } z \in [z_{1,\phi}, z_{0,\phi}] \\ H_1(z) & \text{if } z > z_{0,\phi} \end{cases}$$

$$h(z) = \begin{cases} h_0(z) & \text{if } z < z_{1,\phi} \\ h_0(z) + h_1(z) & \text{if } z \in [z_{1,\phi}, z_{0,\phi}] \\ h_1(z) & \text{if } z > z_{0,\phi} \end{cases}$$

where $H_0(\cdot)$ and $H_1(\cdot)$ are the CDFs of earnings in the presence of linear tax rates τ_0 and τ_1, respectively.

Next, consider choices in the presence of adjustment costs on a budget set with a convex kink. Consider again an initial linear tax schedule with marginal tax rate τ_0. Now, introduce a higher MTR $\tau_1 > \tau_0$ for earnings above z^*. We again assume that the gain to reoptimizing is increasing in initial earnings over the range $[z^*, z^* + \triangle z^*]$. Using the same logic as above—the gain in utility is monotonically increasing in initial earnings, and the cost of adjustment is fixed—there exists a unique level of initial earnings at which the agent is indifferent between adjusting and staying at the initial earnings level. Thus, we have the following result:

Remark 2 (Non-Linear Tax and Adjustment Costs)

When a kink is introduced in the budget set (i.e. a jump in marginal tax rates from τ_0 below z^* to τ_1 above z^*), there is a fixed adjustment cost of ϕ^*, and $z^* \geq z_{1,\phi}$,[75] *then:*

1. *Individuals with initial earnings below a unique threshold $\underline{z}$ do not adjust their earnings.*
2. *The threshold level of earnings is implicitly defined by the following:*

$$u((1-\tau_1)z^* + R_1, z^*) - u((1-\tau_1)\underline{z} + R_1, \underline{z}) \equiv \phi^* \qquad \text{(E.3)}$$
$$z^* \leq \underline{z} \leq z^* + \triangle z^*.$$

3. *Individuals with intial earnings in $[\underline{z}, z^* + \triangle z^*]$ bunch at the kink point z^*.*
4. *Individuals with initial earnings above $z^* + \triangle z^*$ reduce their earnings to a new level of earnings higher than z^*.*
5. *The ex post distribution of earnings is:*

$$H(z) = \begin{cases} H_0(z) & \text{if } z < z^* \\ H_0(z) + H_1(z) - H_0(\underline{z}) & \text{if } z \in [z^*, \underline{z}] \\ H_1(z) & \text{if } z > \underline{z} \end{cases}$$

$$h(z) = \begin{cases} h_0(z) & \text{if } z < z^* \\ h_0(z) + h_1(z) & \text{if } z \in [z^*, \underline{z}] \\ h_1(z) & \text{if } z > \underline{z} \end{cases}$$

[75] $z_{1,\phi}$ is again the *ex post* level of earnings for the individual who initially locates at $z_{0,\phi}$ – where $z_{0,\phi}$ is the initial earnings level over which individuals adjust their earnings defined above in Remark (1). Note that $\underline{z}$ denotes this threshold in the non-linear budget set case, whereas $z_{0,\phi}$ denotes this threshold in the linear budget set case.

6. *Excess bunching at z^* is given by:*

$$B = \int_{\underline{z}}^{z^*+\triangle z^*} h_0(\zeta)\, d\zeta$$

If the kink point z^* is lower than $z_{1,\phi}$, then:

1. *Individuals only adjust their earnings if their initial earnings level is above the threshold $z_{0,\phi}$.*
2. *There is no bunching at z^*.*
3. *The ex post distribution of earnings is the same as in the case of a change in a linear tax rate from τ_0 to τ_1 described in Remark (1).*

E.6 Derivation of Closed-Form Solution for Elasticity and Adjustment Cost (Comparative Static Approach)

As we discuss in Remark (2) above and in Section 7 in the text, the amount of bunching in the presence of a fixed adjustment cost is equal to the integral of the initial earnings density over the range $[\underline{z}, z^* + \Delta z^*]$:

$$B(\boldsymbol{\tau}, z^*; \varepsilon, \phi^*) = \int_{\underline{z}}^{z^*+\Delta z^*} h(\zeta)\, d\zeta, \tag{E.4}$$

where $\boldsymbol{\tau} \equiv (\tau_0, \tau_1)$. If the density is locally uniform, the integral in (E.4) is:

$$B(\boldsymbol{\tau}, z^*; \varepsilon, \phi^*) \approx h(\underline{z})(z^* + \Delta z^* - \underline{z}) \tag{E.5}$$

Taking a first-order Taylor approximation of $u\left((1-\tau_1)\underline{z} + R_1, \underline{z}, \underline{n}\right)$ and $u\left((1-\tau_1)z^* + R_1, z^*, \underline{n}\right)$ at $\left((1-\tau_0)\underline{z} + R_0, \underline{z}, \underline{n}\right)$, and using the first order condition for initial earnings, $(1-\tau_0)u_c = -u_z$, we have from (E.3):

$$\begin{aligned} \phi^* &\approx u_c \cdot (1-\tau_1)\left[z^* - \underline{z}\right] + u_z \cdot (z^* - \underline{z}) \\ \Rightarrow \underline{z} &\approx z^* + \frac{\phi^*/u_c}{(\tau_1 - \tau_0)} \\ &= z^* + \frac{\phi}{d\tau_1}, \end{aligned}$$

where $d\tau_1 = \tau_1 - \tau_0$ and $\phi = \phi^*/u_c$ is the dollar equivalent of the disutility associated with adjusting earnings. Substituting this expression for $\underline{z}$ into (E.5), we have

$$B(\boldsymbol{\tau}, z^*; \varepsilon, \phi) = h(\underline{z})(\Delta z - \phi/d\tau_1),$$

where bunching now depends on the dollar-denominated cost of adjusting, rather than the utility cost. Finally, for small $d\tau$, Δz is small and $h(\underline{z}) \approx h(z^* + \Delta z) \approx h(z^*)$. Let $b \equiv B/h(z^*)$, and note that $\Delta z^* = z^* \left(d\tau_1/\left(1-\tau_0\right)\right)\varepsilon$. The excess mass at the kink can now be expressed as a linear function of the parameters:

$$b(\boldsymbol{\tau}, z^*; \varepsilon, \phi) = \varepsilon\left(z^* \frac{d\tau_1}{1-\tau_0}\right) - \phi\left(\frac{1}{d\tau_1}\right). \tag{E.6}$$

E.6.1 Derivation of Formula for Bunching with Heterogeneity

We derive the formula for bunching B in the presence of heterogeneity under the Comparative Static approach as follows:

$$\begin{aligned}
B &= \iiint_{\underline{z}}^{z^*+\Delta z^*} \tilde{h}\left(\zeta,\epsilon,\varphi\right) d\zeta d\epsilon d\varphi \\
&\approx \iint \left[z^* + \Delta z^* - \underline{z}\right] \tilde{h}\left(z^*,\epsilon,\varphi\right) d\epsilon d\varphi \\
&\approx \iint \left[\epsilon\left(z^* \frac{d\tau_1}{1-\tau_0}\right) - \varphi\left(\frac{1}{d\tau_1}\right)\right] \tilde{h}\left(z^*,\epsilon,\varphi\right) d\epsilon d\varphi \\
&= h\left(z^*\right) \cdot \left[\left(\iint \epsilon \frac{\tilde{h}\left(z^*,\epsilon,\varphi\right)}{h\left(z^*\right)} d\epsilon d\varphi\right)\left(z^* \frac{d\tau_1}{1-\tau_0}\right) - \left(\iint \varphi \frac{\tilde{h}\left(z^*,\epsilon,\varphi\right)}{h\left(z^*\right)} d\epsilon d\varphi\right)\left(\frac{1}{d\tau_1}\right)\right] \\
&= h\left(z^*\right) \cdot \left[\bar{\varepsilon}\left(z^* \frac{d\tau_1}{1-\tau_0}\right) - \bar{\phi}\left(\frac{1}{d\tau_1}\right)\right],
\end{aligned}$$

where we have used the assumption of constant $\tilde{h}\left(\cdot\right)$ and the approximations for Δz^* and $\underline{z}$ in Section 7. Here $h\left(z^*\right) = \iint \tilde{h}\left(z^*,\epsilon,\varphi\right) d\epsilon d\varphi$, and $\bar{\varepsilon}$ and $\bar{\phi}$ are the average elasticity and adjustment cost, respectively.

E.6.2 Linear Adjustment Costs

We now introduce an adjustment cost that increases linearly in the size of the adjustment. Assume that given an initial level of earnings z_0, agents must pay a cost of $\phi^* \cdot |z - z_0|$ when they change their earnings to a new level z. Utility $\tilde{u}$ at the new earnings level can be represented as:

$$\tilde{u}\left(c, z; n, z_0\right) = u\left(c, z; n\right) - \phi^* \cdot \left|z - z_0\right|.$$

The first order condition for earnings can be characterized as:

$$\begin{aligned}
-\frac{u_z\left(c,z;n\right)}{u_c\left(c,z;n\right)} &= \left(1 - \tau - \phi^*/\lambda^* \cdot \operatorname{sgn}\left(z - z_0\right)\right) \\
&= \begin{cases} \left(1-\tau-\phi\right) & \text{if } z > z_0 \\ \left(1-\tau+\phi\right) & \text{if } z < z_0 \end{cases},
\end{aligned}$$

where $\lambda^* = u_c\left(c^*, z^*; n\right)$ is the Lagrange multiplier and $\phi = \phi^*/\lambda^*$ is the dollar equivalent of the linear adjustment cost ϕ^*.

The individual chooses earnings as if he faces an effective marginal tax rate of $\tilde{\tau} = \tau + \phi \cdot \operatorname{sgn}(z - z_0)$. It follows that our predictions about earnings adjustment are similar to our previous predictions, except that the effective marginal tax rate $\tilde{\tau}$ appears, rather than τ. Thus, we can solve for the elasticity of earnings as a function of the change in earnings $\triangle z^*$

due to introduction of a kink in the tax schedule and the jump in marginal tax rate $d\tau_1$:

$$\begin{aligned}\varepsilon &= \frac{\triangle z^*/z^*}{d\tilde{\tau}_1/\left(1-\tilde{\tau}_0\right)} \\ &= \frac{\triangle z^*/z^*}{\left(d\tau_1-2\phi\right)/\left(1-\tau_0-\phi\right)}.\end{aligned}$$

Since the right-hand side is increasing in ϕ, the estimate of the elasticity increases as the linear adjustment cost increases. This makes intuitive sense: the adjustment cost attenuates bunching, so holding constant the level of bunching, the elasticity must be higher as the adjustment cost increases.

Now assume that when an individual adjusts his earnings, he incurs a linear adjustment cost ϕ^{*L} for every unit of change in earnings, as well as a fixed cost ϕ^{*F} associated with any change in earnings. Consider again bunching at z^*, with a tax rate jump of $d\tau_1 = \tau_1 - \tau_0$ at earnings level z^*. We have the following set of expressions for excess mass:

$$\begin{aligned}B &= \int_{\underline{z}}^{z^*+\triangle z^*} h\left(\zeta\right) d\zeta \\ \varepsilon &= \frac{\triangle z^*/z^*}{\left(d\tau_1-2\phi^L\right)/\left(1-\tau_0-\phi^L\right)} \\ \phi^{*F}+\phi^{*L}\cdot\left(\underline{z}-z^*\right) &= u\left(\left(1-\tau_1\right)z^*+R',z^*;\underline{n}\right)-u\left(\left(1-\tau_1\right)\underline{z}+R',\underline{z};\underline{n}\right).\end{aligned}$$

Using a left rectangle approximation for the integral, we have:

$$\begin{aligned}b &\equiv B/h\left(z^*\right) \\ &= z^*+\triangle z^*-\underline{z} \\ &= z^*\left(\frac{d\tau_1-2\phi^L}{1-\tau_0-\phi^L}\varepsilon+1\right)-\underline{z}.\end{aligned}$$

We can further apply an approximation for $\underline{z}$ similar to the approximation we used in Section 7, *i.e.* $\underline{z} = z^* + \phi^F/\left(d\tau_1 - 2\phi^L\right)$. Thus, the expression for bunching can be simplified to:

$$b = \varepsilon\left(z^*\frac{d\tau_1-2\phi^L}{1-\tau_0-\phi^L}\right)-\frac{\phi^F}{\left(d\tau_1-\phi^L\right)},$$

where $\left(\phi^F,\phi^L\right) = \left(\phi^{*F}/\lambda^*,\phi^{*L}/\lambda^*\right)$. In this case, we need at least three kinks to separately identify $\left(\varepsilon,\phi^F,\phi^L\right)$. Because we do not examine a setting in which one can compare bunching under three different positive tax rates, we are not able to estimate these parameters using data (or to fruitfully estimate the parameters in the non-linear case since we do not have a credible source of variation to identify them).

E.7 Derivation of Formula for Bunching with a Pre-Existing Kink (Sharp Change Approach)

The Comparative Static approach abstracts from a key feature of our empirical setting. In particular, the Comparative Static approach models the transition from a budget set with no

kink to one with a kink. This approach facilitates our basic intution and provides a transparent bridge between our approach and existing bunching methods in the presence of a kink. However, in our context, we conduct analysis using data just before and just after the benefit reduction rate was decreased (in 1990 from 50 percent to 33.33 percent for 66-69 year olds; or from 33.33 percent to zero for this group in 2000; or from 33.33 percent for 69-year-olds to zero for 70-year-olds in 1990-1999). These changes involve moving from an initial state with a kink to a new state with a smaller kink. In a frictionless model, the distinction is immaterial. However, as we show, this matters in the presence of a fixed adjustment cost. In particular, when the kink becomes more muted, the change in bunching will be attenuated due to the fixed adjustment cost.

We will assume that in the initial state, bunching is characterized as in Remark (2). Let the initial kink, K_1, be characterized by a lower marginal tax rate, τ_0, to the left of z^*, and a higher marginal tax rate, τ_1, to the right of z^*. The initial level of bunching is:

$$B_1 = \int_{\underline{z}_1}^{z^*+\Delta z_1^*} h\left(\zeta\right) d\zeta$$

Now, consider a change in the kink to K_2, which retains the lower marginal tax rate τ_0 to the left of z^* but reduces the marginal tax rate to the right of z^* to $\tau_2 < \tau_1$. Had we begun with no kink and introduced K_2, bunching would have been:

$$B_2 = \int_{\underline{z}_2}^{z^*+\Delta z_2^*} h\left(\zeta\right) d\zeta$$

Note that relative to K_1, K_2 provides a weaker incentive to bunch, *when starting from a baseline tax schedule with no kink.* Formally, we have $\underline{z}_2 \geq \underline{z}_1$, $\Delta z_2^* < \Delta z_1^*$ and $B_2 \leq B_1$.

E.7.1 Characterizing Bunching

In characterizing bunching when moving from K_1 to K_2, individuals may be separated into several groups based on their optimal level of earnings z_0 in the absence of a kink. First, there are individuals with $z_0 < z^*$. They will locate to the left of the kink under both K_1 and K_2.

Second, we have individuals with $z^* < z_0 \leq \underline{z}_1$ (area i in Figure 6). These individuals would optimize in the presence of K_1 by moving to z^*, were it not for the adjustment cost. Now, with a smaller kink K_2, these individuals continue to remain at the initial earnings level $z_0 > z^*$, as the utility gain to reoptimizing to z^* is even smaller than it was under K_1.

Third, we have those with $\underline{z}_1 < z_0 \leq \underline{z}_2$ (area ii in Figure 6). When moving from no kink to K_1, these individuals locate at the kink, z^*. If the budget set had hypothetically transitioned from no kink to K_2, these individuals would have chosen to remain at z_0, due to the fixed adjustment cost. However, when moving from K_1 to K_2, these agents remain at the kink z^*. The reason is that the frictionless optimum under K_2 is z^* for everyone initially earning in the range $[z^*, z^* + \Delta z_2]$.

Fourth, we have agents with $\underline{z}_2 < z_0 \leq z^* + \Delta z_2$ (area iii in Figure 6). These individuals bunch at z^* when moving from no kink to either K_1 or K_2. Thus, they remain bunching at z^* when moving from K_1 to K_2.

Fifth, we have agents with $z^* + \Delta z_2 < z_0 \leq z^* + \Delta z_1$ (areas iv and v in Figure 6). When starting from a budget set with no kink, these agents bunch under K_1, but not under K_2.

Starting instead from K_1, they must choose between remaining at the kink z^* or moving to the frictionless optimum under K_2, $z_2 > z^*$. We know that at least some of these individuals will remain bunching. To see this, consider an individual with earnings under no kink $z_0 = z^* + \Delta z_2 + \delta_0$. For small enough δ_0 optimal earnings under K_1 is z^*, and optimal earnings under K_2 tends to z^* as δ_0 tends to zero. Likewise, the net utility gain from relocating from z^* to z_2 under K_2 tends to zero as δ_0 tends to zero. However, the fixed adjustment cost remains strictly positive. Therefore, this individual will remain at z^* when moving from K_1 to K_2 for small enough δ_0. In Figure 6, area iv shows those with initial earnings $z^* + \Delta z_2 < z_0 < \bar{z}_0$ who remain bunching at the kink when transitioning from K_1 to K_2. Area v shows those with initial earnings $\bar{z}_0 < z_0 < z^* + \Delta z_1$, who "debunch" from the kink when moving from K_1 to K_2.

When reoptimizing is beneficial for at least some agents in this final group, we will have a reduction in bunching when transitioning from K_1 to K_2.[76] Empirically, we observe such a reduction over time, so this is the case relevant to our setting. In this case, the marginal "de-buncher" will be defined by the following conditions:

$$\begin{aligned} -\frac{u_z\left(c_2, \bar{z}_2; \bar{n}_2\right)}{u_c\left(c_2, \bar{z}_2; \bar{n}_2\right)} &= (1-\tau_2) \\ u\left(\left(1-\tau_2\right)\bar{z}_2 + R_2, \bar{z}_2; \bar{n}_2\right) - u\left(\left(1-\tau_2\right)z^* + R_2, z^*; \bar{n}_2\right) &\equiv \phi^* \\ -\frac{u_z\left(c_0, \bar{z}_0; \bar{n}_2\right)}{u_c\left(c_0, \bar{z}_0; \bar{n}_2\right)} &= (1-\tau_0) \\ \bar{z}_0 &\leq z^* + \Delta z_1^* \end{aligned}$$

In words, the first line indicates that $\bar{z}_2 > z^*$ is the optimal, frictionless level of earnings chosen by the top buncher in the presence of K_2. The second line requires that when facing K_2, this agent is indifferent between remaining at z^* and moving to $\bar{z}_2$ through paying the adjustment cost. The third line defines $\bar{z}_0$ as the initial level of earnings that this individual chooses when facing a constant marginal tax rate of τ_0 and no kink. The fourth line requires that this individual is initially bunching at z^* in response to K_1. If this last inequality is binding, then when moving from K_1 to K_2, none of the bunchers "debunch" and the fraction bunching is unchanged. In that case, we have no variation available to identify both ε and ϕ. As noted above, empirically we do observe that excess bunching falls around 1990 when the BRR falls from 50 percent to 33.33 percent (as well as in the other cases we examine empirically, in which the BRR falls from a positive level to zero). Thus, we restrict attention to the case in which $\bar{z}_0 < z^* + \Delta z_1$.

Bunching at K_2 following K_1, when $\bar{z}_0 < z^* + \Delta z_1$, can therefore be expressed as:

$$\tilde{B}_2 = \int_{\underline{z}_1}^{\bar{z}_0} h\left(\zeta\right) d\zeta$$

We can again solve this system of equations for ϕ^* and ε. Note that ε is still identified by the potential adjustment of the top-most buncher:

$$\varepsilon = \frac{\bar{z}_0 - \bar{z}_2}{\bar{z}_2} \frac{(1-\tau_0)}{d\tau_2}$$

[76] In certain cases, it is possible that reoptimizing away from the kink is not optimal for anyone in the initial earnings range $[z^* + \Delta z_2, z^* + \Delta z_1]$. In that case, there is no change in bunching when moving from K_1 to K_2.

Note that when moving from K_1 to K_2, the change in bunching is smaller than it would be if we had started with steady state bunching at K_1 following no kink (B_1) and then moved to steady state bunching K_2 following no kink (B_2). That is:

$$\begin{aligned} B_1 - \tilde{B}_2 &= \int_{\underline{z}_1}^{z^*+\Delta z_1^*} h(\zeta)\, d\zeta - \int_{\underline{z}_1}^{\bar{z}_0} h(\zeta)\, d\zeta \\ &\leq \int_{\underline{z}_1}^{z^*+\Delta z_1^*} h(\zeta)\, d\zeta - \int_{\underline{z}_2}^{z^*+\Delta z_2^*} h(\zeta)\, d\zeta \\ &= B_1 - B_2, \end{aligned}$$

where the second line follows from the fact that $\bar{z}_0 \geq z^* + \Delta z_2^*$ and $\underline{z}_1 \leq \underline{z}_2$.

E.7.2 Simplified Approximation

We can again build intuition for this result by simplifying the formula for bunching in the second period. Assuming the density is constant over the range $[z^*, z^* + \triangle z_1^*]$, we have:

$$\begin{aligned} \tilde{b}_2 &= \bar{z}_0 - \underline{z}_1 \\ &= (\bar{z}_0 - \bar{z}_2) + \bar{z}_2 - \underline{z}_1 \\ &= \varepsilon \left(\bar{z}_2 \frac{d\tau_2}{1-\tau_0} \right) + \bar{z}_2 - \underline{z}_1 \\ &= \varepsilon \left(\bar{z}_2 \frac{d\tau_2}{1-\tau_0} \right) + (\bar{z}_2 - z^*) - \frac{\phi}{d\tau_1} \end{aligned}$$

where $\tilde{b}_2 \equiv \tilde{B}_2 / h(z^*)$. On the third line, we have used the definition of the elasticity and on the fourth line, we used a first-order approximation to solve for $\underline{z}_1$ as before in Section (E.6). Thus, we can see why bunching when moving from a larger to smaller kink is greater than would be predicted by the Comparative Static method. First, the term multiplying ε has a $\bar{z}_2$ instead of a z^* and there is an additional term $\bar{z}_2 - z^*$ – both of which increase bunching, since $\bar{z}_2 > z^*$. Both of these capture of the excess bunching from above, due to inertia. Finally, the third term has a $d\tau_1$ in the denominator instead of a $d\tau_2$. The larger denominator increases bunching – $d\tau_1 > d\tau_2$ – and captures the fact that there is less attenuation in bunching from below, also due to inertia.

E.7.3 Elasticities Under Frictionless (Saez 2010) Formula

We investigate the results when applying the Saez (2010) formula for estimating elasticities—applicable to a frictionless setting—in a setting in which there are in fact adjustment costs. In other words, we answer the question: if there are adjustment costs and we mis-specify our estimate of the elasticity by assuming that we are in a frictionless setting, in what way do we mis-estimate the elasticity? As we show, if we face an adjustment cost as in the Sharp Change model, but we estimate the elasticity using the Saez (2010) formula applicable to a frictionless setting, we will see the elasticity estimate increase when we move from a larger kink to a smaller kink (as in our empirical application, and as we observe empirically in Figure F.21). Thus, an increase in the Saez (2010) estimate of the elasticity—of the sort that we observe empirically—is a telltale sign that we are operating under a Sharp Change model.

We first present the formula for the elasticity in a frictionless model, as in Saez (2010). Next,

we present the formulas for the elasticities we would estimate if we mis-specified the model as the frictionless (Saez 2010) model, even though we in fact face the Sharp Change model.

Saez (2010) model We assume that tax changes are relatively small and that we can therefore treat the density is constant. (We derive analogous results if we instead use exact formulas under quasilinearity.) Assume that we begin with a more pronounced kink K_1 and then move to a less pronounced kink K_2, by lowering the jump in marginal tax rates at exempt amount from $d\tau_1$ to $d\tau_2$. Assume in each year, we can estimate normalized bunching: $b \equiv B/h\left(z^*\right)$. In a frictionless (Saez 2010) model, we have:

$$\begin{aligned} b_1 &= \triangle z_1^* = \varepsilon z^* \frac{d\tau_1}{1-\tau_0} \\ b_2 &= \triangle z_2^* = \varepsilon z^* \frac{d\tau_2}{1-\tau_0} \end{aligned}$$

where we have used the fact that: $\triangle z^* = \varepsilon z^* d\tau / \left(1-\tau_0\right)$. A natural estimator of the elasticity is the Saez estimator e^S:

$$e^S = \frac{b}{z^*}\frac{\left(1-\tau_0\right)}{d\tau} = \frac{b}{a \cdot d\tau}$$

where $a \equiv z^* / \left(1-\tau_0\right)$.

In each period (denoted by the subscript), we have the following for the Saez estimator when there are no frictions:

$$\begin{aligned} e_1^S &= \frac{b_1}{a \cdot d\tau_1} = \varepsilon \\ e_2^S &= \frac{b_2}{a \cdot d\tau_2} = \varepsilon \end{aligned}$$

Thus, $e_1^S = e_2^S$. Here e_1^S denotes the Saez (2010) estimate of the elasticity in period 1 (under K_1), and e_2^S denotes this elasticity in period 2 (under K_2).

Sharp Change Model By contrast, in the Sharp Change model, we start at kink K_1 and then move straight to K_2, once again estimating normalized bunching. We have the following results here:

$$\begin{aligned} b_1 &= \triangle z_1^* + z^* - \underline{z}_1 \\ b_2 &= \bar{z}_0 - \underline{z}_1 \end{aligned}$$

Rewrite b_1 and b_2 as follows:

$$\begin{aligned} b_1 &= \triangle z_1^* - \left(\underline{z}_1 - z^*\right) \\ b_2 &= \triangle z_2^* + \left(\bar{z}_0 - \triangle z_2^* - \underline{z}_1\right) \end{aligned}$$

The Saez estimators now return:

$$\begin{aligned} e_1^S &= \frac{b_1}{a \cdot d\tau_1} = \varepsilon - \frac{\left(\underline{z}_1 - z^*\right)}{a \cdot d\tau_1} \\ e_2^S &= \frac{b_2}{a \cdot d\tau_2} = \varepsilon + \frac{\left(\bar{z}_0 - \triangle z_2^* - \underline{z}_1\right)}{a \cdot d\tau_2} \end{aligned}$$

The change in Saez estimators is:

$$\begin{aligned} e_2^S - e_1^S &= \frac{(\bar{z}_0 - \triangle z_2^* - \underline{z}_1)}{a \cdot d\tau_2} + \frac{(\underline{z}_1 - z^*)}{a \cdot d\tau_1} \\ &\geq \frac{(\bar{z}_0 - \triangle z_2^* - \underline{z}_1)}{a \cdot d\tau_1} + \frac{(\underline{z}_1 - z^*)}{a \cdot d\tau_1} \\ &= \frac{\bar{z}_0 - \triangle z_2^* - \underline{z}_1 + \underline{z}_1 - z^*}{a \cdot d\tau_1} \\ &= \frac{\bar{z}_0 - \triangle z_2^* - z^*}{a \cdot d\tau_1} \\ &\geq \frac{\triangle z_2^* + z^* - \triangle z_2^* - z^*}{a \cdot d\tau_1} \\ &= 0 \end{aligned}$$

where in the second line, we use the fact that $d\tau_1 > d\tau_2$ and in the second-to-last line, we use the fact that $\bar{z}_0 \geq \triangle z_2^* + z^*$.

Thus, $e_2^S - e_1^S$ is weakly greater than zero: the Saez (2010) frictionless elasticity estimate weakly increases (as we observe empirically). The pattern we observe empirically—an upward jump in the Saez (2010) estimate of the elasticity when the policy change occurs in 1990, as shown in Figure F.21—is a telltale sign that we are operating in the Sharp Change model.

E.7.4 Sharp Change Approach with Linear Adjustment Cost

Note that under the Sharp Change approach, with a linear adjustment cost we can derive an approximation for bunching (analogously to Section E.6.2) to show that:

$$\tilde{b}_2 = \varepsilon \left(\bar{z}_2 \frac{d\tau_2 - 2\phi^L}{1 - \tau_0 - \phi^L} \right) - (\bar{z}_2 - z^*) - \frac{\phi^F}{d\tau_1 - \phi^L}.$$

In our empirical application, we apply the Sharp Change approach in estimating elasticities and adjustment costs using data on individuals in different years (in the baseline specification, 1989 and 1990). Our empirical approach is applicable in the case in which individuals make year-by-year static earnings decisions; in the baseline, this effectively assumes that individuals weigh the cost of adjustment against the benefits in 1990. If instead individuals compare the costs of adjustment in 1990 to the benefits of adjustment in 1990 and subsequent years, then the benefits and therefore the estimated cost of adjustment would likely be larger. In this case, our estimated cost of adjustment could be considered a lower bound. Our estimates demonstrate the applicability of the methodology, including in settings in which the benefits may be realized over more years (as these discounted benefits would then be weighed against the costs). As we discuss, we view our static approach as a natural first step toward estimating elasticities and adjustment costs, but developing a dynamic model of adjustment frictions represents an important next step.

E.8 Estimating the Elasticity and Adjustment Cost

In this section, we describe in more detail how we use data on the amount of bunching to estimate the elasticity and adjustment cost. Let $b = (b_1, b_2, \ldots, b_K)$ be a vector of (estimated) bunching amounts normalized by the density at the kink, using the method described in Section

(3). Let $\boldsymbol{\tau} = (\boldsymbol{\tau}_1, \ldots, \boldsymbol{\tau}_K)$ be the tax schedule at each kink. The triplet $\boldsymbol{\tau}_k = (\tau_0^k, \tau_1^k, \tau_2^k)$ denotes the tax rate below (τ_0^k) and above (τ_1^k) the kink k; when using the Sharp Change method, τ_2^k denotes the *ex post* marginal tax rate above the kink after it has been reduced, as in Section (E.7). Let $z^* = (z_1^*, \ldots, z_K^*)$ be the earnings levels associated with each kink. To estimate (ε, ϕ), we seek the values of the parameters that make predicted bunching $\hat{b}$ and actual (estimated) bunching b as close as possible on average.

Letting $\hat{b}(\varepsilon, \phi) = (\hat{b}(\boldsymbol{\tau}_1, z_1^*, \varepsilon, \phi), \ldots, \hat{b}(\boldsymbol{\tau}_K, z_K^*, \varepsilon, \phi))$, our estimator is:

$$\left(\hat{\varepsilon}, \hat{\phi}\right) = \text{argmin}_{(\varepsilon,\phi)} \left(\hat{b}(\varepsilon, \phi) - b\right)' W \left(\hat{b}(\varepsilon, \phi) - b\right), \tag{E.7}$$

where W is a $K \times K$ diagonal matrix whose diagonal entries are the inverse of the variances of the estimates of the b_k.

We obtain our estimates by minimizing equation (E.7) numerically. Solving this problem requires evaluating $\hat{b}$ at each trial guess of (ε, ϕ).[77] Recall that in general bunching takes the form:

$$B_k(\boldsymbol{\tau}_k, z_k^*; \varepsilon, \phi^*) = \int_{z_k^{lb}}^{z_k^{ub}} h\left(\zeta\right) d\zeta,$$

where $\left(z_k^{lb}, z_k^{ub}\right)$ are the ex-ante earnings levels of the lowest and highest earning bunchers, in the presence of linear tax at the lower tax rate, τ_0^k. Define $z_k^* + \Delta z_1^k$ as the *ex ante* earnings level for the highest earning buncher – in the absence of frictions – when the size of the kink is $d\tau_1^k = \tau_1^k - \tau_0^k$. As in the main text, we continue to assume that $h(\cdot)$ is uniform in $[z_k^*, z^* + \Delta z_1^k]$, so that

$$b_k(\boldsymbol{\tau}_k, z_k^*; \varepsilon, \phi^*) = z_k^{ub} - z_k^{lb},$$

where $b = B/h\left(z_k^*\right)$. The definitions of $\left(z_k^{lb}, z_k^{ub}\right)$ vary depending on the setting and are defined as follows. In the frictionless case (Saez 2010), we have:

$$\begin{aligned} z_k^{lb} &= z_k^* \\ z_k^{ub} &= z_k^* + \Delta z_1^k. \end{aligned}$$

In the presence of a fixed adjustment cost, we have under the Comparative Static approach:

$$\begin{aligned} z_k^{lb} &= \underline{z}_1^k \\ z_k^{ub} &= z_k^* + \Delta z_1^k, \end{aligned}$$

where $\underline{z}_1^k$ is the ex-ante earnings of the marginal buncher from below given a fixed cost of adjustment. This is defined in the indifference condition above in equation (E.3). Finally, under the Sharp Change method, we have:

$$\begin{aligned} z_k^{lb} &= \underline{z}_1^k \\ z_k^{ub} &= \bar{z}_0^k, \end{aligned}$$

where $\underline{z}_1^k$ is similarly the *ex ante* earnings of the marginal buncher from below (calculated using

[77] In solving problem (E.7), we impose that $\phi \geq 0$. When $\phi < 0$, every individual adjusts her earnings by at least some arbitrarily small amount, regardless of the size of ϕ. This implies that ϕ is not identified if it is less than zero.

a kink with $d\tau_1^k = \tau_1^k - \tau_0^k$). The *ex ante* earnings of the marginal buncher from above, $\bar{z}_0^k$, is defined in Section (E.7) where $d\tau_1^k = \tau_1^k - \tau_0^k$ and $d\tau_2^k = \tau_2^k - \tau_0^k$.

Our estimator assumes a quasilinear utility function, $u(c, z; n) = c - \frac{n}{1+1/\varepsilon}\left(\frac{z}{n}\right)^{1+1/\varepsilon}$, following Saez (2010), Chetty *et al.* (2011) and Kleven and Waseem (forthcoming). (In order to relax this assumption empirically, we would have to observe wealth, which is not available in the data.) Note that because we have assumed quasilinearity, $\phi^* = \phi$, $\Delta z_1^k = z_k^* \left(\left(\frac{1-\tau_1^k}{1-\tau_0^k}\right)^{\varepsilon} - 1\right)$ and $n = z(\tau)/(1-\tau)^{\varepsilon}$, where $z(\tau)$ are the optimal, interior earnings under a linear tax of τ. However, there typically is not a closed form solution for the $\left(z_k^{lb}, z_k^{ub}\right)$ in other cases. Instead, given ε and ϕ, we find $\left(z_k^{lb}, z_k^{ub}\right)$ numerically as the solution to relevant indifference condition. For example, $\underline{z}_1^k$ is defined implicitly by:

$$\underbrace{u((1-\tau_1^k)z_k^* + R_1^k, z_k^*; \underline{z}_1^k/(1-\tau_0^k)^{\varepsilon})}_{\text{utility from adjusting to kink}} - \underbrace{u((1-\tau_1^k)\underline{z}_1^k + R_1^k, \underline{z}_1^k; \underline{z}_1^k/(1-\tau_0^k)^{e})}_{\text{utility from not adjusting}} = \phi,$$

The equation is continuously differentiable and has a unique solution for $\underline{z}_1^k$. As such, Newton-type solvers are able to find $\underline{z}_1^k$ accurately. Note that some combinations of $\boldsymbol{\tau}_k, z_k^*, \varepsilon$, and ϕ imply $z_k^{lb} > z_k^{ub}$. In this case, the lowest-earning adjuster does not adjust to the kink, and whenever this happens we set $\hat{b}_k = 0$. The predicted amount of bunching is therefore:

$$\hat{b}_k(\boldsymbol{\tau}_k, z_k^*; \varepsilon, \phi) = \max(z_k^{ub} - z_k^{lb}, 0).$$

We have also shown a robustness check in our Tables in which we assume that the earnings distribution is lognormal, rather than assuming that that $h(\cdot)$ is uniform in $[z_k^*, z_k^* + \Delta z_1^k]$. Specifically, we use the distribution of earnings at age 61 over 1986-1988 and 1992-1994 to estimate the parameters of a lognormal earnings distribution, (μ_z, σ_z), using maximum likelihood. (Individuals age 61 are not subject to the AET but are not far removed in age from those at retirement age, making this a reasonable counterfactual earnings density for those subject to the AET). We then solve for bunching using:

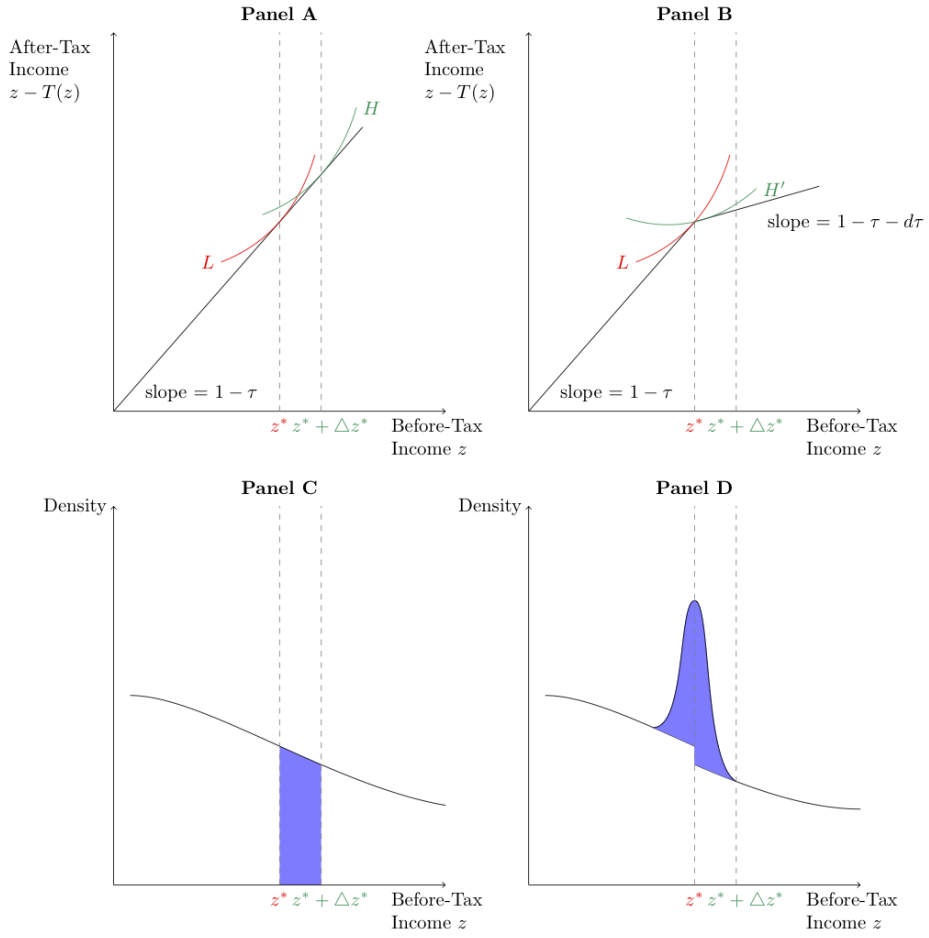
$$B_k(\boldsymbol{\tau}_k, z_k^*; \varepsilon, \phi^*) = \Phi\left(\frac{\log z_k^{ub} - \mu_z}{\sigma_z}\right) - \Phi\left(\frac{\log z_k^{lb} - \mu_z}{\sigma_z}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF.

We estimate bootstrapped standard errors. Observe that the estimated vector of parameters $(\hat{\varepsilon}, \hat{\phi})$ is a function of the estimated amount of bunching; call this function $\theta(b)$. To compute bootstraped standard errors, we use the bootstrap procedure of Chetty *et al.* (2011) to obtain 200 bootstrap samples of b. For each bootstrap sample, we compute $\hat{\varepsilon}$ and $\hat{\phi}$ as the solution to (E.7). The standard deviation of $\hat{\varepsilon}$ and $\hat{\phi}$ across bootstrap samples is the bootstrap standard error, and we compute confidence intervals analogously. We estimate whether an estimate is significantly different from zero by assessing how frequently the constraint $\phi \geq 0$ binds in our estimation. Given this constraint, p-values are from a one-sided test of equality with zero. We have also estimated the standard errors using the delta method and obtained similar results.

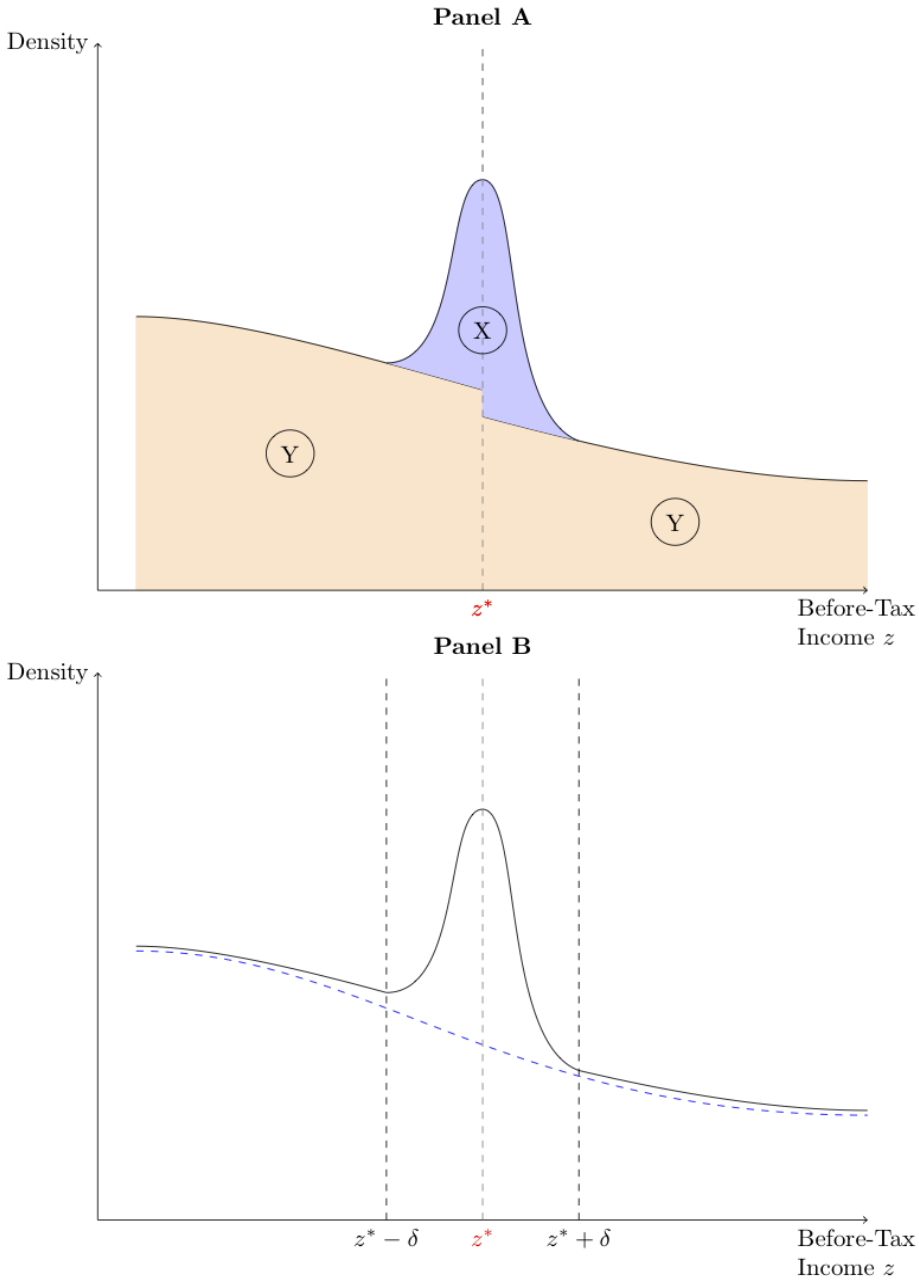
F Appendix: Additional Figures

Figure F.1: Bunching Response to a Convex Kink (Frictionless Case)



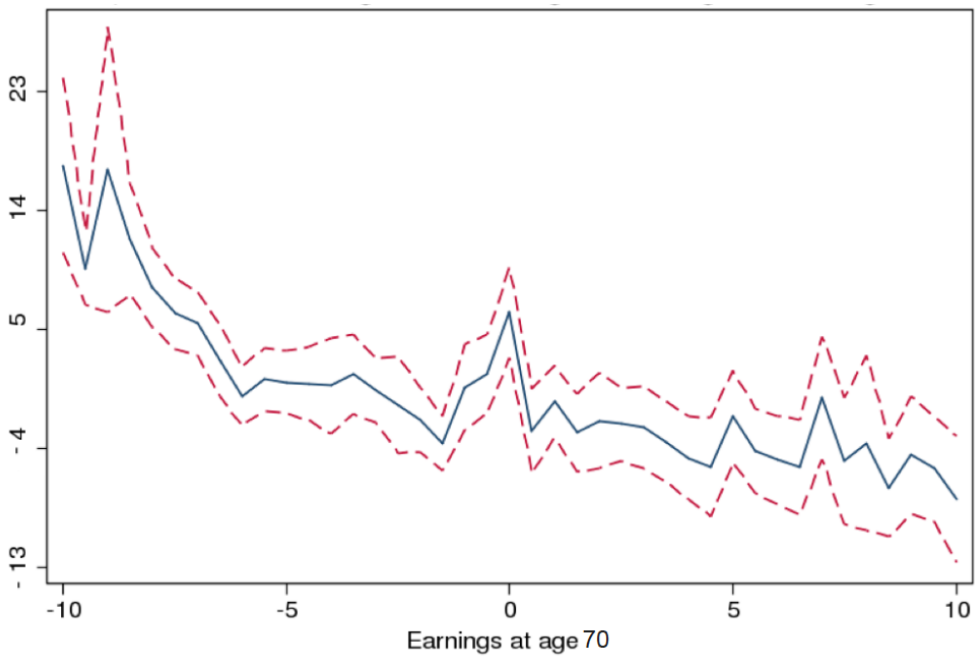
Note: When we move from a linear budget constraint (Panel A) to a convex kink (B), individuals with initial earnings between z^* and $z^* + \Delta\, z^*$ relocate to the kink. As we move from a linear budget constraint (Panel C) to a convex kink (Panel D), a spike in the earnings density appears at the kink, corresponding to the density that was initially located between z^* and $z^* + \Delta\, z^*$. The spike is spread out in the vicinity of the kink in Panel D; this may result from several factors discussed in Saez (2010), such as inability to control earnings precisely.

Figure F.2: Bunching Estimation Methodology



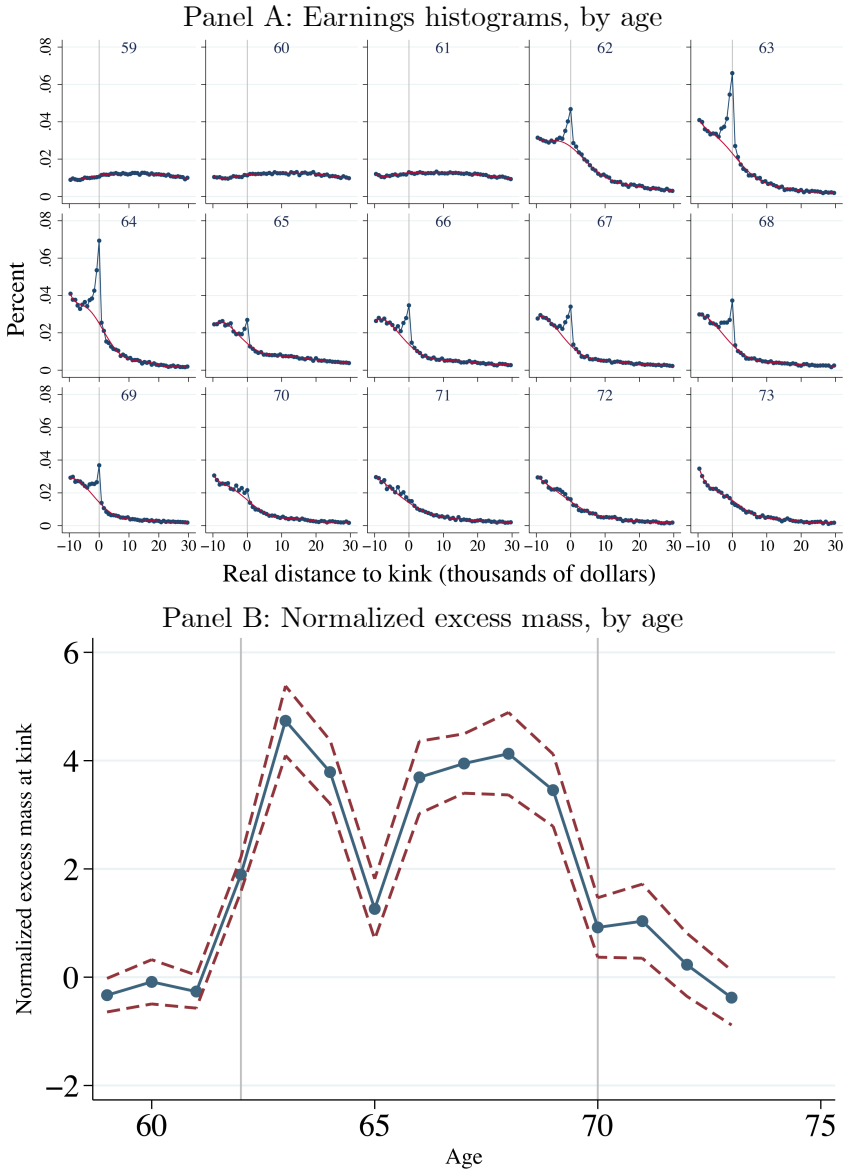
Note: Panel A decomposes the ex-post earnings distribution shown in Appendix Figure F.1 Panel D into two groups. The bunchers, group X, are those who bunch at the kink in the presence of the higher marginal tax rate $\tau + d\tau$ but not at the lower marginal tax rate τ. The non-bunchers, group Y, are comprised of those who locate to the left of the kink under the initial lower marginal tax rate τ, and those who locate to the right of the kink under the higher marginal tax rate $\tau + d\tau$. Panel B demonstrates how the distribution of earnings in the absence of the kink is estimated to recover the share of bunchers, by excluding data in a neighborhood of z^*.

Figure F.3: Mean Percentage Change in Earnings from Age 70 to 71, by Earnings at 70, 1990-1998



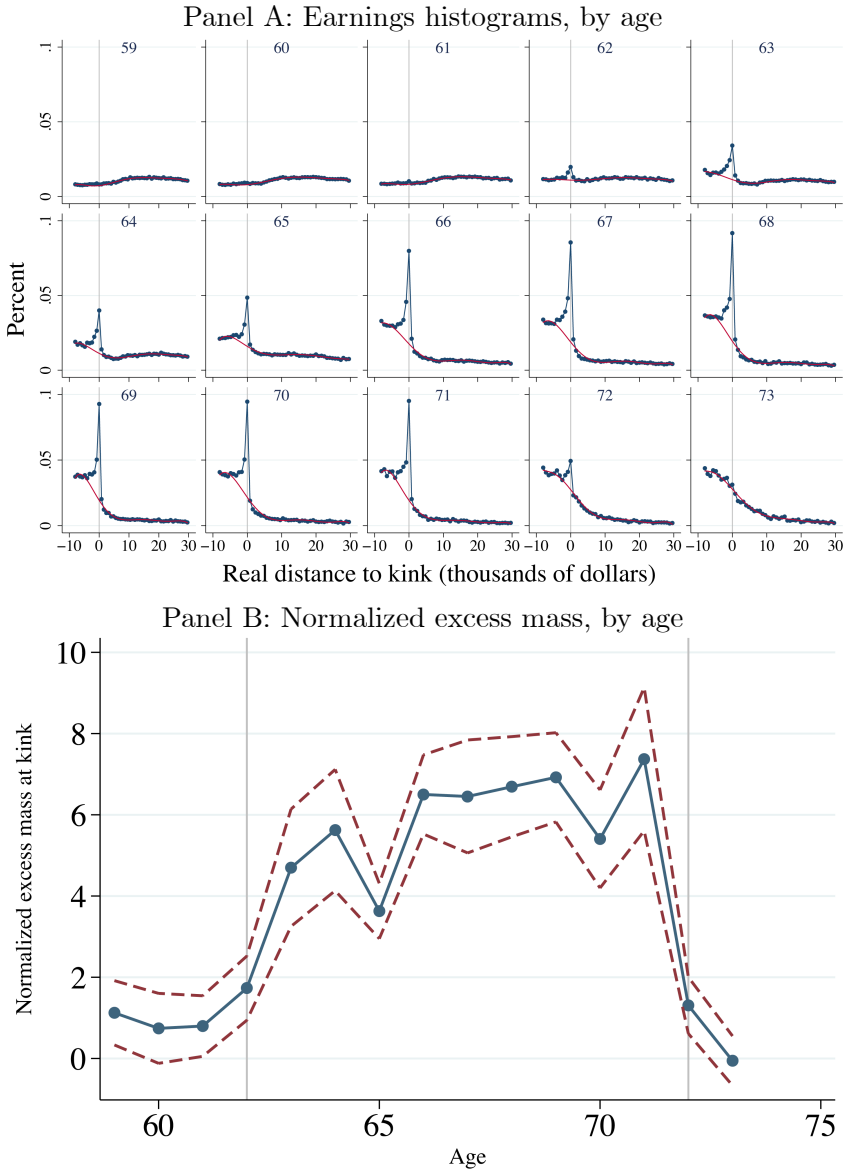
Note: The figure shows the mean percentage change in earnings from age 70 to age 71 (y-axis), against earnings at age 70 (x-axis). Earnings are measured relative to the kink, shown at zero on the x-axis. The data are a 20 percent random sample of 70-year-olds in the LEHD in 1990-1998. We exclude 1999 as a base year in this and similar graphs because the AET is eliminated for those over NRA in 2000. Higher earnings growth far below the kink reflects mean reversion visible in this part of the earnings distribution at all ages. We also find a spike in mean earnings growth from age 70 to age 71 among those born in January.

Figure F.4: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-old OASI Claimants, 1990-1999



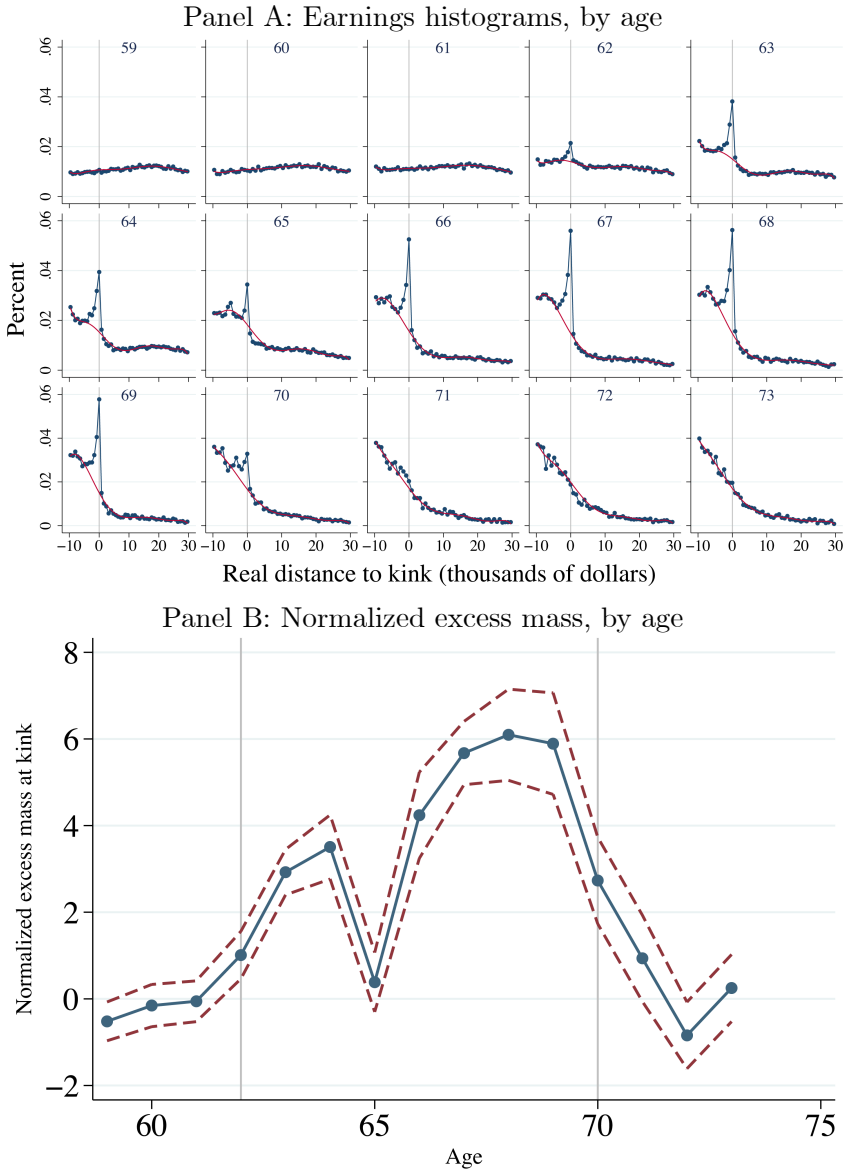
See notes to Figures 2 and 3. This figure differs from Figures 2 and 3 only because the sample in year t consists only of people who have claimed OASI in year t or before (whereas in Figures 2 and 3 it consists of all those who claimed by age 65).

Figure F.5: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 1972-1982



See notes to Figures 2 and 3. This figure differs from Figures 2 and 3 only because the years examined are 1972-1982 (whereas in Figures 2 and 3 the years examined are 1990-1999).

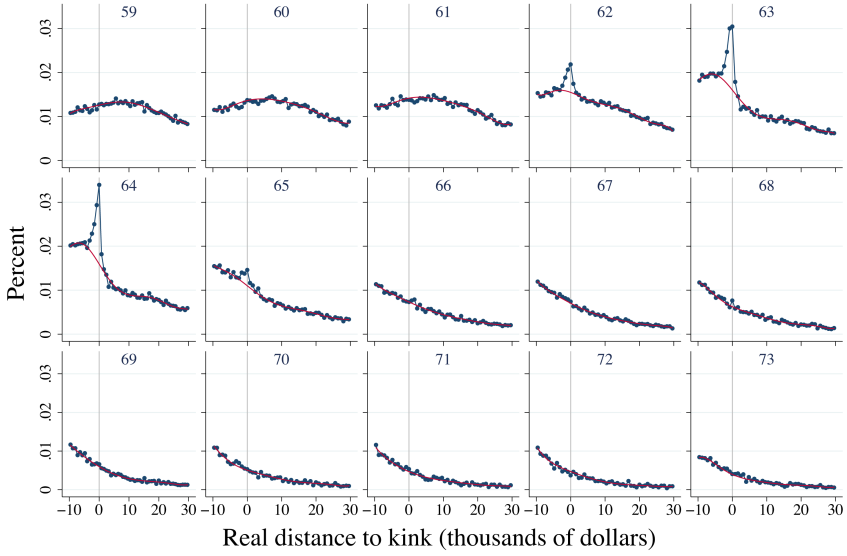
Figure F.6: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 1983-1989



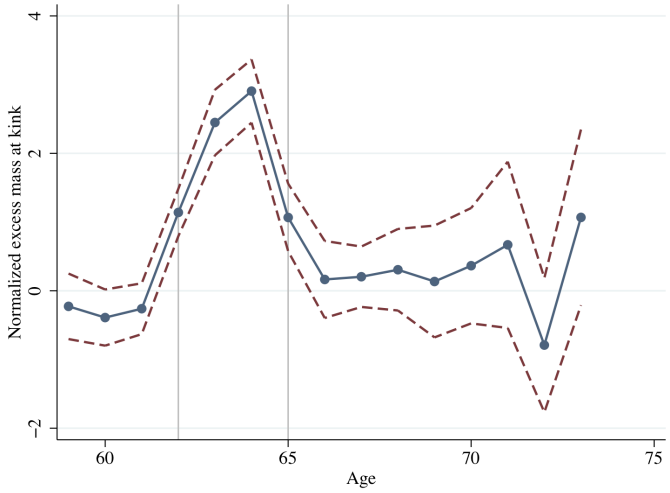
See notes to Figures 2 and 3. This figure differs from Figures 2 and 3 only because the years examined are 1983-1989 (whereas in Figures 2 and 3 the years examined are 1990-1999).

Figure F.7: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 2000-2006

Panel A: Earnings histograms, by age

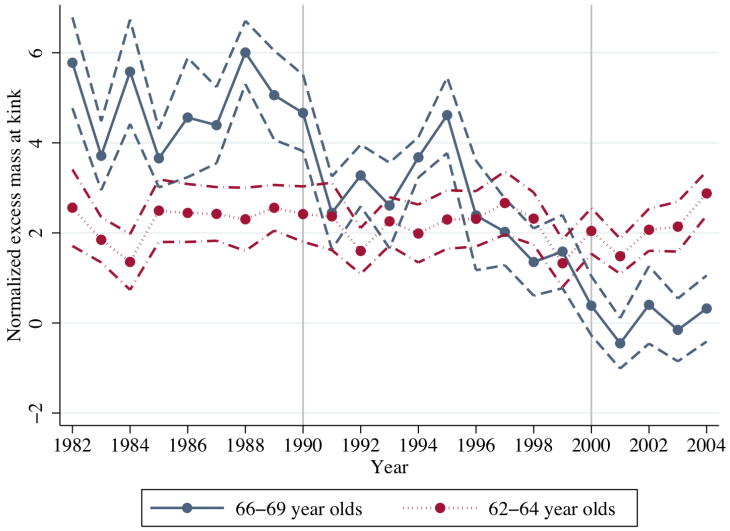


Panel B: Normalized excess mass, by age



See notes to Figures 2 and 3. This figure differs from Figures 2 and 3 only because the years examined are 2000-2006 (whereas in Figures 2 and 3 the years examined are 1990-1999). As explained in the main text, the NRA slowly rose from 65 for cohorts that reached age 62 during this period; the results are extremely similar when the sample is restricted to those who claimed by 66, instead of 65. In the year of attaining NRA, the AET applies for months prior to such attainment.

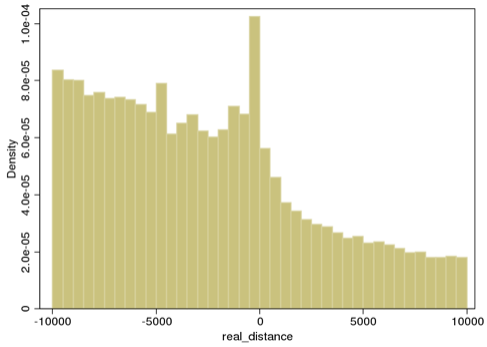
Figure F.8: Comparison of Normalized Excess Bunching Among 62-64 Year-Olds and 66-69 Year-Olds, 1982-2004



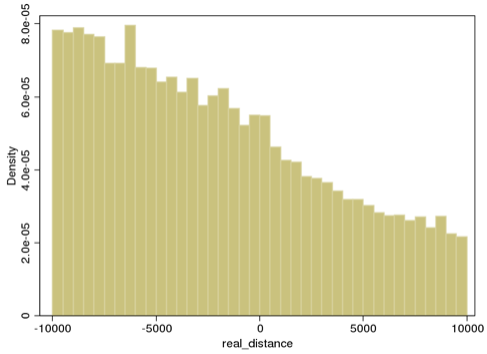
Note: the figure shows excess normalized bunching among 62-64 year-olds and 66-69 year-olds in each year from 1982 to 2004. Note the caveat that the 62-64 year-old group faces a notch at the exempt amount, as opposed to the kink faced by those 66-69.

Figure F.9: Adjustment Across Ages: Histograms of Earnings, 66-69 Year-Olds, 1999-2001

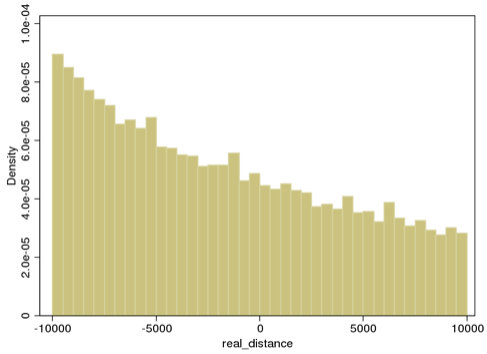
Panel A: Earnings histogram, 66-69 year-olds, 1999



Panel B: Earnings histogram, 66-69 year-olds, 2000

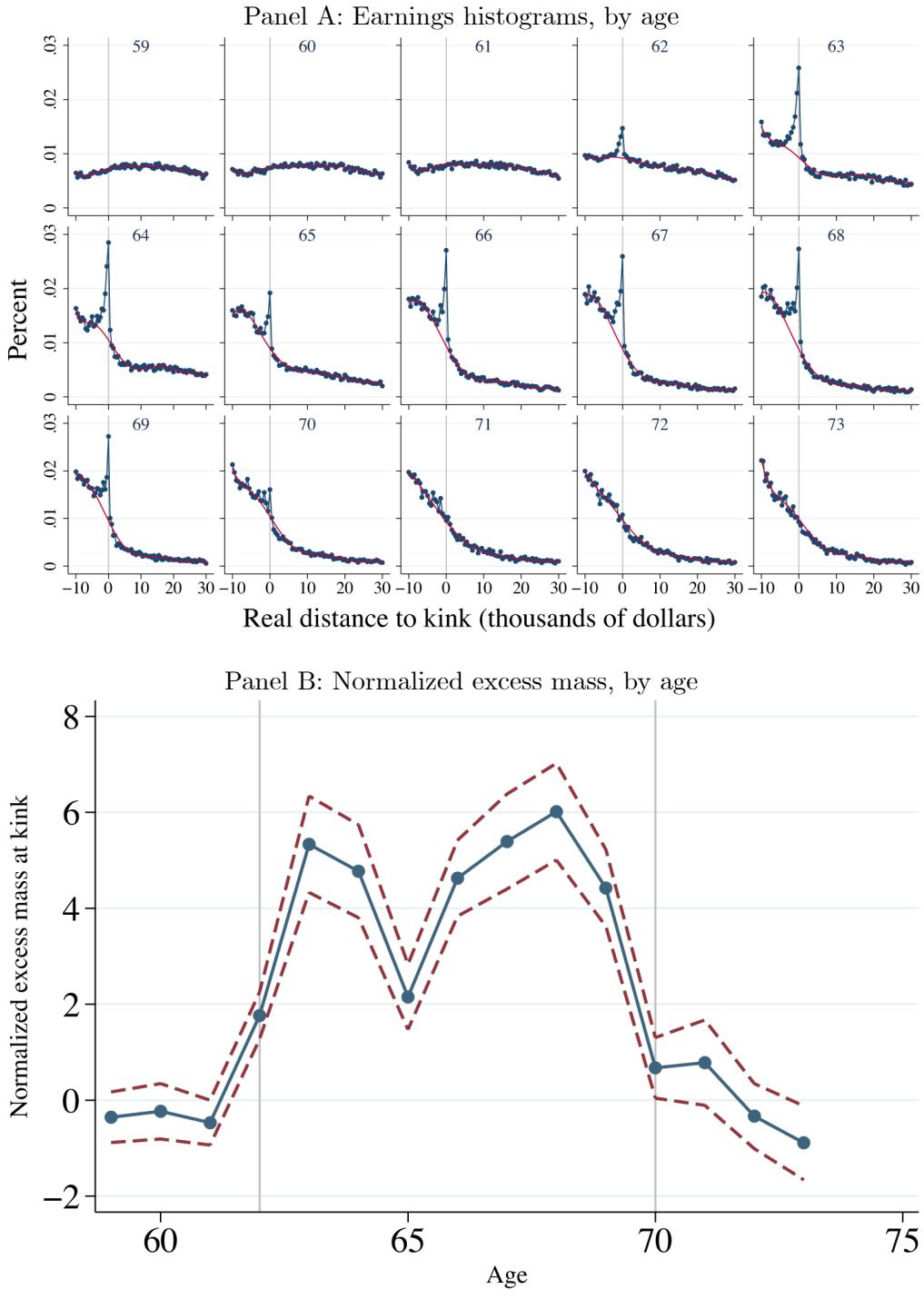


Panel C: Earnings histogram, 66-69 year-olds, 2001



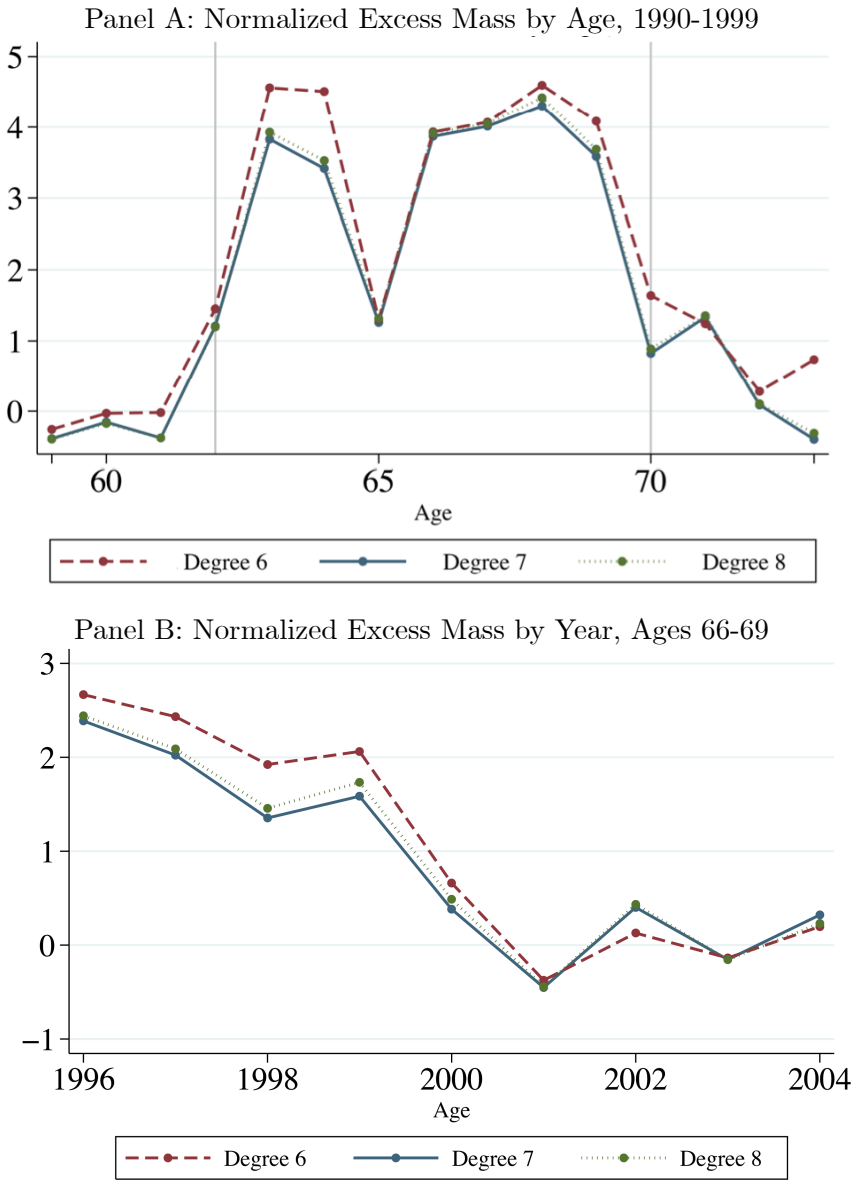
Note: the figure shows a histogram of earnings in 1999, 2000, and 2001, using LEHD data on 66-69 year-olds. Earnings are measured relative to the kink, shown at zero.

Figure F.10: Adjustment Across Ages: Histograms of Earnings and Normalized Excess Mass, 59-73-year-olds Claiming OASI by Age 65, 1990-1999



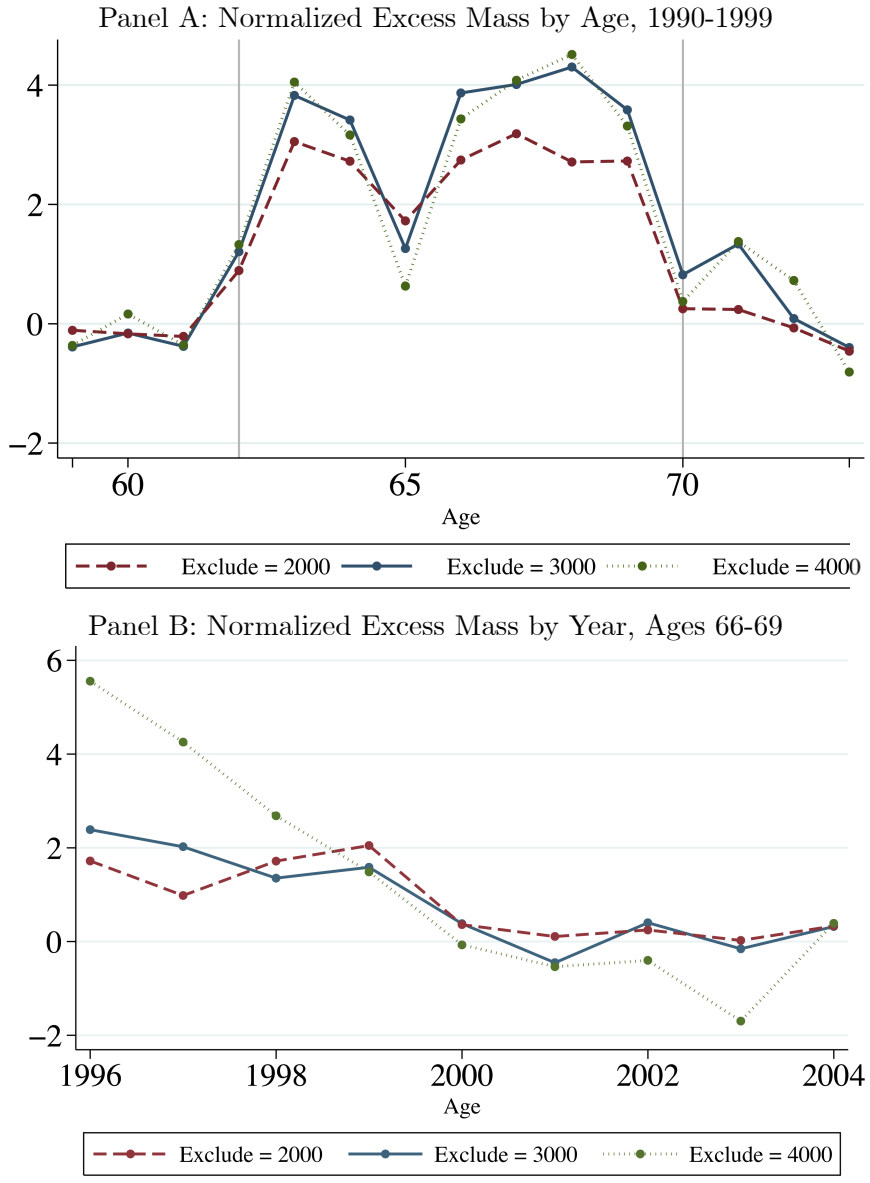
See notes to Figures 2 and 3. This figure differs from Figures 2 and 3 only because the bandwidth is $500 (whereas in Figures 2 and 3 it is $800).

Figure F.11: Robustness to Polynomial Degree: Normalized Excess Mass by Age and Year, OASI Claimants by 65



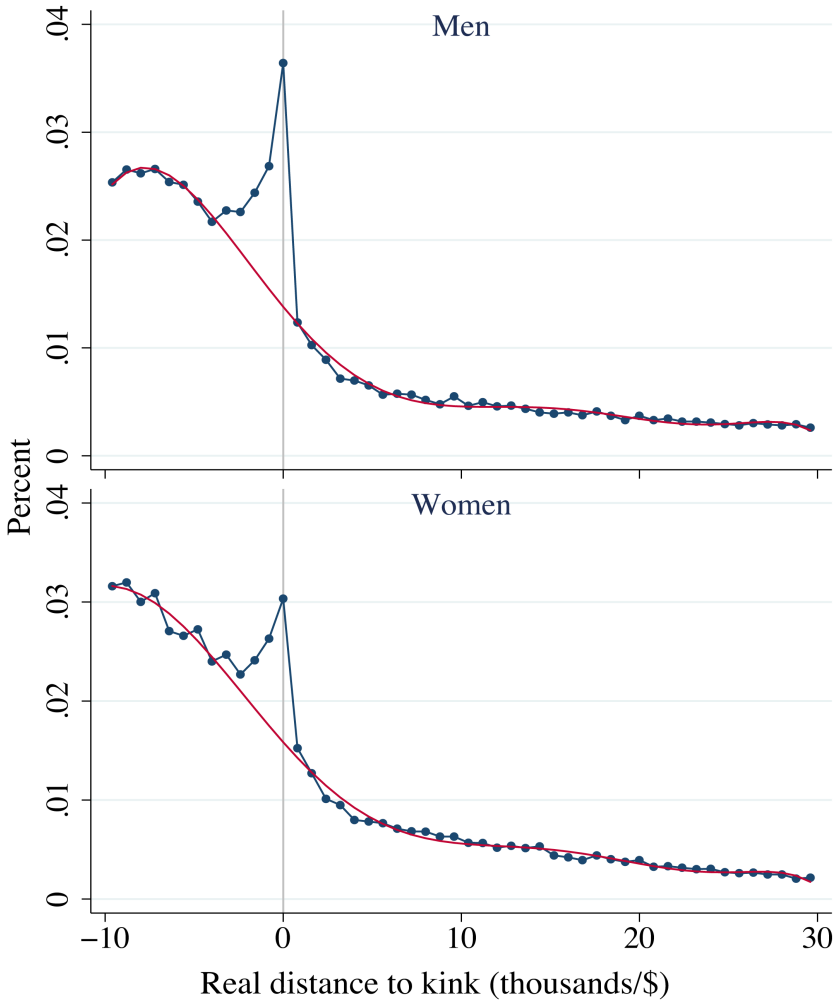
Notes: The figure shows the difference in estimates of normalized excess bunching as we vary the degree of the polynomial used. For additional notes on the samples see Figure 3 for Panel A and Figure 4 for Panel B.

Figure F.12: Robustness to the Excluded Region: Normalized Excess Mass by Age and Year, OASI Claimants by 65



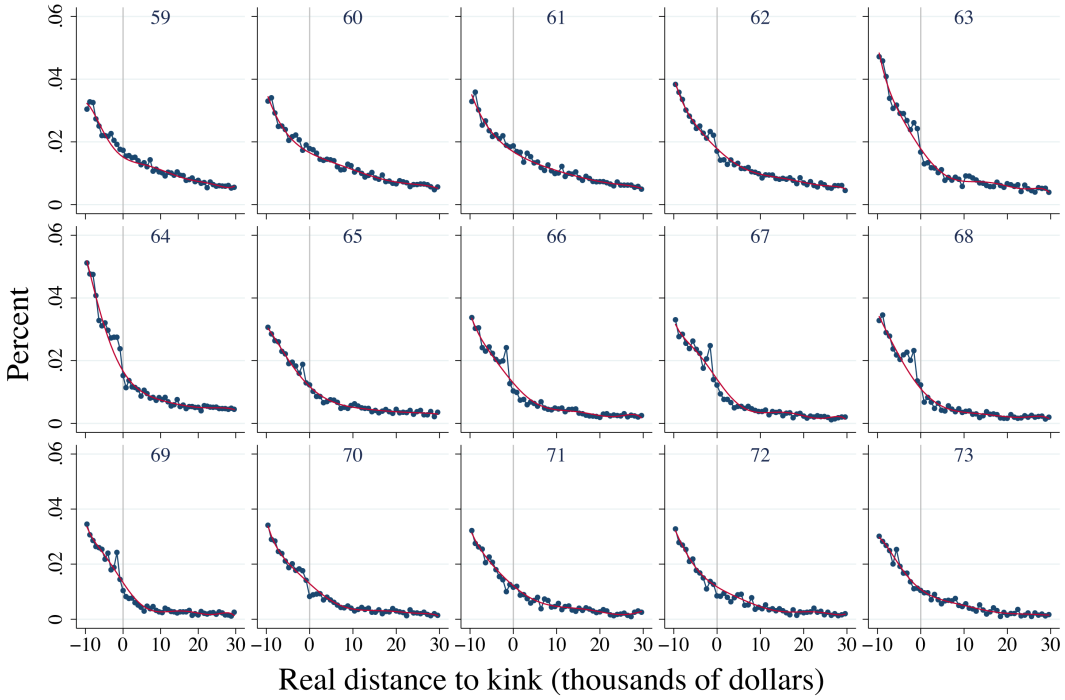
Notes: The figure shows the difference in estimates of normalized excess bunching as we vary the region about the kink that is "dummied out" in the polynomial estimation. For additional notes on the samples see Figure 3 for Panel A and Figure 4 for Panel B.

Figure F.13: Adjustment by Sex: Histograms of Earnings, 59-73-year-olds Claiming OASI by Age 65, 1990-1999



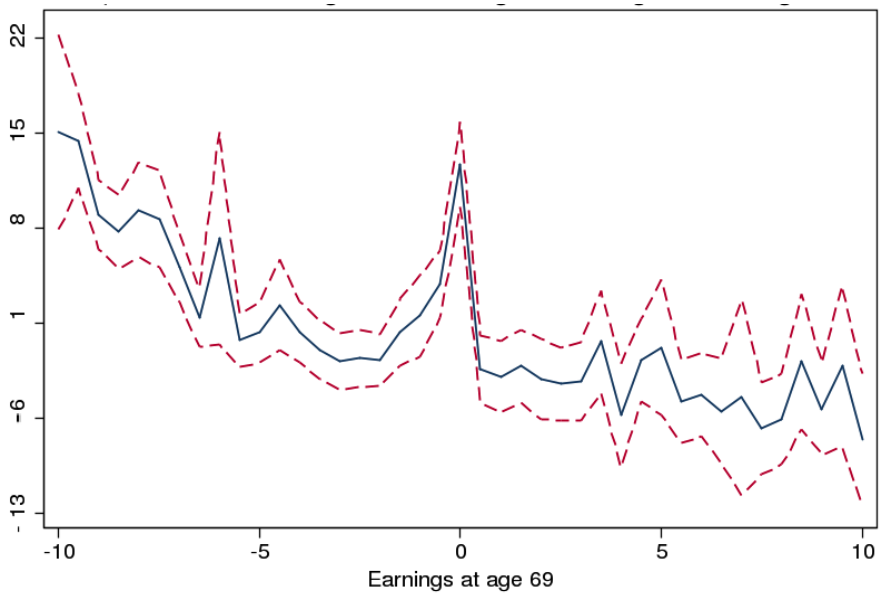
See notes to Figure 2. The sample examined is the same as in Figure 2 but examines men and women separately.

Figure F.14: Adjustment Across Ages: Histograms of Earnings, 59-73-year-olds Claiming OASI by Age 65 with Self-Employment Income, 1990-1999



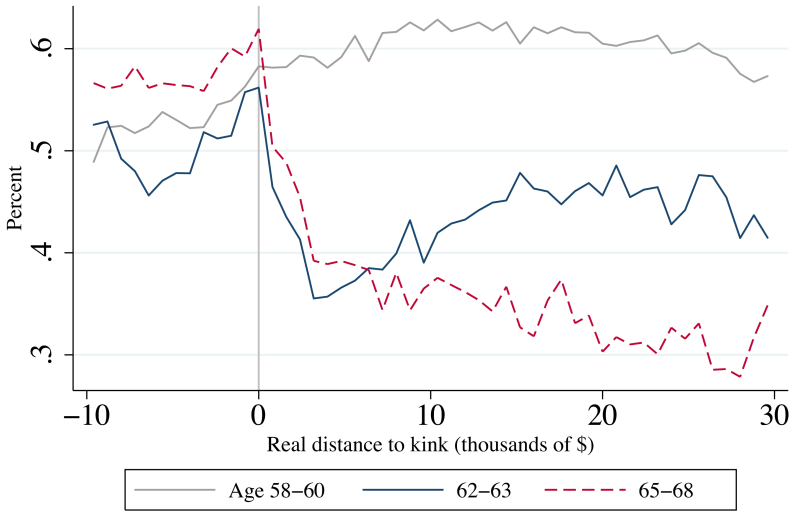
See notes to Figure 2. The figure differs from Figure 2 only because the sample consists of those with positive self-employment income (whereas in Figure 2 those with positive self-employment income are excluded).

Figure F.15: Mean Percentage Change in Earnings from Age 69 to 70, by Earnings at 69, 1990-1998



Note: The figure shows the mean percentage change in earnings from age 69 to age 70 (y-axis), against earnings at age 69 (x-axis). Earnings are measured relative to the kink, shown at zero on the x-axis. The data are a 20 percent random sample of 69-year-olds in the LEHD in 1990-1998. We exclude 1999 as a base year in this and similar graphs because the AET is eliminated for those over NRA in 2000.

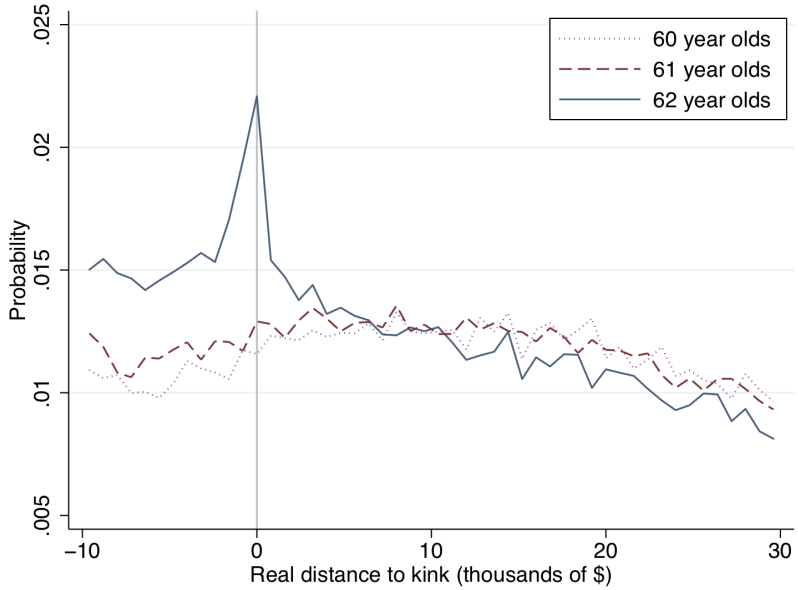
Figure F.16: Probability that Earnings Move with Kink, 1990-1998



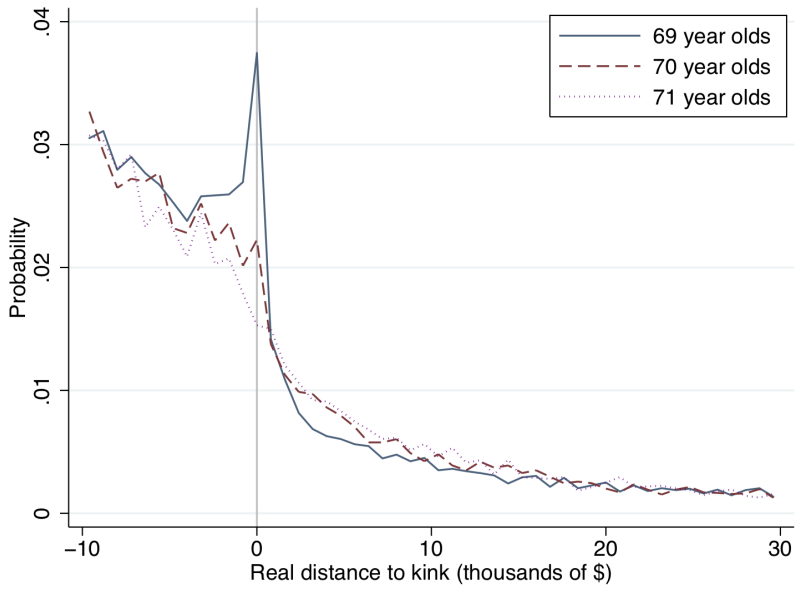
Note: The figure shows the probabiliy that individual earnings move with the kink from year to year (i.e. the probability that an individual locates at the kink in year $t+1$, conditional on locating at the kink in year t), for age groups 58 to 60, 62 to 63, and 65 to 68. (Each of these ages refers to age in year t.) Results are similar when considering similar age bins. The kink is defined as the region within $2800 of the exempt amount. We exclude the year 1999 because the AET is eliminated for those over NRA in 2000. See other notes to Figure 2.

Figure F.17: Earnings Distributions by Age, OASI Claimants by Age 65, 1990-1999

Panel A: Earnings Distributions by Age, 60-62

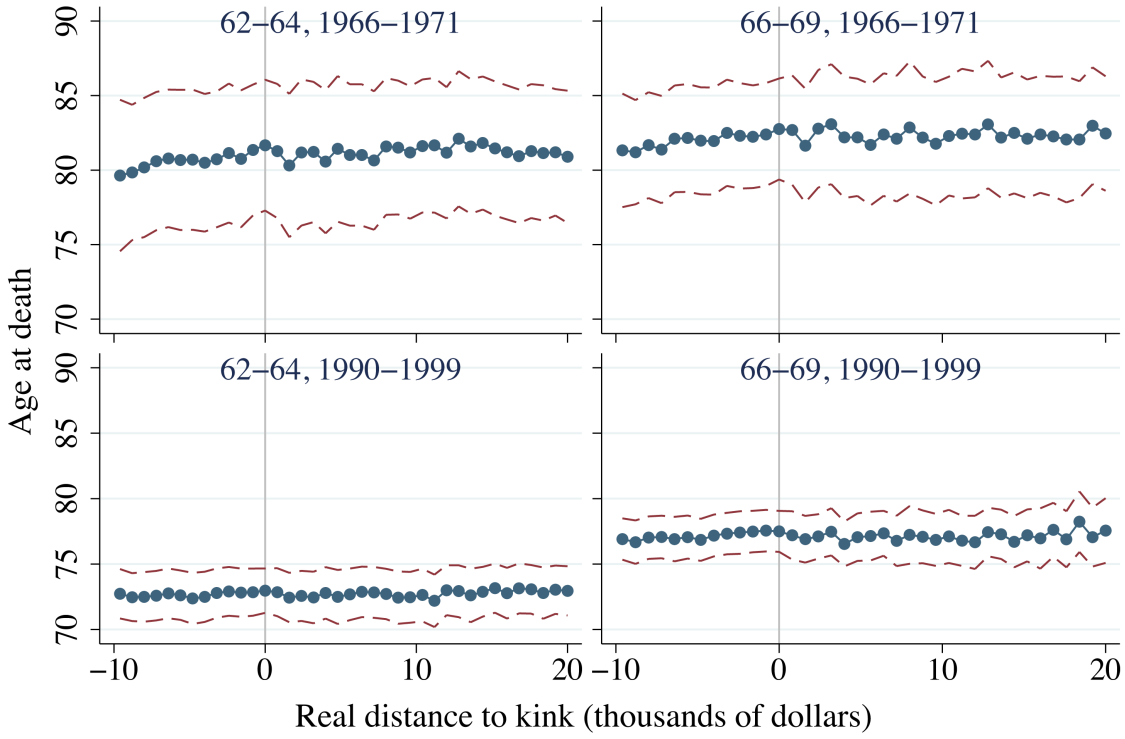


Panel B: Earnings Distributions by Age, 69-71



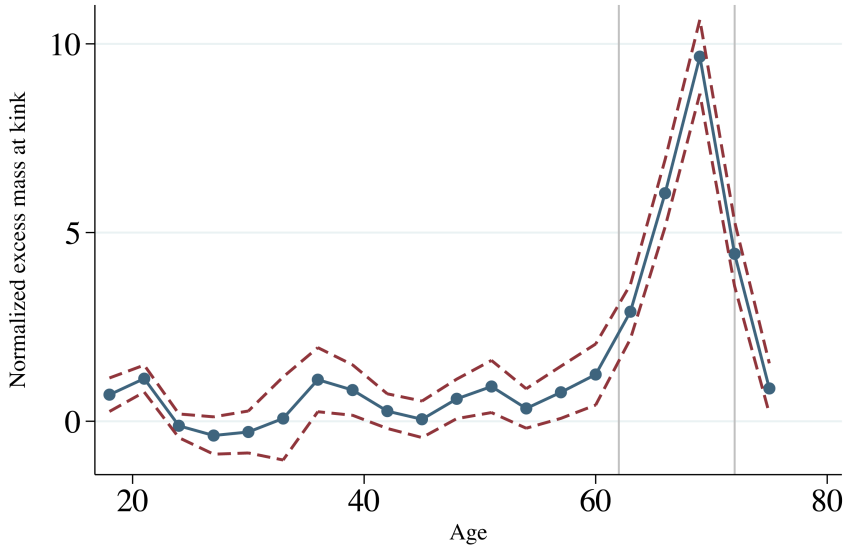
Notes: The figure shows earnings distributions at ages 60, 61, and 62 (Panel A) and at ages 69, 70, and 71 (Panel B).

Figure F.18: Mortality Analysis: Mean Age at Death, 62-69-year-olds Claiming OASI by Age 65, 1966-1971 and 1990-1999



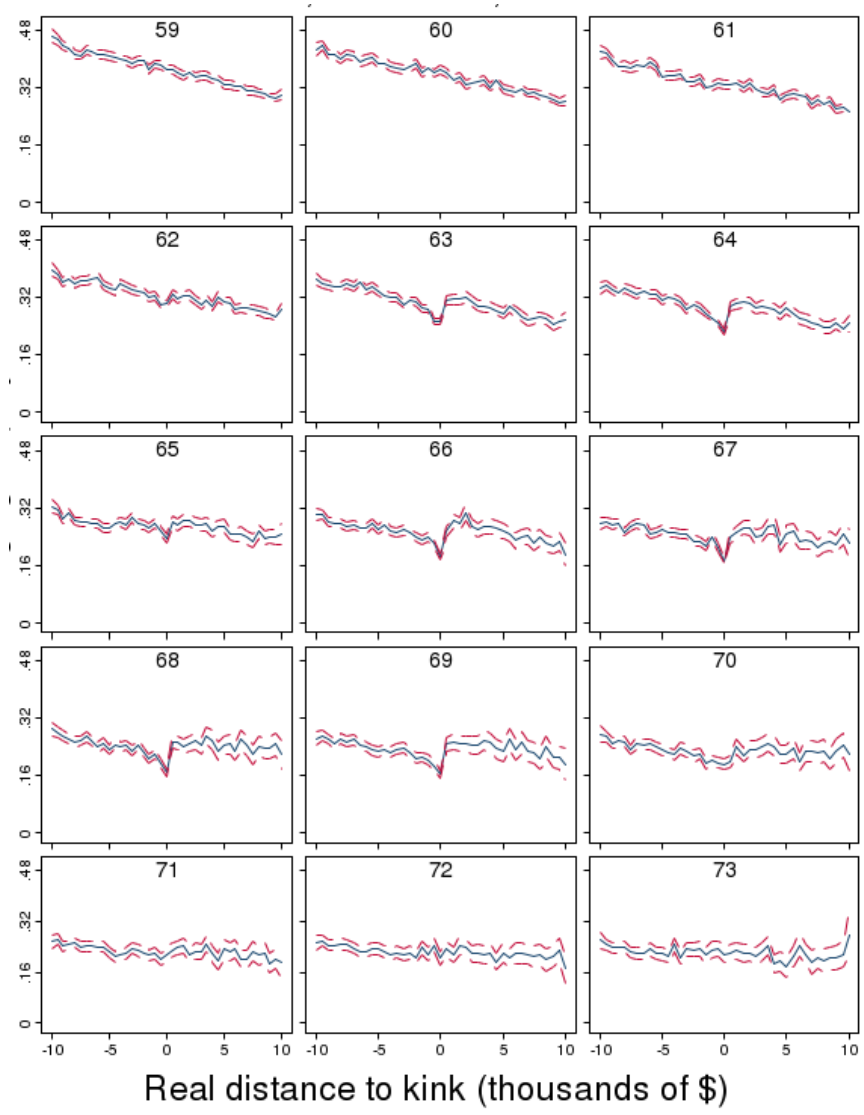
Note: The figure shows mean age at death from a one percent random sample of SSA administrative data on individuals aged 59-73, claiming OASI by age 65, between 1966 and 1971 (inclusive) in the top panel, and between 1990 and 1999 (inclusive) in the bottom panel. The figure shows no clearly noticeable patterns at the kink that are different from those away from the kink. This holds true for those 62-64 and 66-69, in a period prior to the introduction of the Delayed Retirement Credit (*i.e.* 1966-1971) and subsequent to its introduction (*i.e.* 1990-1999). Results are similar for other time periods.

Figure F.19: Normalized Excess Mass at Kink by Age, 1966-1971



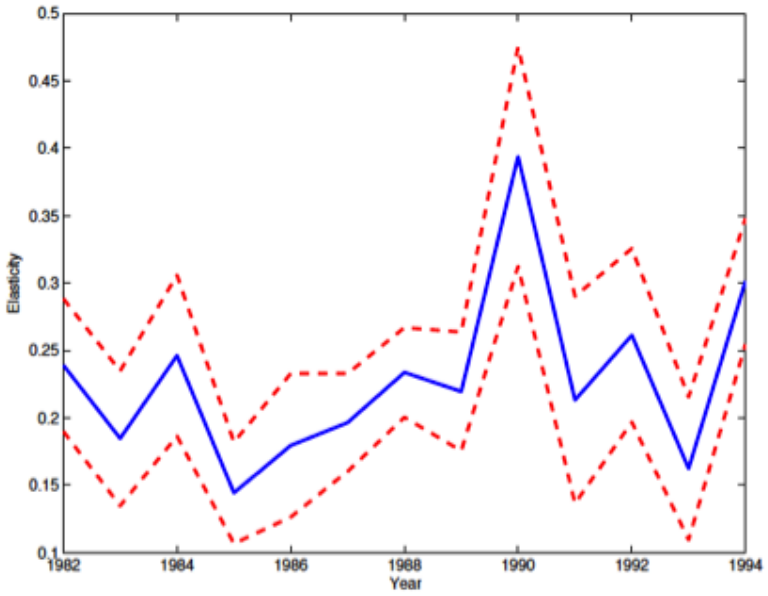
Note: The figure shows normalized excess bunching, ages 18 to 75, 1966-1971. We group ages into three-year bins. See other notes to Figure 3.

Figure F.20: Fraction of Workers Changing Employers from Age t to Age t+1, by Age t Earnings, 1990-1998



Note: The figure shows the fraction of workers who change employers from age t to age $t+1$ (y-axis), plotted against earnings at age 69 (x-axis). For example, 0.16 on the y-axis implies that 16 percent of workers change employers from age t to age $t+1$. Earnings are measured relative to the kink, shown at zero on the x-axis. The data are a 20 percent random sample of the LEHD in 1990-1998. The bin width is \$800. Solid (dotted) lines show show point estimates (95 percent confidence intervals).

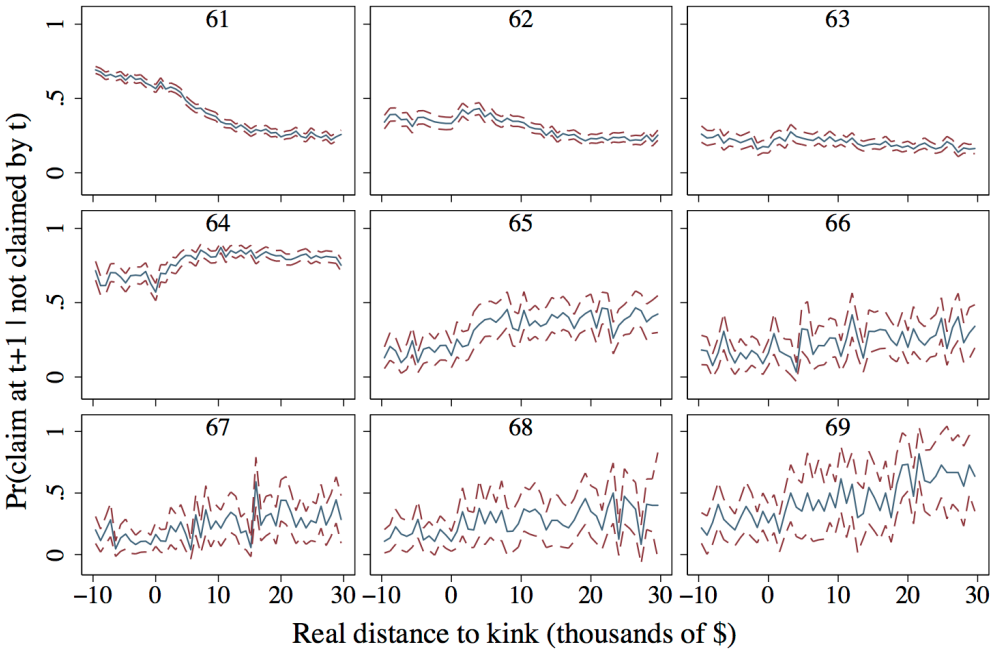
Figure F.21: Elasticity Estimates by Year, Saez (2010) Method, 1982-1994



Note: The figure shows elasticities estimated using the Saez (2010) method, by year from 1982 to 1994, among 67-68 year-old OASI claimants. We use our methods for estimating normalized excess bunching but use Saez' (2010) formula to calculate elasticities, under a constant density. This method yields the following formula:

$$\varepsilon = \left[\log\left(\frac{b}{z^*} + 1\right)\right] \Big/ \left[\log\left(\frac{1-\tau_0}{1-\tau_1}\right)\right]$$

Figure F.22: Probability of claiming OASI in year t+1 among 61-68 year-olds in year t who are not claiming, 1990-1998



Note: The figure shows the probabiliy that an individual claims OASI in year $t+1$, conditional on not claiming OASI in year t, for those ages 61-68 in year t from 1990-1998.

G Appendix: Additional Estimates of ε and ϕ

Table G.1: Estimates of Elasticity and Adjustment Cost Using Sharp Change Method and Disappearance of Kink at Age 70

	(1)	(2)	(3)
	ε	ϕ	$\varepsilon \vert \phi = 0$, Age 69
Basic	0.28	89.93	0.25
	[0.25, 0.34]***	[25.51, 240.00]***	[0.22, 0.30]***
Lognormal	0.30	\$87.19	0.27
	[0.26, 0.36]***	[31.88, 256.39]***	[0.23, 0.32]***
Benefit Enhancement	0.42	\$58.01	0.38
	[0.37, 0.50]***	[16.70, 146.86]***	[0.33, 0.45]***
Excluding FICA	0.36	\$83.01	0.33
	[0.31, 0.43]***	[23.79, 213.17]***	[0.28, 0.38]***
Bandwidth = \$500	0.20	\$17.05	0.26
	[0.22, 0.32]***	[0.69, 76.00]***	[0.16, 0.23]***
68-70 year-olds	0.30	\$79.10	0.28
	[0.27, 0.35]***	[24.61, 189.34]***	[0.25, 0.32]***

Note: The table estimates elasticities and adjustment costs using the removal of the AET at age 70, using data on 69-71 year-olds. We cannot estimate the constrained elasticity using only data on age 70 because the benefit reduction rate is zero at that age. We show these results in the Appendix, rather than the main text, because the estimates of excess bunching at age 70 are potentially affected by the coarse measure of age that we use, as explained above in the Appendix. To address this issue, we use both age 70 and age 71 in estimating these results. Using only age 70—or alternatively using only age 71—both show very similar results, which is unsurprising because Figure 3 shows that normalized excess bunching is similar at ages 70 and 71. The row labeled "68-70 year-olds" uses data from ages within this range. See also notes to Table 2.

Table G.2: Estimates of Elasticity and Adjustment Cost Using Sharp Change Method and 1990 Policy Change, Assuming no Pre-Period Bunching Attenuation

	(1)	(2)	(3)	(4)
	ε	ϕ	$\varepsilon \vert \phi = 0$	
			1990	1989
Baseline	0.22	\$139.53	0.39	0.22
	[0.18, 0.26]***	[46.25, 359.40]***	[0.33, 0.48]***	[0.18, 0.26]***
Lognormal density	0.24	\$149.87	0.43	0.24
	[0.19, 0.29]***	[42.55, 426.32]***	[0.35, 0.54]***	[0.19, 0.29]***
Benefit Enhancement	0.37	\$76.24	0.59	0.37
	[0.30, 0.43]***	[17.28, 237.63]***	[0.49, 0.73]***	[0.30, 0.43]***
Excluding FICA	0.30	\$119.84	0.50	0.30
	[0.25, 0.35]***	[32.33, 344.45]***	[0.42, 0.62]***	[0.25, 0.35]***
Bandwidth = \$500	0.24	\$83.80	0.38	0.24
	[0.19, 0.30]***	[5.86, 296.54]***	[0.30, 0.50]***	[0.19, 0.30]***
68-70 year-olds	0.31	\$106.32	0.50	0.31
	[0.27, 0.34]***	[25.21, 291.94]***	[0.42, 0.62]***	[0.27, 0.34]***

Note: The table applies the Sharp Change method to the 1990 policy change, using data from 1989 and 1990, but assumes that bunching in 1989 is not attenuated by adjustment frictions. The constrained estimate of bunching using data only from 1989 is mechanically the same as the unconstrained estimate, as both rely on the Saez (2010) formula for bunching. See also notes to Appendix Table G.1.

Table G.3: Estimates of Elasticity and Adjustment Cost Using Sharp Change Method and Elimination of Earnings Test in 2000 for 66-69 Year-Olds

	(1)	(2)	(3)
	ε	ϕ	$\varepsilon \vert \phi = 0$, 1999
Baseline	0.10	\$23.23	0.09
	[0.03, 0.19]***	[0.00, 331.52]*	[0.04, 0.14]***
Lognormal density	0.10	\$20.17	0.09
	[0.03, 0.21]***	[0.00, 346.90]*	[0.04, 0.14]***
Benefit Enhancement	0.15	\$14.93	0.14
	[0.04, 0.36]***	[0.00, 350.46]*	[0.07, 0.21]***
Excluding FICA	0.13	\$21.31	0.12
	[0.03, 0.28]***	[0.00, 402.69]*	[0.06, 0.18]***
Bandwidth = \$500	0.10	\$56.48	0.09
	[0.07, 0.21]***	[0.03, 367.05]**	[0.05, 0.13]***
1998 to 2000	0.12	\$23.09	0.11
	[0.08, 0.55]***	[0.00, 957.13]*	[0.07, 0]***

Note: The table applies the Sharp Change method to the 2000 policy change, using data on 1999 and 2000. It is not possible to perform the constrained estimate using a cross-section from the year 2000 because the marginal tax rate is zero in this year, implying that the elasticity is undefined. The final row uses data from 1998 and 2000, rather than 1999 and 2000. Note that the left endpoint of the confidence interval is sometimes at 0.00 even though the p-value indicates significance at the 5 percent level; this is because a few of the bootstrap replications show point estimates that are positive and very small so round to 0.00. See also notes to Appendix Table G.1.

Table G.4: Estimates of Elasticity and Adjustment Cost Using Comparative Static Method and 1990 Policy Change

	(1)	(2)	(3)	(4)
	ε	ϕ	$\varepsilon \vert \phi = 0$	
			1993	1989
Baseline	0.21	\$0.00	0.21	0.21
	[0.18, 0.34]***	[0.00, 902.56]	[0.14, 0.31]***	[0.17, 0.26]***
Lognormal density	0.24	\$33.35	0.22	0.23
	[0.20, 0.36]***	[0.00, 577.01]	[0.15, 0.34]***	[0.19, 0.29]***
Benefit Enhancement	0.37	\$78.09	0.32	0.36
	[0.30, 0.50]***	[0.00, 352.03]	[0.21, 0.47]***	[0.29, 0.43]***
Excluding FICA	0.30	\$58.54	0.27	0.29
	[0.25, 0.42]***	[0.00, 492.47]	[0.18, 0.40]***	[0.24, 0.35]***
Bandwidth = \$500	0.25	\$0.00	0.25	0.25
	[0.21, 0.80]***	[0.00, 4188.55]	[0.20, 0.35]***	[0.19, 0.32]***
1988 and 1993	0.26	\$146.33	0.21	0.25
	[0.23, 0.33]***	[0.00, 408.86]	[0.14, 0.31]***	[0.22, 0.28]***
Linear Approximation	0.46	\$379.10	0.30	0.39
	[0.58, 0.60]***	[351.27, 406.93]***	[0.19, 0.41]***	[0.31, 0.47]***

Note: The table applies the Comparative Static method to the 1990 policy change, using data from 1989 and 1993. "Linear approximation" refers to estimates based on the linearized formulas presented in the text. See also notes to Appendix Table G.1.